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UGC Pattern

Calculus of Variations

*(For B.A./B.Sc.(Hons.) & M.A./M.Sc., of all Indian Universities and Other
Competitive Examinations viz. N.E.T., G.A.T.E. etc.)*



By

Mukesh Kumar Singh

M.Sc. (Gold Medalist), J.R.F. (N.E.T.), Ph.D.

**Head, Department of Basic Science,
Uttar Pradesh Textile Technology Institute,
(Formerly Known as G.C.T.I.), Kanpur (U.P.)**



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About the Book



The Book “**Calculus of Variations**” lucidly covers topics as under described:

- ⊙ Variational problems with fixed boundaries,
- ⊙ Variational problems with moving boundaries,
- ⊙ Sufficient conditions for an extremum,
- ⊙ Variational method for boundary value problem (for ordinary and partial differential equation).

Key Features:

- ⊙ Objective Questions along with Hints and Solutions for N.E.T. and G.A.T.E. examination.
- ⊙ Miscellaneous Exercises at the end of each Chapter.
- ⊙ A large number of Problems solved in step by step manner.

About the Author

Dr. Mukesh Kumar Singh is presently **Head Department of Basic Science, Uttar Pradesh Textile Technology Institute** (formerly known as **G.C.T.I., Kanpur**). Prior to this he was lecturer, Mathematics at **UPIC, Varanasi**. He has an appreciable academic carrier. He obtained his **B.Sc.** and **M.Sc.** degree from **University of Allahabad** and was awarded **Gold Medal in M.Sc.** .

He has also qualified **J.R.F. (N.E.T.)** of **C.S.I.R.**. He was awarded **Ph.D.** in the field of **Fluid Dynamics**. He has authored several other Books in the field of Mathematics.

Preface



“**Calculus of Variations**” forms an important branch of Mathematics which plays a vital role in **Applied Mathematics**.

This text has been designed to meet the requirements of the **Undergraduate** and **Postgraduate** Students as well as those appearing for N.E.T., G.A.T.E. and several other **Competitive Examinations**.

The present book titled “**Calculus of Variations**” presents a theoretical foundation of subject matter coupled with its applications. A number of worked out examples have been given and sufficient number of problems are included in the exercise to cover the topics dealt in the book.

I am greatly indebted to my friends and well wishers, without their valuable support and encouragement this dream venture would remain unfulfilled.

The author wish to express their thanks to **Mr. S.K. Rastogi**, *Managing Director*; **Mr. Sugam Rastogi**, *Executive Director*, **Mrs. Kanupriya Rastogi** *Director* and **entire team** of **KRISHNA Prakashan Media (P) Ltd.**, **Meerut** for bringing out this book in the present nice form.

Any suggestion or comments from the readers for the improvement of the book will be gratefully acknowledged.

July, 2014



— *The Author*

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Leonhard Euler (1707-1783)

Leonhard Euler was born in based **Switzerland** on **15 April 1707**. His father **Paul Euler** was a Pastor of the reformed church. He studied in the University of Basel during the period 1720-1723 and got **Doctoral Degree** under the supervision of **Johann Bernoulli**. In 1730, he became a professor of Physics and later of Mathematics in **St. Petersburg, Russia**.

Euler worked in almost all areas of Mathematics such as Geometry, Trigonometry, Calculus, Algebra Number Theory as well as Fluid Mechanics and other areas of Physics, even after he became totally blind in 1766.

Euler is the only mathematician to have two numbers named after him: the important Euler's number in Calculus, e approximately equal to 2.71828, and the Euler Constant γ (**gamma**) approximately equal to 0.57721.

Due to the Pioneering studies of Euler, the development of "**Calculus of Variations**" as an independent mathematical discipline along with its own method of investigation was possible.

In 1783, Euler passed away at age of 76. Few of the notable student, he gave to the world are **Nicolas fuss**, **Johann Hennert** and **Joseph Louis Lagrange**.

Chapter 1



Variational Problems with Fixed Boundaries

1.1 Calculus of Variation

The calculus of variation is a powerful tool of modern mathematics for the solution of problems in dynamics of rigid body, optimization of orbits and vibration problems.

The origin of calculus of variation is the generalization of the elementary theory of maxima and minima of function of single and more variable.

The main aspects of subject “Calculus of variations” is as follows :

1. To explore the maximum or minimum of functionals defined over a class of functions.
 2. **Geodesic curve** : To find the geodesics curve on the surface which means that finding the shortest curve joining two point on the surface.
 3. **Brachistochrone problem (Bernoulli–1696)** : If a smooth body is allowed to slide down a smooth curve from point A to B under gravity, then to determine the curve along which the time taken will be the least.
-

4. **Minimal surface** : To determine which curve will yield the least area of surface of revolution.

5. **Iso-perimetric problem** : In this problem, our aim is to find what a curve of given perimeter will enclose the maximum area.

1.2 Functionals

A function whose values are determined by one or several function is called **functional**.

i.e., A function of function is called **functional**.

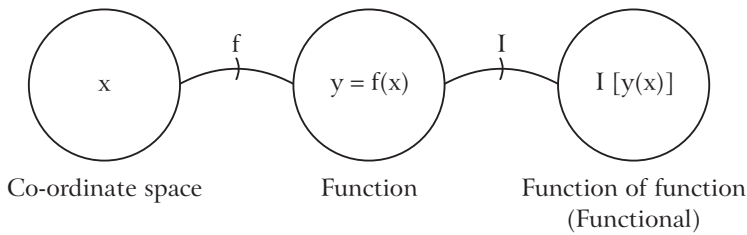


Fig. 1.1

1.2.1 Example of Functional

Example : A simple example of functional is the shortest length of a curve through the point (x_0, y_0) and (x_1, y_1) .

For, let $y = y(x)$ be curve in xy -plane joining the point (x_0, y_0) and (x_1, y_1) .

Let ds be elementary arc length of curve, then

$$ds^2 = dx^2 + dy^2$$

or

$$ds^2 = \left[1 + \left(\frac{dy}{dx} \right)^2 \right] dx^2$$

or

$$ds = \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{1/2} dx$$

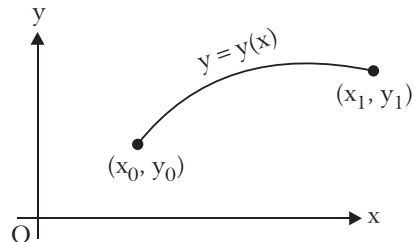


Fig. 1.2

i.e.,

$$ds = \sqrt{1 + y'^2} dx \quad \text{where } y' = \frac{dy}{dx}$$

Hence total length of curve is

$$L[y(x)] = \int_{x_0}^{x_1} \sqrt{1 + y'^2} dx$$

This is a functional.

The another example of functional is the area S of a surface bounded by given curve C . Similarly the co-ordinates of centre of gravity of a curve, and time of transit between two given points of a curve are examples of functionals.

Remark : In general, it is required to find the curve $y = y(x)$ where $y(x_0) = y_0$ and $y(x_1) = y_1$ such that the given function $F(x, y, y')$, the definite integral

$$I[y(x)] = \int_{x_0}^{x_1} F(x, y, y') dx$$

is maximum or minimum *i.e.*, extremum.

This integral is known as **functional**.

Note : In differential calculus, we find the maximum or minimum value of functions.

But calculus of variation deals with the problems of maxima or minima of functionals.

Definition : A function $I[y(x)]$ is said to be linear if it satisfies

- (i) $I[y_1(x) + y_2(x)] = I[y_1(x)] + I[y_2(x)]$
- (ii) $I[c y(x)] = c I[y(x)]$.

Definition : A function $I[y(x)]$ is maximum on a curve $y = y_1(x)$ if the value of $I[y(x)]$ on any curve close to $y = y_1(x)$ do not exceed $I[y_1(x)]$.

It means that

$$\Delta I = I[y(x)] - I[y_1(x)] \leq 0$$

and

$$\Delta I = 0 \text{ on } y = y_1(x)$$

In case of minimum of $I[y(x)]$

$$\Delta I \geq 0.$$

1.3 Extremal

A function $y = y(x)$ which extremizes a functional is called extremal or extremizing function.

1.4 Euler's Equation

[C.C.S. University, Meerut 2008]

(Necessary condition for existence of extremal) :

The necessary condition for functional

$$I[y(x)] = \int_{x_0}^{x_1} F(x, y, y') dx$$

to be maximum or minimum is that

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

where F is twice continuously differentiable function and value of $x_0, x_1, y(x_0)$ and $y(x_1)$ are prescribed.

Proof. Consider the functional

$$I[y(x)] = \int_{x_0}^{x_1} F(x, y, y') dx \quad \dots(1)$$

Let $y = y(x)$ be extremal of functional and $\bar{y}(x)$ be neighbourhood of $y(x)$ such that

$$\bar{y}(x) = y(x) + \varepsilon \eta(x) \quad \dots(2)$$

where ε is small parameter and $\eta(x)$ be an arbitrary function such that

$$\eta(x_0) = \eta(x_1) = 0. \quad \dots(3)$$

From (1),

$$I[\bar{y}(x)] = \int_{x_0}^{x_1} F(x, \bar{y}, \bar{y}') dx.$$

Using (2), we get

$$I[\bar{y}(x)] = \int_{x_0}^{x_1} F(x, y + \varepsilon \eta, y' + \varepsilon \eta') dx$$

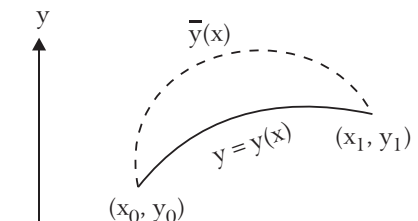


Fig. 1.3

which is function of ε .

$$\therefore I(\epsilon) = \int_{x_0}^{x_1} F(x, y + \epsilon\eta, y' + \epsilon\eta') dx$$

$$\text{or } I(\epsilon) = \int_{x_0}^{x_1} \left[F(x, y, y') + \epsilon\eta \frac{\partial F}{\partial y} + \epsilon\eta' \frac{\partial F}{\partial y'} + 0(\epsilon^2) \right] dx$$

where $0(\epsilon^2)$ is term containing ϵ^2 and higher power of ϵ^2 ,

it is called Big Oh

[by Taylor's theorem of function of several variable]

$$\therefore \frac{dI}{d\epsilon} = \int_{x_0}^{x_1} \left[0 + \eta \frac{\partial F}{\partial y} + \eta' \frac{\partial F}{\partial y'} + 0(\epsilon) \right] dx$$

$$\text{or } \frac{dI}{d\epsilon} = \int_{x_0}^{x_1} \left[\eta \frac{\partial F}{\partial y} + \eta' \frac{\partial F}{\partial y'} + 0(\epsilon) \right] dx.$$

The necessary condition for existence of extremal is

$$\left(\frac{dI}{d\epsilon} \right)_{\epsilon=0} = 0$$

$$\Rightarrow \int_{x_0}^{x_1} \left[\eta \frac{\partial F}{\partial y} + \eta' \frac{\partial F}{\partial y'} \right] dx = 0$$

$$\Rightarrow \int_{x_0}^{x_1} \eta \frac{\partial F}{\partial y} dx + \int_{x_0}^{x_1} \eta' \frac{\partial F}{\partial y'} dx = 0$$

$$\Rightarrow \int_{x_0}^{x_1} \eta \frac{\partial F}{\partial y} dx + \left[\eta(x) \frac{\partial F}{\partial y'} \right]_{x_0}^{x_1} - \int_{x_0}^{x_1} \eta(x) \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) dx = 0$$

$$\Rightarrow \int_{x_0}^{x_1} \eta \frac{\partial F}{\partial y} dx + 0 - \int_{x_0}^{x_1} \eta(x) \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) dx \quad [\because \eta(x_0) = \eta(x_1) = 0]$$

$$\Rightarrow \int_{x_0}^{x_1} \eta(x) \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right] dx = 0.$$

Since $\eta(x)$ is an arbitrary. Therefore, we have

$$\boxed{\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0}$$

$$\text{or } F_y - \frac{d}{dx} (F_{y'}) = 0$$

which is required **Euler's equation**.

Note : It is also known as Euler-Lagrange's equation.

1.5 Other Form of Euler's Equation

$$1. \quad \frac{d}{dx} F(x, y, y') = \frac{\partial F}{\partial x} \frac{dx}{dx} + \frac{\partial F}{\partial y} \frac{dy}{dx} + \frac{\partial F}{\partial y'} \frac{dy'}{dx}$$

$$\text{or} \quad \frac{dF}{dx} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} y' + \frac{\partial F}{\partial y'} y'' \quad \dots(1)$$

$$\text{But} \quad \frac{d}{dx} \left(y' \frac{\partial F}{\partial y'} \right) = y' \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{\partial F}{\partial y'} y'' \quad \dots(2)$$

On subtracting (2) from (1), we have

$$\frac{dF}{dx} - \frac{d}{dx} \left(y' \frac{\partial F}{\partial y'} \right) = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} y' - y' \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right)$$

$$\begin{aligned} \text{or} \quad \frac{d}{dx} \left[F - y' \frac{\partial F}{\partial y'} \right] - \frac{\partial F}{\partial x} &= y' \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right] \\ &= y(0) \quad [\text{by Euler's equation}] \\ &= 0 \end{aligned}$$

Hence

$$\frac{d}{dx} \left[F - y' \frac{\partial F}{\partial y'} \right] - \frac{\partial F}{\partial x} = 0$$

which is an another form of Euler's equation.

2. We know that $\frac{\partial F}{\partial y'}$ is also a function of x, y, y' say $\phi(x, y, y')$

$$\therefore \quad \frac{\partial F}{\partial y'} = \phi(x, y, y').$$

Differentiate w.r.t. x , we get

$$\begin{aligned} \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) &= \frac{d}{dx} \phi(x, y, y') \\ &= \frac{\partial \phi}{\partial x} \frac{dx}{dx} + \frac{\partial \phi}{\partial y} \frac{dy}{dx} + \frac{\partial \phi}{\partial y'} \frac{dy'}{dx} \\ &= \frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} y' + \frac{\partial \phi}{\partial y'} y'' \end{aligned}$$

$$\begin{aligned}
&= \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial y'} \right) + \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial y'} \right) y' + \frac{\partial}{\partial y'} \left(\frac{\partial F}{\partial y'} \right) y'' \\
&= \frac{\partial^2 F}{\partial x \partial y'} + y' \frac{\partial^2 F}{\partial y \partial y'} + y'' \frac{\partial^2 F}{\partial y'^2}
\end{aligned}$$

Hence, Euler equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0 \text{ becomes}$$

$$\frac{\partial F}{\partial y} - \frac{\partial^2 F}{\partial x \partial y'} - y' \frac{\partial^2 F}{\partial y \partial y'} - y'' \frac{\partial^2 F}{\partial y'^2} = 0$$

or

$$F_y - F_{xy'} - y' F_{yy'} - y'' F_{y'^2} = 0$$

This is third form of Euler's equation.

1.6 Solutions of Euler's Equation

Every solution of Euler's equation which satisfies the boundary conditions is called an **Extremal** or a stationary function of the problem.

1.7 Particular Cases of Euler's Equation

Case I : When $F = F(x, y)$.

$$\therefore \frac{\partial F}{\partial y'} = 0.$$

Then Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0 \text{ becomes}$$

$$\frac{\partial F}{\partial y} - \frac{d}{dx} (0) = 0$$

$$\frac{\partial F}{\partial y} = 0$$

Example 1 : Extremize

$$I[y(x)] = \int_0^1 (x \sin y + \cos y) dx$$

$$y(0) = 0, \quad y(1) = \pi/2.$$

Solution : Here

$$F(x, y, y') = x \sin y + \cos y.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow x \cos y - \sin y - \frac{d}{dx} (0) = 0$$

$$\Rightarrow x \cos y - \sin y = 0$$

$$\Rightarrow x \cos y = \sin y.$$

$$\therefore \tan y = x$$

 $\Rightarrow$

$$y = \tan^{-1} x$$

$$\therefore y(0) = \tan^{-1} 0 = 0, \quad y(1) = \tan^{-1} 1 = \frac{\pi}{2}.$$

This is required extremal which satisfying given boundary conditions.

Example 2 : Extremize

$$I[y(x)] = \int_1^e (xe^y - ye^x) dx$$

$$y(1) = 1, \quad y(e) = e.$$

Solution : Here $F = xe^y - ye^x$.

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow xe^y - e^x - \frac{d}{dx}(0) = 0$$

$$\Rightarrow xe^y - e^x = 0$$

$$\Rightarrow xe^y = e^x$$

$$\therefore e^y = \frac{e^x}{x}.$$

Taking log both sides, we get

$$y \log e = \log \left(\frac{e^x}{x} \right)$$

$$\Rightarrow y = \log e^x - \log x$$

i.e.,

$$y(x) = x - \log x$$

$$\text{Here, } y(1) = 1 - \log 1 = 1 - 0 = 1$$

$$y(e) = e - \log e = e - 1.$$

Hence this is not extremal because it does not satisfy given boundary condition.

It will be extremal if

$$y(1) = 1, \quad y(e) = e - 1.$$

$$\text{Case II : When } F = M(x, y) + y' N(x, y).$$

$$\therefore \frac{\partial F}{\partial y} = \frac{\partial M}{\partial y} + y' \frac{\partial N}{\partial y}$$

$$\text{and } \frac{\partial F}{\partial y'} = N(x, y).$$

Hence Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0 \text{ becomes}$$

$$\frac{\partial M}{\partial y} + y' \frac{\partial N}{\partial y} - \frac{d}{dx} [N(x, y)] = 0$$

$$\Rightarrow \frac{\partial M}{\partial y} + y' \frac{\partial N}{\partial y} - \left[\frac{\partial N}{\partial x} + y' \frac{\partial N}{\partial y} \right] = 0$$

$$\Rightarrow \frac{\partial M}{\partial y} + y' \frac{\partial N}{\partial y} - \frac{\partial N}{\partial x} - y' \frac{\partial N}{\partial y} = 0$$

$$\Rightarrow \frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 0$$

i.e.,

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

This is condition of exactness of differential equation

$$M(x, y) dx + N(x, y) dy = 0.$$

Example 3 : Extremize

$$I[y(x)] = \int_0^1 (y^2 + y' x^2) dx$$

$$y(0) = 0, \quad y(1) = 1.$$

Solution : Here

$$F = y^2 + y' x^2.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 2y - \frac{d}{dx} (x^2) = 0$$

$$\Rightarrow 2y - 2x = 0$$

i.e.,

$$y = x$$

This is required extremal which satisfy given boundary conditions.

Case III : When $F = F(y')$.

$$\therefore \frac{\partial F}{\partial y} = 0.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 0 - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

On integrating, we get

$$\frac{\partial F}{\partial y'} = \text{constant}$$

or $F_{y'} = \text{constant}$

Example 4 : Find the curve of least length joining two points in a plane.

[C.C.S. Uni. Meerut 2008, 2009]

Solution : Consider the functional

$$\begin{aligned} I[y(x)] &= \int_{x_0}^{x_1} ds \\ &= \int_{x_0}^{x_1} \sqrt{1 + y'^2} dx \end{aligned}$$

$$[\because ds = \sqrt{1 + y'^2} dx]$$

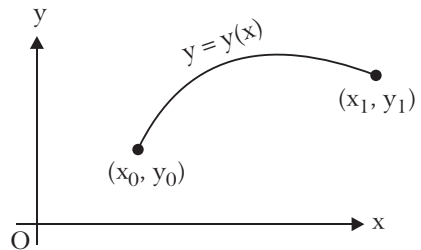


Fig. 1.4

Here $F = \sqrt{1 + y'^2}$.

$$\therefore \frac{\partial F}{\partial y} = 0, \quad \frac{\partial F}{\partial y'} = \frac{2y'}{2\sqrt{1 + y'^2}} = \frac{y'}{\sqrt{1 + y'^2}}.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 0 - \frac{d}{dx} \left(\frac{y'}{\sqrt{1 + y'^2}} \right) = 0$$

$$\Rightarrow \frac{d}{dx} \left(\frac{y'}{\sqrt{1+y'^2}} \right) = 0$$

On integrating, we get

$$\frac{y'}{\sqrt{1+y'^2}} = \text{constant} = c_1$$

$$\text{i.e.,} \quad y' = c_1 \sqrt{1+y'^2}.$$

Squaring, we get

$$y'^2 = c_1^2 (1+y'^2)$$

$$\Rightarrow (1-c_1^2) y'^2 = c_1^2$$

$$\text{or} \quad y'^2 = \frac{c_1^2}{1-c_1^2}$$

$$\text{or} \quad y' = \frac{c_1}{\sqrt{1-c_1^2}} = c \text{ (constant)}$$

$$\text{i.e.,} \quad \frac{dy}{dx} = c.$$

Integrating, we get

$$y = cx + d$$

This is required extremal which represent a straight line.

Case IV : When $F = F(x, y')$.

$$\therefore \frac{\partial F}{\partial y} = 0.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$



$$\Rightarrow 0 - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\text{i.e.,} \quad \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

On integrating, we get

$$\frac{\partial F}{\partial y'} = \text{constant}$$

$$\text{or} \quad F_{y'} = \text{constant}.$$

Example 5 : Find the curve, the time taken along which the least, when velocity at any point of it is $v = x$.

Solution : Consider the functional

$$\begin{aligned} I[y(x)] &= \int_{x_0}^{x_1} \frac{dS}{v} \\ &= \int_{x_0}^{x_1} \frac{\sqrt{1 + y'^2}}{x} dx \end{aligned}$$

$$[\because ds = \sqrt{1 + y'^2} dx]$$

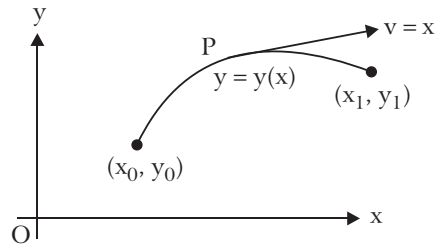


Fig. 1.5

$$\text{Here} \quad F = F(x, y') = \frac{\sqrt{1 + y'^2}}{x}.$$

$\therefore$ By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 0 - \frac{d}{dx} \left(\frac{y'}{x \sqrt{1 + y'^2}} \right) = 0$$

$$\text{or} \quad \frac{d}{dx} \left(\frac{y'}{x \sqrt{1 + y'^2}} \right) = 0.$$

On integrating, we get

$$\frac{y'}{x \sqrt{1 + y'^2}} = \text{constant} = c_1' \text{ (say).}$$

$$\therefore \frac{y'}{x \sqrt{1 + y'^2}} = c_1'. \quad \dots(1)$$

Put $y' = \tan t$, i.e., $\frac{dy}{dx} = \tan t \quad \dots(2)$

We get $\frac{\tan t}{x \sec t} = c_1'$

$$\Rightarrow x = \frac{1}{c_1'} \sin t$$

$$\Rightarrow x = c_1 \sin t, \text{ where } c_1 = \frac{1}{c_1'}. \quad \dots(3)$$

Now since

$$\begin{aligned} \frac{dy}{dt} &= \frac{dy}{dx} \cdot \frac{dx}{dt} \\ &= \tan t \cdot c_1 \cos t \end{aligned}$$

i.e., $\frac{dy}{dt} = c_1 \sin t.$

Integrating, we get

$$y = -c_1 \cos t + c_2$$

or $y - c_2 = -c_1 \cos t. \quad \dots(4)$

From (3) and (4),

$$\begin{aligned} x^2 + (y - c_2)^2 &= c_1^2 \sin^2 t + c_1^2 \cos^2 t \\ &= c_1^2 (\sin^2 t + \cos^2 t) \\ &= c_1^2 \end{aligned}$$

$$\therefore \boxed{x^2 + (y - c_2)^2 = c_1^2}$$

This is required extremal which represent circles.

Case V : When $F = F(y, y')$.

[C.C.S. University Meerut 2006, 2010]

Now
$$\frac{dF}{dx} = \frac{\partial F}{\partial y} \frac{dy}{dx} + \frac{\partial F}{\partial y'} \frac{dy'}{dx}$$

or
$$\frac{dF}{dx} = y' \frac{\partial F}{\partial y} + y'' \frac{\partial F}{\partial y'}$$

or
$$\frac{d}{dx} (F) = \left(y' \frac{\partial}{\partial y} + y'' \frac{\partial}{\partial y'} \right) F.$$

$\therefore \frac{d}{dx} = y' \frac{\partial}{\partial y} + y'' \frac{\partial}{\partial y'}. \quad \dots(1)$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

or
$$F_y - \left(y' \frac{\partial}{\partial y} + y'' \frac{\partial}{\partial y'} \right) F_{y'} = 0 \quad [\text{By (1)}]$$

or
$$F_y - y' \frac{\partial}{\partial y} (F_{y'}) - y'' \frac{\partial}{\partial y'} (F_{y'}) = 0$$

or
$$F_y - y' F_{yy'} - y'' F_{y'y'} = 0$$

Now, multiplying both sides by y' , we get

$$y' F_y - y'^2 F_{yy'} - y' y'' F_{y'y'} = 0$$

or
$$\frac{d}{dx} [F - y' F_{y'}] = 0.$$

Integrating, we get

$$F - y' F_{y'} = \text{constant}$$

This is required condition for necessary condition of existence of extremal.

Example 6 : (Brachis to Chrono's Problem). Find the curve on which a particle will slide from one point to another point in the shortest time under the action of gravity. (Friction and resistance of media are ignored).

[C.C.S. University Meerut 2006, 2007, 2008; CSJM Kanpur M. Sc. 2010]

Solution : Let a particle slide on the curve OA from O with zero velocity.

At time t , let $P(x, y)$ be position of particle.

Then velocity (v) at $P(x, y)$ is given by

$$v^2 = u^2 + 2gh$$

$$\therefore v^2 = 0 + 2gy$$

$$[\because u = 0, h = y]$$

$$\text{or } v = \sqrt{2gy}.$$

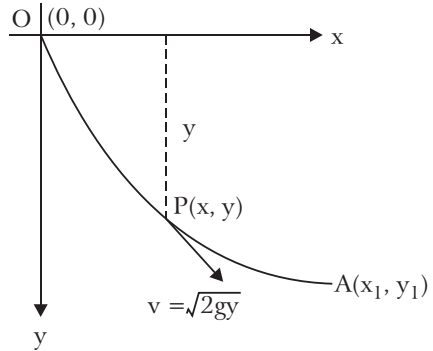


Fig. 1.6

$\therefore$ Time taken by the particle from O to A is

$$\begin{aligned} T &= \int_0^{x_1} \frac{ds}{v} \\ &= \int_0^{x_1} \frac{\sqrt{1+y'^2}}{\sqrt{2gy}} dx \quad [\because ds = \sqrt{1+y'^2} dx] \end{aligned}$$

Here $F \equiv F(y, y') = \frac{\sqrt{1+y'^2}}{\sqrt{2gy}}.$

$\therefore$ Necessary condition for existence of extremal is

$$F - y' F_{y'} = \text{constant}$$

or
$$\frac{\sqrt{1+y'^2}}{\sqrt{2gy}} - y' \frac{y'}{\sqrt{1+y'^2} \sqrt{2gy}} = \text{constant}$$

or
$$\frac{\sqrt{1+y'^2}}{\sqrt{y}} - \frac{y'^2}{\sqrt{y} \sqrt{1+y'^2}} = \text{constant}$$

or
$$\frac{1+y'^2 - y'^2}{\sqrt{y} \sqrt{1+y'^2}} = \text{constant}$$

or
$$\frac{1}{\sqrt{y} \sqrt{1+y'^2}} = \text{constant.} \quad \dots(1)$$

Put $y' = \cot t$

i.e., $\frac{dy}{dx} = \cot t. \quad \dots(2)$

Then (1) becomes

$$\frac{1}{\sqrt{y} \operatorname{cosec} t} = c_1'$$

or $\sqrt{y} = \frac{1}{c_1'} \sin t.$

Squaring both sides, we get

$$y = \frac{1}{c_1'^2} \sin^2 t$$

or $y = c_1 \sin^2 t, \text{ where } c_1 = \frac{1}{c_1'^2} = \text{constant}$

or $y = \frac{c_1}{2} (1 - \cos 2t). \quad \dots(3)$

Now $\frac{dx}{dt} = \frac{dx}{dy} \frac{dy}{dt}$

$$= \tan t \frac{c_1}{2} 2 \sin 2t \quad [\text{By (2) and (3)}]$$

$$= c_1 \frac{\sin t}{\cos t} 2 \sin t \cos t$$

$$= c_1 \cdot 2 \sin^2 t$$

$$\frac{dx}{dt} = c_1 (1 - \cos 2t).$$

Integrating w.r.t. t , we get

$$x = c_1 \left(t - \frac{\sin 2t}{2} \right)$$

or $x = \frac{c_1}{2} (2t - \sin 2t). \quad \dots(4)$

From (3) and (4), we get parametric equation of curve

$$x = \frac{c_1}{2} (2t - \sin 2t)$$

$$y = \frac{c_1}{2} (1 - \cos 2t).$$

Take $\frac{c_1}{2} = a, 2t = \theta$

$\therefore$

$$x = a (\theta - \sin \theta)$$

$$y = a (1 - \cos \theta)$$

which is cycloid.

Hence extremals are cycloid.

Example 7 : (Minimum Surface of Revolution) : *Show that the area of the surface of revolution of a curve $y = y(x)$ is*

$$2\pi \int_{x_1}^{x_2} y \sqrt{1 + y'^2} dx$$

Hence, show that for the area to be minimum, the curve must be catenary.

[CSJM Kanpur M. Sc. 2010; C.C.S. Uni. Meerut 2010]

Solution : The area of surface of revolution of curve

$$y = f(x)$$

joining the point (x_0, y_0) and (x_1, y_1) is

$$\begin{aligned} S[y(x)] &= \int_{x_0}^{x_1} 2\pi y ds \\ &= \int_{x_0}^{x_1} 2\pi y \sqrt{1 + y'^2} dx. \end{aligned}$$

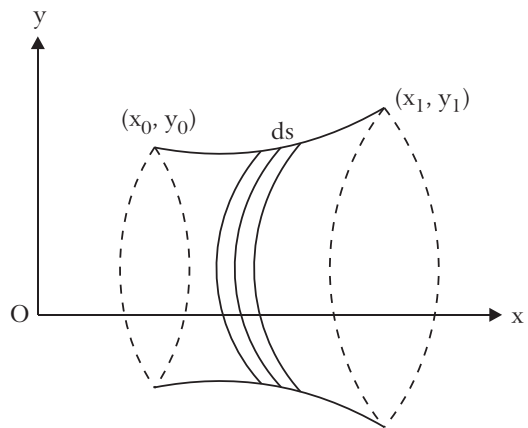


Fig. 1.7

Here $F - F(y, y') = 2\pi y \sqrt{1 + y'^2}$.

∴ The necessary condition for existence of extremal is

$$F - y' F_{y'} = \text{constant}$$

$$\text{i.e.,} \quad 2\pi y \sqrt{1 + y'^2} - y' \cdot 2\pi y \frac{y'}{\sqrt{1 + y'^2}} = \text{constant}$$

$$\text{or} \quad y \sqrt{1 + y'^2} - \frac{yy'^2}{\sqrt{1 + y'^2}} = \text{constant}$$

$$\text{or} \quad \frac{y(1 + y'^2) - yy'^2}{\sqrt{1 + y'^2}} = \text{constant}$$

$$\text{or} \quad \frac{y}{\sqrt{1 + y'^2}} = \text{constant} = c_1 \text{ (say).}$$

$$\therefore \quad \frac{y}{\sqrt{1 + y'^2}} = c_1 \quad \dots(1)$$

$$\text{Put} \quad y' = \sinh t$$

$$\text{i.e.,} \quad \frac{dy}{dx} = \sinh t. \quad \dots(2)$$

Therefore (1) becomes

$$\frac{y}{\cosh t} = c_1$$

$$\Rightarrow \quad y = c_1 \cosh t. \quad \dots(3)$$

$$\text{Now} \quad \frac{dx}{dt} = \frac{dx}{dy} \frac{dy}{dt}$$

$$= \operatorname{cosech} t c_1 \sinh t \quad [\text{By (2) and (3)}]$$

$$\therefore \quad \frac{dx}{dt} = c_1.$$

Integrating w.r.t. t , we get

$$x = c_1 t + c_2$$

$$\Rightarrow \quad t = \frac{x - c_2}{c_1}.$$

Put in (3), we get

$$y = c_1 \cosh\left(\frac{x - c_2}{c_1}\right)$$

which is the equation of catenary.

This is required extremal.

Problem Set

Problem 1 : Find the extremal of the functional

$$I[y(x)] = \int_a^b (y^2 + y'^2 + 2ye^x) dx.$$

[CSJM, Kanpur M. Sc. 2009]

Solution : Here $F = y^2 + y'^2 + 2ye^x$

$$\therefore \frac{\partial F}{\partial y} = 2y + 2e^x$$

$$\frac{\partial F}{\partial y'} = 2y'.$$

Therefore, by Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 2y + 2e^x - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow y + e^x - \frac{d^2 y}{dx^2} = 0$$

$$\text{or } \frac{d^2 y}{dx^2} - y = e^x$$

$$\text{or } (D^2 - 1)y = e^x. \quad \dots(1)$$

Auxiliary equation is

$$m^2 - 1 = 0$$

$$\Rightarrow m = \pm 1.$$

$$\therefore \quad \text{C. F.} = c_1 e^x + c_2 e^{-x}$$

$$\text{P. I.} = \frac{1}{D^2 - 1} e^x$$

$$= x \frac{1}{2D} e^x$$

$$= \frac{x}{2} \frac{1}{D} e^x$$

$$= \frac{x}{2} \frac{1}{1} \cdot e^x$$

$$= \frac{x}{2} e^x.$$

Hence solution of (1) is

$$y = \text{C. F.} + \text{P. I.}$$

i.e.,

$$y = c_1 e^x + c_2 e^{-x} + \frac{x}{2} e^x$$

This is required extremal.

Problem 2 : On what curves can the functional

$$I[y(x)] = \int_0^1 (y'^2 + 12xy) dx$$

$$y(0) = 0, y(1) = 1$$

be extremized ?

[CSJM, Kanpur M. Sc. 2009, 2014]

Solution : Here $F = y'^2 + 12xy$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

i.e., $12x - \frac{d}{dx} (2y') = 0$

or $6x - \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$

or
$$6x - \frac{d^2 y}{dx^2} = 0$$

or
$$\frac{d^2 y}{dx^2} = 6x.$$

Integrating w.r.t. x , we get

$$\frac{dy}{dx} = \frac{6x^2}{2} + c_1$$

i.e.,
$$\frac{dy}{dx} = 3x^2 + c_1.$$

Again integrate w.r.t. x , we get

$$y = \frac{3x^3}{3} + c_1 x + c_2$$

or
$$y = x^3 + c_1 x + c_2 \quad \dots(1)$$

Now
$$y(0) = 0 \Rightarrow 0 + 0 + c_2 = 0$$

$$\Rightarrow c_2 = 0$$

$$y(1) = 1 \Rightarrow 1 + c_1 + c_2 = 1$$

$$\Rightarrow 1 + c_1 = 1$$

$$\Rightarrow c_1 = 0.$$

Putting value of c_1 and c_2 in (1), we get

$$y = x^3$$

This is required extremal.

Problem 3 : Show that the general solution of the Euler equation for the functional

$$\int_a^b \frac{1}{y} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

is
$$(x - h)^2 + y^2 = k^2. \quad [\text{CSJM, Kanpur M. Sc. 2010, 2012}]$$

Solution : Given the functional is

$$\begin{aligned} I[y(x)] &= \int_a^b \frac{1}{y} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\ &= \int_a^b \frac{\sqrt{1 + y'^2}}{y} dx. \end{aligned}$$

Here $F \equiv F(y, y') = \frac{\sqrt{1 + y'^2}}{y}.$

$\therefore$ The necessary condition of existence of extremal is

$$F - y' F_{y'} = \text{constant}$$

i.e.,
$$\frac{\sqrt{1 + y'^2}}{y} - y' \frac{y'}{y\sqrt{1 + y'^2}} = \text{constant}$$

or
$$\frac{\sqrt{1 + y'^2}}{y} - \frac{y'^2}{y\sqrt{1 + y'^2}} = \text{constant}$$

or
$$\frac{(1 + y'^2) - y'^2}{y\sqrt{1 + y'^2}} = \text{constant}$$

i.e.,
$$\frac{1}{y\sqrt{1 + y'^2}} = \text{constant} = c_1 \text{ (say)} \quad \dots(1)$$

Put $y' = \tan t$

i.e.,
$$\frac{dy}{dx} = \tan t. \quad \dots(2)$$

Then (1) becomes

$$\frac{1}{y \sec t} = c_1$$

$\Rightarrow y = \frac{1}{c_1} \cos t$

or $y = k \cos t, \quad \text{where } k = \frac{1}{c_1} \quad \dots(3)$

Now

$$\frac{dx}{dt} = \frac{dx}{dy} \cdot \frac{dy}{dt}$$

$$= -\cot t \, k \sin t$$

[By (2) and (3)]

$$\frac{dx}{dt} = -k \cos t.$$

Integrate, we get

$$x = -k \sin t + h, \quad \text{where } h = \text{constant}$$

or

$$x - h = -k \sin t \quad \dots(4)$$

From (3) and (4), we have

$$(x - h)^2 + y^2 = k^2 (\cos^2 t + \sin^2 t)$$

i.e.,

$$(x - h)^2 + y^2 = k^2$$

This is required extremal.

Hence extremals are circles.

Problem 4 : Find the curves on which the functional

$$\int_1^2 \frac{x^3}{y'^2} dx$$

with $y(1) = 0$ and $y(2) = 3$ can be extremized.

[CSJM, Kanpur M. Sc. 2011, 2014]

Solution : Given functional is

$$I[y(x)] = \int_1^2 \frac{x^3}{y'^2} dx.$$

Here

$$F = \frac{x^3}{y'^2}.$$

$\therefore$ By Euler equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$



$$\Rightarrow 0 - \frac{d}{dx} \left(-\frac{2x^3}{y'^3} \right) = 0$$

$$\text{or } 2 \frac{d}{dx} \left(\frac{x^3}{y'^3} \right) = 0$$

$$\text{i.e., } \frac{d}{dx} \left(\frac{x^3}{y'^3} \right) = 0.$$

Integrating w.r.t. x , we get

$$\frac{x^3}{y'^3} = \text{constant} = c_1^3 \text{ (say)}$$

$$\text{or } y'^3 = \frac{1}{c_1^3} x^3$$

$$\text{i.e., } y' = \frac{1}{c_1} x$$

$$\text{or } y' = cx \text{ where } c = \frac{1}{c_1}$$

$$\text{i.e., } \frac{dy}{dx} = cx.$$

Integrate, we get

$$y = \frac{cx^2}{2} + d, \text{ where } c \text{ and } d \text{ are constant.} \quad \dots(1)$$

$$\text{Now, } y(1) = 0 \Rightarrow \frac{c}{2} + d = 0 \quad \dots(2)$$

$$y(2) = 3 \Rightarrow 2c + d = 3 \quad \dots(3)$$

Solving (2) and (3), we get

$$c = 2 \text{ and } d = -1.$$



Putting value of c and d in (1), we get

$$y = \frac{2x^2}{2} - 1$$

or

$$y = x^2 - 1$$

This is required extremal.

Problem 5 : Find the extremals of the functional

$$I[y(x)] = \int_{x_0}^{x_1} \frac{1+y^2}{y'^2} dx. \quad [\text{CSJM, Kanpur M. Sc. 2011}]$$

Solution : Here $F \equiv F(y, y') = \frac{1+y^2}{y'^2}.$

The necessary condition for existence of extremal is

$$F - y' F_{y'} = \text{constant}$$

$$\text{i.e.,} \quad \frac{1+y^2}{y'^2} - y' (1+y^2) \left(\frac{-2}{y'^3} \right) = \text{constant}$$

$$\text{or} \quad \frac{1+y^2}{y'^2} + \frac{2(1+y^2)}{y'^2} = \text{constant}$$

$$\text{or} \quad \frac{3(1+y^2)}{y'^2} = \text{constant}$$

$$\text{or} \quad \frac{1+y^2}{y'^2} = \text{constant} = c_1' \text{ (say)}$$

$$\text{or} \quad y'^2 = \frac{1}{c_1'} (1+y^2)$$

$$\text{or} \quad y' = \frac{1}{\sqrt{c_1'}} \sqrt{1+y^2}$$

$$\text{or} \quad y' = c_1 \sqrt{1+y^2}, \text{ where } c_1 = \frac{1}{\sqrt{c_1'}}$$

$$\text{i.e.,} \quad \frac{dy}{dx} = c_1 \sqrt{1+y^2}$$

$$\text{or} \quad \frac{dy}{\sqrt{1+y^2}} = c_1 dx.$$

Integrating, we get

$$\sinh^{-1} y = c_1 x + c_2$$

or

$$y = \sinh (c_1 x + c_2)$$

This is required extremal.

Problem 6 : Extremize the functional

$$I[y(x)] = \int_{x_0}^{x_1} (y'^2 - y^2 + 4y \sin x) dx.$$

Solution : Here $F = y'^2 - y^2 + 4y \sin x$.

$$\therefore \frac{\partial F}{\partial y} = -2y + 4 \sin x$$

$$\frac{\partial F}{\partial y'} = 2y'.$$

$\therefore$ By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow -2y + 4 \sin x - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow -y + 2 \sin x - \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

$$\Rightarrow \frac{d^2 y}{dx^2} + y = 2 \sin x$$

$$\text{or} \quad (D^2 + 1)y = 2 \sin x. \quad \dots(1)$$

Auxiliary equation is

$$m^2 + 1 = 0 \Rightarrow m = \pm i$$

$$\therefore \quad \text{C. F.} = c_1 \cos x + c_2 \sin x$$

$$\begin{aligned} \text{Now,} \quad \text{P. I.} &= \frac{1}{D^2 + 1} 2 \sin x \\ &= 2 \frac{1}{D^2 + 1} \sin x \\ &= 2 \frac{1}{2D} \sin x \\ &= \frac{1}{D} \sin x \\ &= \int \sin x \, dx = -\cos x. \end{aligned}$$

Hence solution is

$$y = \text{C. F.} + \text{P. I.}$$

$$\Rightarrow \quad y = c_1 \cos x + c_2 \sin x - \cos x.$$

This is required extremal.

Problem 7 : Find the extremal of the functional

$$V[y(x)] = \int_0^\pi [(y')^2 - y^2 + 4y \cos x] \, dx$$

with $y(0) = 0$, $y(\pi) = 0$.

[CSJM, Kanpur M. Sc. 2013]

Solution : Here $F = (y')^2 - y^2 + 4y \cos x$.

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$i.e., \quad -2y + 4 \cos x - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow \quad -2y + 4 \cos x - 2 \frac{d^2 y}{dx^2} = 0$$

or
$$\frac{d^2 y}{dx^2} + y = -2 \cos x$$

or
$$(D^2 + 1) y = -2 \cos x, \quad \text{where } D \equiv \frac{d}{dx} \quad \dots(1)$$

$\therefore$ Auxiliary equation is

$$m^2 + 1 = 0$$

$\Rightarrow m = \pm i$

$\therefore$ C. F. = $c_1 \cos x + c_2 \sin x$

Now P. I. = $\frac{1}{D^2 + 1} (-2 \cos x)$

$$= -2 \frac{1}{D^2 + 1} \cos x$$

$$= -2x \frac{1}{2D} \cos x$$

$$= -x \int \cos x \, dx$$

$$= -x \sin x.$$

Hence solution of (1) is

$$y = \text{C.F.} + \text{P.I.}$$

i.e., $y = c_1 \cos x + c_2 \sin x - x \sin x. \quad \dots(2)$

Now $y(0) = 0 \Rightarrow c_1 = 0$

$$y(\pi) = 0 \Rightarrow -c_1 + c_2 \cdot 0 - 0 = 0$$

$$\Rightarrow c_2 \cdot 0 = 0$$

$$\Rightarrow c_2 \text{ is arbitrary.}$$

Therefore (2) becomes

$$y = c_2 \sin x - x \sin x$$

or

$$y = c \sin x - x \sin x \quad \text{where } c_2 = c = \text{constant}$$

This is required extremal.

Problem 8 : Extremize the functional

$$I[y(x)] = \int_0^{2\pi} \left[\left(\frac{d}{dx} y \right)^2 - y^2 \right] dx$$

$$y(0) = 1, y(2\pi) = 1.$$

[CSIR NET, June 2011]

Solution : Here

$$F = \left(\frac{dy}{dx} \right)^2 - y^2$$

$$i.e., \quad F = y'^2 - y^2.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$i.e., \quad -2y - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow \quad y + \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

$$i.e., \quad \frac{d^2 y}{dx^2} + y = 0 \quad \dots(1)$$

Auxiliary equation is

$$m^2 + 1 = 0$$

$$\Rightarrow \quad m = \pm i.$$

$\therefore$ Solution of (1) is

$$y = c_1 \cos x + c_2 \sin x \quad \dots(2)$$

Now,

$$y(0) = 1 \Rightarrow c_1 = 1$$

$$y(2\pi) = 1 \Rightarrow c_1 \cos 2\pi + c_2 \sin 2\pi = 1$$

$$\Rightarrow 1 + c_2 \cdot 0 = 1$$

$$\Rightarrow c_2 \cdot 0 = 0$$

$$\Rightarrow c_2 \text{ is arbitrary,}$$

Therefore (2) becomes

$$y = \cos x + c_2 \sin x$$

This is required extremal.

This shows that extremal has **infinitely** many solutions.

Problem 9 : Show that the variational problem of extremizing the functional

$$I[y(x)] = \int_1^3 y(3x - y) dx$$

$$y(3) = 4, \quad y(1) = 1$$

has no solution.

[CSIR NET, June 2012]

Solution : Here

$$F = y(3x - y)$$

or

$$F = 3xy - y^2.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 3x - 2y - \frac{d}{dx} (0) = 0$$

$$\Rightarrow 3x - 2y = 0$$

$$\Rightarrow y = \frac{3}{2}x.$$

This is not extremal because it does not satisfy the given boundary condition

$$y(1) = 1.$$

Hence given functional does not have any solution.

Problem 10 : Test for an extremum of functional

$$I[y(x)] = \int_0^1 (xy + y^2 - 2y^2 y') dx$$

$$y(0) = 1, y(1) = 2.$$

Solution : Here

$$F = xy + y^2 - 2y^2 y'.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow x + 2y - 4yy' - \frac{d}{dx} (-2y^2) = 0$$

$$\text{or } x + 2y - 4yy' + 4yy' = 0$$

$$\text{or } x + 2y = 0$$

$$\Rightarrow y = -\frac{x}{2}.$$

$$\text{Now, } y(0) = 0, y(1) = -1/2.$$

This extremal does not satisfy given boundary condition $y(0) = 1, y(1) = 2$.

Hence there is not extremal.

Problem 11 : Find the stationary value or extremal of the functional

$$\int_0^{\pi/2} (y'^2 - y^2 + 2xy) dx$$

with $y(0) = 0$ and $y(\pi/2) = 0$.

Solution : Here $F = y'^2 - y^2 + 2xy$(1)

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

From (1),

$$\Rightarrow -2y + 2x - \frac{d}{dx} (2y') = 0$$

or
$$-y + x - \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

or
$$\frac{d^2 y}{dx^2} + y = x$$

or
$$(D^2 + 1) y = x. \quad \dots(2)$$

Auxiliary equation is

$$m^2 + 1 = 0$$

$$\Rightarrow m = \pm i.$$

$\therefore$ C. F. = $c_1 \cos x + c_2 \sin x$.

Now, P. I. = $\frac{1}{D^2 + 1} x$

$$= (1 + D^2)^{-1} x$$

$$= (1 - D^2 + D^4 - \dots) x$$

$$= x - D^2 x + D^4 x - \dots$$

$$= x - 0 + 0 - \dots$$

$$= x.$$

Hence solution of (2) is

$$y = \text{C. F.} + \text{P. I.}$$

i.e., $y = c_1 \cos x + c_2 \sin x + x. \quad \dots(3)$

Now $y(0) = 0 \Rightarrow c_1 = 0$

$$y(\pi/2) = 0 \Rightarrow 0 + c_2 + \pi/2 = 0$$

$$\Rightarrow c_2 = -\pi/2.$$

Putting value of c_1 and c_2 in (3), we get

$$y = \frac{-\pi}{2} \sin x + x$$

i.e., $y = x - \frac{\pi}{2} \sin x$

This is required extremal.

Problem 12 : Find the stationary value or extremal of the functional

$$\int_1^2 \frac{\sqrt{1+y'^2}}{x} dx$$

with $y(1) = 0, y(2) = 1.$

Solution : Here $F = \frac{\sqrt{1+y'^2}}{x}. \quad \dots(1)$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

From (1),

$$\Rightarrow 0 - \frac{d}{dx} \left(\frac{y'}{x\sqrt{1+y'^2}} \right) = 0$$

or
$$\frac{d}{dx} \left(\frac{y'}{x\sqrt{1+y'^2}} \right) = 0.$$

Integrate, we get

$$\frac{y'}{x\sqrt{1+y'^2}} = \text{constant} = c_1 \text{ (say)}$$

$$\text{or} \quad y'^2 = c_1^2 x^2 (1 + y'^2)$$

$$\text{or} \quad (1 - c_1^2 x^2) y'^2 = c_1^2 x^2$$

$$\text{or} \quad y'^2 = \frac{c_1^2 x^2}{1 - c_1^2 x^2}$$

$$\text{or} \quad y' = \frac{c_1 x}{\sqrt{1 - c_1^2 x^2}}.$$

$$\therefore \quad \frac{dy}{dx} = \frac{c_1 x}{\sqrt{1 - c_1^2 x^2}}.$$

Integrating, we get

$$y = -\frac{1}{c_1} \sqrt{1 - c_1^2 x^2} + c_2. \quad \dots(2)$$

$$\text{Now} \quad y(1) = 0 \Rightarrow -\frac{1}{c_1} \sqrt{1 - c_1^2} + c_2 = 0 \quad \dots(3)$$

$$y(2) = 1 \Rightarrow -\frac{1}{c_1} \sqrt{1 - 4c_1^2} + c_2 = 1. \quad \dots(4)$$

Now from (3),

$$\frac{1}{c_1^2} (1 - c_1^2) = c_2^2$$

$$\text{or} \quad \frac{1}{c_1^2} - 1 - c_2^2 = 0. \quad \dots(5)$$

Now from (4),

$$(1 - c_2)^2 = \frac{1}{c_1^2} (1 - 4c_1^2)$$

$$\text{or} \quad 1 + c_2^2 - 2c_2 = \frac{1}{c_1^2} - 4$$

$$\text{or} \quad \frac{1}{c_1^2} - c_2^2 - 1 = 4 - 2c_2. \quad \dots(6)$$

From (5) and (6), we get

$$4 - 2c_2 = 0 \Rightarrow c_2 = 2.$$

Putting value of c_2 in (5), we get

$$c_1 = \frac{1}{\sqrt{5}}.$$

Hence, putting value of c_1 and c_2 in (2), we get

$$y = -\sqrt{5} \sqrt{1 - \frac{x^2}{5}} + 2$$

or
$$(y - 2)^2 = 5 \left(1 - \frac{x^2}{5} \right)$$

or
$$(y - 2)^2 = 5 - x^2$$

$\therefore$

$$x^2 + (y - 2)^2 = 5$$

This is required extremal.

Problem 13 : Find the extremals of the following functionals :

(i) $\int_{x_0}^{x_1} (x + y') y' dx.$

(ii) $\int_{x_0}^{x_1} (1 + x^2 y') y' dx.$

(iii) $\int_{x_0}^{x_1} \frac{y'^2}{x^3} dx.$

Solution : (i) Consider

$$I[y(x)] = \int_{x_0}^{x_1} (x + y') y' dx.$$

Here,

$$F = (x + y') y'. \quad \dots(1)$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

From (1),

$$\Rightarrow 0 - \frac{d}{dx} (x + 2y') = 0$$

$$\text{or } \frac{d}{dx} (x + 2y') = 0.$$

Integrate, we get

$$x + 2y' = \text{constant} = c$$

$$\text{or } x + 2 \frac{dy}{dx} = c$$

$$\text{or } \frac{dy}{dx} = \frac{c}{2} - \frac{x}{2}$$

$$\text{or } \frac{dy}{dx} = c_1 - \frac{x}{2}, \quad \text{where } c_1 = c / 2$$

Integrate w.r.t. x , we get

$$y = c_1 \cdot x - \frac{x^2}{4} + c_2$$

i.e.,

$$y = -\frac{x^2}{4} + c_1 x + c_2$$

This is required extremal.

$$(ii) \quad \text{Consider } I[y(x)] = \int_{x_0}^{x_1} (1 + x^2 y') y' dx.$$

$$\text{Here } F = (1 + x^2 y') y'.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 0 - \frac{d}{dx} (1 + 2x^2 y') = 0$$

$$\text{or } \frac{d}{dx} (1 + 2x^2 y') = 0.$$

Integrate, we get

$$1 + 2x^2 y' = \text{constant} = c$$

or
$$1 + 2x^2 \frac{dy}{dx} = c$$

or
$$\frac{dy}{dx} = \frac{c-1}{2x^2}.$$

Integrate, we get

$$y = \frac{1-c}{2} \frac{1}{x} + c_2$$

or
$$y = \frac{c_1}{x} + c_2 \quad \text{where } c_1 = \frac{1-c}{2}$$

This is required extremal.

(iii) Consider
$$I[y(x)] = \int_{x_0}^{x_1} \frac{y'^2}{x^3} dx.$$

Here
$$F = \frac{y'^2}{x^3}.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

or,
$$0 - \frac{d}{dx} \left(\frac{2y'}{x^3} \right) = 0$$

or
$$\frac{d}{dx} \left(\frac{2y'}{x^3} \right) = 0.$$

Integrate, we get

$$\frac{2y'}{x^3} = \text{constant} = c_1$$

or
$$\frac{dy}{dx} = c_1 \frac{x^3}{2}.$$

Integrate

$$y = c_1 \frac{x^4}{8} + c_2$$

This is required extremal.

Problem 14 : Find the extremals of the functional and extremum value of the following :

(i) $I[y(x)] = \int_{1/2}^1 x^2 y'^2 dx$

with condition $y\left(\frac{1}{2}\right) = 1, y(1) = 2.$

(ii) $I[y(x)] = \int_0^2 (x - y')^2 dx$

subject to $y(0) = 0, y(2) = 4.$

Solution : (i) $I[y(x)] = \int_{1/2}^1 x^2 y'^2 dx. \quad \dots(1)$

Here $F = x^2 y'^2.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 0 - \frac{d}{dx} (2x^2 y') = 0$$

or $\frac{d}{dx} (x^2 y') = 0.$

Integrate, we get

$$x^2 y' = \text{constant}$$

i.e., $x^2 \frac{dy}{dx} = c$

or $dy = c \frac{dx}{x^2}.$

Integrating, we get

$$y = -\frac{c}{x} + d. \quad \dots(2)$$

Now

$$y(1/2) = 1 \Rightarrow -2c + d = 1$$

$$y(1) = 2 \Rightarrow -c + d = 2.$$

Solving, we get

$$c = 1, \quad d = 3.$$

Therefore (2) becomes

$$y = -\frac{1}{x} + 3$$

This is required extremal.

Extremum value :

Since $y = -\frac{1}{x} + 3$

$\therefore y' = \frac{1}{x^2}.$

From (1), extremum value

$$\begin{aligned} &= \int_{1/2}^1 x^2 y'^2 dx \\ &= \int_{1/2}^1 x^2 \frac{1}{x^4} dx \\ &= \int_{1/2}^1 \frac{1}{x^2} dx \\ &= \left[-\frac{1}{x} \right]_{1/2}^1 = -1 + 2 = 1. \end{aligned}$$

$\therefore$

$$\text{Extremum value} = 1$$

$$(ii) \quad I[y(x)] = \int_0^2 (x - y')^2 dx. \quad \dots(1)$$

Here $F = (x - y')^2.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow \quad 0 - \frac{d}{dx} \{-2(x - y')\} = 0$$

$$\text{or} \quad \frac{d}{dx} (x - y') = 0$$

$$\text{or} \quad 1 - \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

$$\text{or} \quad 1 - \frac{d^2 y}{dx^2} = 0$$

$$\text{or} \quad \frac{d^2 y}{dx^2} = 1.$$

Integrating, we get

$$y = \frac{x^2}{2} + cx + d. \quad \dots(2)$$

$$\text{Now} \quad y(0) = 0 \Rightarrow d = 0$$

$$y(2) = 4 \Rightarrow 2 + 2c + d = 0.$$

$$\therefore \quad c = -1, d = 0.$$

Putting these values in (2), we get

$$y = \frac{x^2}{2} - x$$

This is required extremal.

Extremum value :

Now
$$y = \frac{x^2}{2} - x.$$

$$\therefore y' = x - 1.$$

From (1), extremum value

$$\begin{aligned} &= \int_0^2 (x - y')^2 dx \\ &= \int_0^2 [x - (x - 1)]^2 dx \\ &= \int_0^2 dx = [x]_0^2 = 2. \end{aligned}$$

$\therefore$ Extremum value = 2.

Problem 15 : A light travel in a medium from one point to another point, so that the time of travel given by $\int \frac{ds}{v(x, y)}$ where s is arc length and $v(x, y)$ is the velocity of light in the medium, is maximum, show that path of travel is

$$v \frac{d^2 y}{dx^2} + \left[1 + \left(\frac{dy}{dx} \right)^2 \right] \frac{\partial v}{\partial y} - \frac{dy}{dx} \left[1 + \left(\frac{dy}{dx} \right)^2 \right] \frac{\partial v}{\partial x} = 0.$$

Solution : Here, functional is

$$\begin{aligned} I &= \int \frac{ds}{v(x, y)} \\ &= \int_{x_1}^{x_2} \frac{\sqrt{1 + y'^2}}{v(x, y)} dx \end{aligned}$$

$$\therefore F = \frac{\sqrt{1 + y'^2}}{v}$$

$$\Rightarrow \frac{\partial F}{\partial y} = - \frac{\sqrt{1 + y'^2}}{v^2} \frac{\partial v}{\partial y}$$

and
$$\frac{\partial F}{\partial y'} = \frac{y'}{v\sqrt{1 + y'^2}}.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow \frac{\partial F}{\partial y} = \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right)$$

$$\Rightarrow -\frac{\sqrt{1+y'^2}}{v^2} \frac{\partial v}{\partial y} = \frac{d}{dx} \left[\frac{y'}{v\sqrt{1+y'^2}} \right]$$

$$\Rightarrow -\frac{\sqrt{1+y'^2}}{v^2} \frac{\partial v}{\partial y} = \left[\frac{v\sqrt{1+y'^2} y'' - y' \frac{d}{dx} \{v\sqrt{1+y'^2}\}}{v^2 (1+y'^2)} \right]$$

$$\Rightarrow -(1+y'^2)^{3/2} \frac{\partial v}{\partial y} = v\sqrt{1+y'^2} y'' - y' \frac{d}{dx} \{v\sqrt{1+y'^2}\}$$

$$\Rightarrow -(1+y'^2)^{3/2} \frac{\partial v}{\partial y} = v\sqrt{1+y'^2} y'' - y' \sqrt{1+y'^2} \frac{dv}{dx} + y' \frac{vy' y''}{\sqrt{1+y'^2}}$$

$$\Rightarrow -(1+y'^2)^2 \frac{\partial v}{\partial y} = vy'' (1+y'^2) - y' (1+y'^2) \frac{dv}{dx} + vy'^2 y''$$

$$\Rightarrow -(1+y'^2)^2 \frac{\partial v}{\partial y} = vy'' - y' (1+y'^2) \frac{dv}{dx}. \quad \dots(1)$$

Since

$$v = v(x, y).$$

$$\therefore \frac{dv}{dx} = \frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} \frac{dy}{dx}$$

$$\text{or} \quad \frac{dv}{dx} = \frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} y'.$$

Put in (1), we get

$$-(1+y'^2) \frac{\partial v}{\partial y} = vy'' - y' (1+y'^2) \left\{ \frac{\partial v}{\partial x} + y' \frac{\partial v}{\partial y} \right\}$$

$$\Rightarrow vy'' - y' (1+y'^2) \left\{ \frac{\partial v}{\partial x} + y' \frac{\partial v}{\partial y} \right\} + (1+y'^2)^2 \frac{\partial v}{\partial y} = 0$$

$$\Rightarrow \quad \nu y'' - y' (1 + y'^2) \frac{\partial \nu}{\partial x} - y'^2 (1 + y'^2) \frac{\partial \nu}{\partial y} + (1 + y'^2)^2 \frac{\partial \nu}{\partial y} = 0$$

$$\text{or} \quad \nu y'' + (1 + y'^2) \frac{\partial \nu}{\partial y} - y' (1 + y'^2) \frac{\partial \nu}{\partial x} = 0$$

$$\text{i.e.,} \quad \nu \frac{d^2 y}{dx^2} + \left[1 + \left(\frac{dy}{dx} \right)^2 \right] \frac{\partial \nu}{\partial y} - \frac{dy}{dx} \left[1 + \left(\frac{dy}{dx} \right)^2 \right] \frac{\partial \nu}{\partial x} = 0. \quad \dots(2)$$

Problem 16 : In the above problem if $\nu = y$, find the nature of curve.

Solution : If $\nu = y$.

$$\therefore \quad \frac{\partial \nu}{\partial y} = 1 \quad \text{and} \quad \frac{\partial \nu}{\partial x} = 0.$$

Then differential equation (2) (in above problem) becomes

$$y \frac{d^2 y}{dx^2} + 1 + \left(\frac{dy}{dx} \right)^2 = 0$$

$$\text{or} \quad y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 + 1 = 0. \quad \dots(1)$$

In (1), put

$$p = \frac{dy}{dx}.$$

$$\therefore \quad \frac{d^2 y}{dx^2} = \frac{dp}{dx} = \frac{dp}{dy} \frac{dy}{dx} = p \frac{dp}{dy}.$$

Therefore (1) becomes

$$py \frac{dp}{dy} + p^2 + 1 = 0$$

$$\text{or} \quad \frac{p}{1 + p^2} dp + \frac{dy}{y} = 0.$$

Integrating, we get

$$\frac{1}{2} \log (1 + p^2) + \log y = k, \quad \text{where } k \text{ is constant.}$$

$$\text{or} \quad \log (1+p^2) + 2 \log y = 2k$$

$$\text{or} \quad \log (1+p^2) y^2 = 2k$$

$$y^2 (1+p^2) = e^{2k} = a^2 \text{ (say)}$$

$$\text{or} \quad p^2 = \frac{a^2 - y^2}{y^2}$$

$$\text{or} \quad p = \frac{dy}{dx} = \frac{\sqrt{a^2 - y^2}}{y}$$

$$\text{or} \quad dx = \frac{y}{\sqrt{a^2 - y^2}} dy.$$

Integrating, we get

$$x + b = \int \frac{y}{\sqrt{a^2 - y^2}} dy.$$

$$\text{Put} \quad a^2 - y^2 = z.$$

$$\Rightarrow \quad y dy = -\frac{1}{2} dz.$$

$$\therefore \quad x + b = -\frac{1}{2} \int \frac{dz}{\sqrt{z}} = -\sqrt{z} = -\sqrt{a^2 - y^2}$$

$$\text{or} \quad (x + b)^2 = a^2 - y^2$$

$$\text{or} \quad (x + b)^2 + y^2 = a^2.$$

Hence required path will be an arc of a circle.

Problem 17 : Show that the shortest line between any two points on a cylinder is a Helix.

Solution : If L be length of any line on the cylinder between p_1 and p_2 , then

$$L = \int_{p_1}^{p_2} \left[1 + r^2 \left(\frac{d\theta}{dz} \right)^2 \right]^{1/2} dz \quad \dots(1)$$

where r, θ, z are cylindrical co-ordinate with $r = \text{constant}$.

Now we have to extremize (1), we have

$$F = \left[1 + r^2 \left(\frac{d\theta}{dz} \right)^2 \right]^{1/2}.$$

or $F = [1 + r^2 \theta'^2]^{1/2}$ where $\theta' = \frac{d\theta}{dz}$.

$\therefore \frac{\partial F}{\partial \theta} = 0, \frac{\partial F}{\partial \theta'} = \frac{r^2 \theta'}{(1 + r^2 \theta'^2)^{1/2}}.$

By Euler's equation

$$\frac{\partial F}{\partial \theta} - \frac{d}{dz} \left(\frac{\partial F}{\partial \theta'} \right) = 0$$

$\Rightarrow 0 - \frac{d}{dz} \left[\frac{r^2 \theta'}{\sqrt{1 + r^2 \theta'^2}} \right] = 0$

$\Rightarrow \frac{d}{dz} \left[\frac{r^2 \theta'}{\sqrt{1 + r^2 \theta'^2}} \right] = 0.$

On integrating, we get

$$\frac{r^2 \theta'}{\sqrt{1 + r^2 \theta'^2}} = \text{constant} = c_1$$

$\Rightarrow r^2 \theta' = c_1 \sqrt{1 + r^2 \theta'^2}.$

Squaring, we get

$$r^4 \theta'^2 = c_1^2 (1 + r^2 \theta'^2)$$

or $r^2 \theta'^2 (r^2 - c_1^2) = c_1^2$

or $r^2 \theta'^2 = \frac{c_1^2}{r^2 - c_1^2}$

or $r \theta' = \frac{c_1}{\sqrt{r^2 - c_1^2}} = c_2 \text{ (say)}$

or $r \frac{d\theta}{dz} = c_2.$

Integrating w.r.t. z , we get

$$r\theta = c_2 z + c_3 \quad \dots(2)$$

which is equation of **helix**.

If at $p_1, \theta = 0, z = 0$, we have $c_3 = 0$.

Assuming that at $p_2, \theta = \theta_2$ and $z = z_2$, we get from (2)

$$c_2 = \frac{r\theta_2}{z_2}.$$

Using this condition, the equation of helix (2) becomes

$$r\theta = \left(\frac{r\theta_2}{z_2} \right) z.$$

1.8 Geodesics

Geodesics on a surface is a curve along which the distance between two points on the surface is a minimum.

To find the geodesics on surface is a variational problem

$$I = \int_{p_1}^{p_2} F(u, v, v') du$$

e.g., Geodesics on a plane are straight lines.

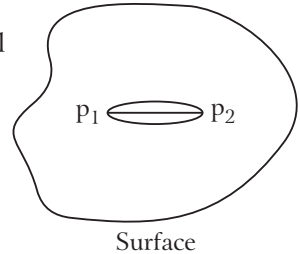


Fig. 1.8

Problem 18 : Find the Geodesics on

(i) Right circular cylinder

(ii) Right circular cone.

Solution : (i) Right circular cylinder : The element of arc on cylinder is

$$ds^2 = (dr)^2 + (r d\theta)^2 + (dz)^2$$

i.e., $r = a = \text{radius of cylinder.}$

$$\therefore ds^2 = 0 + (a d\theta)^2 + (dz)^2$$

or
$$ds^2 = \left[a^2 + \left(\frac{dz}{d\theta} \right)^2 \right] (d\theta)^2$$

$\Rightarrow ds = \sqrt{a^2 + z'^2} d\theta, \text{ where } z' = \frac{dz}{d\theta}.$

$\therefore$ Length of any line on the cylinder between p_1 and p_2 is

$$\begin{aligned} L &= \int_{p_1}^{p_2} ds \\ &= \int_{p_1}^{p_2} \sqrt{a^2 + z'^2} d\theta. \end{aligned}$$

Here
$$F = F(\theta, z, z') = \sqrt{a^2 + z'^2}.$$

By Euler's equation

$$\frac{\partial F}{\partial z} - \frac{d}{d\theta} \left(\frac{\partial F}{\partial z'} \right) = 0$$

$\Rightarrow 0 - \frac{d}{d\theta} \left(\frac{z'}{\sqrt{a^2 + z'^2}} \right) = 0$

or
$$\frac{d}{d\theta} \left[\frac{z'}{\sqrt{a^2 + z'^2}} \right] = 0.$$

Integrating, we get

$$\frac{z'}{\sqrt{a^2 + z'^2}} = c$$

$\Rightarrow z'^2 = c^2 (a^2 + z'^2)$

$\Rightarrow (1 - c^2) z'^2 = c^2 a^2$

or
$$z' = \frac{c}{\sqrt{1 - c^2}} = k \text{ (say); } k = \text{constant}$$

i.e.,
$$\frac{dz}{d\theta} = k.$$

Integrating, we get

$$z = k\theta + b$$

This curve is circular helix. Hence, extremal is circular helix.

(ii) Right circular cone

Using spherical polar co-ordinate (r, θ, ϕ) .

The element of arc on the cone is

$$ds^2 = dr^2 + r^2 \sin^2 \alpha d\phi^2$$

or
$$ds = \left[\left(\frac{dr}{d\phi} \right)^2 + r^2 \sin^2 \alpha \right]^{1/2} d\phi.$$

Hence length of any line on the cone from p_1 to p_2 is

$$\begin{aligned} L &= \int_{p_1}^{p_2} ds \\ &= \int_{p_1}^{p_2} \sqrt{r^2 \sin^2 \alpha + r'^2} d\phi. \end{aligned}$$

Here

$$F = F(r, r') = \sqrt{r^2 \sin^2 \alpha + r'^2}.$$

Therefore by Euler's equation

$$F - r' \frac{\partial F}{\partial r'} = \text{constant} = c \text{ (say)}$$

$$\Rightarrow \sqrt{r^2 \sin^2 \alpha + r'^2} - r' \frac{r'}{\sqrt{r^2 \sin^2 \alpha + r'^2}} = c$$

or
$$(r^2 \sin^2 \alpha + r'^2) - r'^2 = c \sqrt{r^2 \sin^2 \alpha + r'^2}$$

$$r^2 \sin^2 \alpha = c \sqrt{r^2 \sin^2 \alpha + r'^2}.$$

Squaring, we get

$$r^4 \sin^4 \alpha = c^2 (r^2 \sin^2 \alpha + r'^2)$$

$$\text{or} \quad r^4 \sin^2 \alpha - c^2 r^2 \sin^2 \alpha = c^2 r'^2$$

$$\Rightarrow \quad r' = \frac{dr}{d\phi} = \frac{1}{c} r \sin \alpha \sqrt{r^2 \sin^2 \alpha - c^2}$$

$$\text{or} \quad \frac{dr}{r \sin \alpha \sqrt{r^2 \sin^2 \alpha - c^2}} = \frac{1}{c} d\phi.$$

On integrating, it gives required geodesics is

$$r \sin \alpha = c \sec (\phi \sin \alpha + \beta)$$

where α and β are constants.

Exercise 1.1

1. Find the extremal of the functional

$$(i) \quad \int_{x_0}^{x_1} \frac{y'^2}{x^3} dx$$

$$(ii) \quad \int_0^1 (y^2 + y'^2) dx, \text{ where } y(0) = 0, y(1) = 1.$$

$$(iii) \quad \int_1^3 y(3x - y) dx \text{ with } y(1) = 1, y(3) = \frac{9}{2}.$$

2. Find the extremals of the functional

$$I[y(x)] = \int_0^1 (y'^2 - y^2) dx$$

subject to the following boundary conditions

$$y(0) = 0, y(1) = 1$$

3. Find the extremals of $\int_0^1 xy'^3 dx$ with conditions

$$y(0) = 0, y(1) = 1.$$

4. Test the extremal of the functional

$$I[y(x)] = \int_0^1 (xy + y^2 - 2y^2 y') dx$$

under the conditions

$$y(0) = 1, \quad y(1) = 2.$$

5. Find the extremal for the functional

$$I[y(x)] = \int_0^{\pi/2} (y'^2 - y^2) dx$$

with $y(0) = 0$, $y(\pi/2) = 1$.

6. Extremize the functional

$$I[y(x)] = \int_0^{\pi/8} (y'^2 + 2yy' - 16y^2) dx$$

with $y(0) = 0$, $y(\pi/8) = 1$.

7. Extremize the functional

$$V[y(x)] = \int_0^2 [y'^2 + 6xy + x^3] dx$$

$$y(0) = 0, \quad y(2) = 2.$$

8. Extremize the functional

$$V[y(x)] = \int_{x_0}^x (xy' + y'^2) dx.$$

9. Find the curve on which functional

$$I[y(x)] = \int_1^e (xy'^2 + yy') dx$$

with $y(1) = 0$, $y(e) = 1$

can be extremized.

10. Find the curve on which the functional

$$I[y(x)] = \int_0^1 (2xy + y^2 - y'^2) dx$$

$$y(0) = y(1) = 0$$

can be extremized.

11. Find the number of extremals for the functional

$$I[y(x)] = \int_0^\pi (4y \cos x + y'^2 - y^2) dx$$

with $y(0) = 0$, $y(\pi) = 0$.

12. Extremize the functional

$$I[y(x)] = \int_0^1 (y'^2 + 4y^2) dx$$

with $y(0) = e^2$, $y(1) = 1$.

13. If the extremal of the functional $\int_0^2 \frac{y'^2}{x} dx$ with $y(0) = \alpha$, $y(2) = \beta$ is parabola passing through origin, then find the value of α and β .

14. For what value of α , the extremal problem

$$I[y(x)] = \int_0^\pi \left\{ \left(\frac{dy}{dx} \right)^2 - y^2 \right\} dx$$

$y(0) = 1$, $y(\pi) = \alpha$ has a unique extremal.

15. Find the extremal value of the functional

$$(i) \quad \int_0^{\pi/2} (y'^2 - y^2) dx; y(0) = 0, y\left(\frac{\pi}{2}\right) = 1$$

$$(ii) \quad \int_0^1 (y'^2 + 12xy) dx; y(0) = 0, y(1) = 1.$$

16. Investigate the extremals of the functional

$$I = \int_1^2 (y' + x^2 y'^2) dx$$

under the boundary condition $y(1) = 0, y(2) = 1$.

17. Extremize the functional

$$I = \int_{x_1}^{x_2} x^n \left(\frac{dy}{dx} \right)^2 dx.$$

18. Find the geodesics on sphere.

Answers

- | | |
|-----|---|
| 1. | (i) $y = c_1 \frac{x^4}{4} + c_2$
(ii) $y = \frac{\sinh x}{\sinh 1}$
(iii) No extremal, because it does not satisfy boundary condition. |
| 2. | $y = \frac{\sin x}{\sin 1}$ |
| 3. | $y = x^{3/4}$ |
| 4. | No extremal, because it does not satisfy given boundary condition. |
| 5. | No extremal |
| 6. | $y = \sin 4x$ |
| 7. | $2y = x^3 - 2x$ |
| 8. | $y = c_1 + c_2 x - \frac{x^2}{4}$ |
| 9. | $y = \log x$ |
| 10. | $y = \frac{5}{8} (x - x^2)$ |
| 11. | Infinite |

- | | | |
|-----|--------------------------------------|---------------------|
| 12. | $y = e^{2-2x}$ | |
| 13. | $\alpha = 0, \beta = 1$ | |
| 14. | $\alpha = -1$ | |
| 15. | (i) 0 | (ii) $\frac{21}{5}$ |
| 16. | $y = x^{-1} - 1$ | |
| 17. | $y = \frac{c_1}{1-n} x^{-n+1} + c_2$ | |
| 18. | Geodesics is great circle. | |

1.9 Functional Dependent on Higher Derivatives

1.9.1 Euler-Poisson Equation

[CS]M Kanpur M. Sc. 2009]

Consider the functional

$$I[y(x)] = \int_{x_0}^{x_1} F(x, y, y', y'' \dots, y^n) dx \quad \dots(1)$$

where values of $x_0, x_1, y(x_0), y(x_1), y'(x_0), y'(x_1), \dots, y^{n-1}(x_0), y^{n-1}(x_1)$ are prescribed.

Let $y(x)$ be extremal of (1) and $\bar{y}(x)$ be neighbourhood of $y(x)$ such that

$$\bar{y}(x) = y(x) + \epsilon \eta(x) \quad \dots(2)$$

where ϵ is small parameter and $\eta(x)$ is an arbitrary function such that

$$\left. \begin{aligned} \eta(x_0) &= \eta(x_1) = 0 \\ \eta'(x_0) &= \eta'(x_1) = 0 \\ &\vdots \\ \eta^{n-1}(x_0) &= \eta^{n-1}(x_1) = 0 \end{aligned} \right\} \quad \dots(3)$$

Now from (1),

$$I[\bar{y}(x)] = \int_{x_0}^{x_1} F(x, \bar{y}, \bar{y}', \bar{y}'' \dots \bar{y}^n) dx.$$

Using (2), we get

$$\text{or} \quad I(\epsilon) = \int_{x_0}^{x_1} F(x, y + \epsilon \eta, y' + \epsilon \eta', \dots, y'' + \epsilon \eta'') dx$$

$$\begin{aligned} \text{or} \quad I(\epsilon) = \int_{x_0}^{x_1} & \left[F(x, y, y', y'', \dots, y'') + \epsilon \eta \frac{\partial F}{\partial y} + \epsilon \eta' \frac{\partial F}{\partial y'} \right. \\ & \left. + \dots \epsilon \eta'' \frac{\partial F}{\partial y''} + O(\epsilon^2) \right] dx \quad [\text{By Taylor's theorem}] \end{aligned}$$

The necessary condition for existence of extremal is

$$\left(\frac{dI}{d\epsilon} \right)_{\epsilon=0} = 0$$

$$\Rightarrow \quad \int_{x_0}^{x_1} \left[\eta \frac{\partial F}{\partial y} + \eta' \frac{\partial F}{\partial y'} + \eta'' \frac{\partial F}{\partial y''} + \dots \eta^n \frac{\partial F}{\partial y^n} \right] dx = 0$$

$$\begin{aligned} \Rightarrow \quad \int_{x_0}^{x_1} \eta(x) \frac{\partial F}{\partial y} dx + \int_{x_0}^{x_1} \eta'(x) \frac{\partial F}{\partial y'} dx + \int_{x_0}^{x_1} \eta'' \frac{\partial F}{\partial y''} dx + \dots \\ + \int_{x_0}^{x_1} \eta^n \frac{\partial F}{\partial y^n} dx = 0 \end{aligned}$$

$$\begin{aligned} \text{or} \quad \int_{x_0}^{x_1} \eta(x) \frac{\partial F}{\partial y} dx + \left[\eta(x) \frac{\partial F}{\partial y'} \right]_{x_0}^{x_1} - \int_{x_0}^{x_1} \eta(x) \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) dx \\ + \left[\eta'(x) \frac{\partial F}{\partial y''} \right]_{x_0}^{x_1} - \int_{x_0}^{x_1} \eta'(x) \frac{d}{dx} \left(\frac{\partial F}{\partial y''} \right) dx + \dots \\ + \left[\eta^{n-1}(x) \frac{\partial F}{\partial y^n} \right]_{x_0}^{x_1} - \int_{x_0}^{x_1} \eta^{n-1}(x) \frac{d}{dx} \left(\frac{\partial F}{\partial y^n} \right) dx = 0 \end{aligned}$$

Using (3), we get

$$\begin{aligned} \int_{x_0}^{x_1} \eta(x) \frac{\partial F}{\partial y} dx - \int_{x_0}^{x_1} \eta(x) \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) dx - \int_{x_0}^{x_1} \eta'(x) \frac{d}{dx} \left(\frac{\partial F}{\partial y''} \right) dx + \dots \\ - \int_{x_0}^{x_1} \eta^{n-1}(x) \frac{d}{dx} \left(\frac{\partial F}{\partial y^n} \right) dx = 0 \end{aligned}$$

$$\begin{aligned}
 \text{or} \quad & \int_{x_0}^{x_1} \eta(x) \frac{\partial F}{\partial y} dx - \int_{x_0}^{x_1} \eta(x) \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) dx - \left[\eta(x) \frac{d}{dx} \left(\frac{\partial F}{\partial y''} \right) \right]_{x_0}^{x_1} \\
 & + \int_{x_0}^{x_1} \eta(x) \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) dx \dots - \left[\eta^{n-2}(x) \frac{d}{dx} \left(\frac{\partial F}{\partial y^n} \right) \right]_{x_0}^{x_1} \\
 & + \int_{x_0}^{x_1} \eta^{n-2}(x) \frac{d^2}{dx^n} \left(\frac{\partial F}{\partial y^n} \right) dx = 0.
 \end{aligned}$$

Again using (2) and continuing the process, we get

$$\begin{aligned}
 & \int_{x_0}^{x_1} \eta(x) \frac{\partial F}{\partial y} dx - \int_{x_0}^{x_1} \eta(x) \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) dx + \int_{x_0}^{x_1} \eta(x) \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) dx \dots \\
 & + (-1)^n \int_{x_0}^{x_1} \eta(x) \frac{d^n}{dx^n} \left(\frac{\partial F}{\partial y^n} \right) dx = 0 \\
 \Rightarrow & \int_{x_0}^{x_1} \eta(x) \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) \dots + (-1)^n \frac{d^n}{dx^n} \left(\frac{\partial F}{\partial y^n} \right) \right] dx = 0
 \end{aligned}$$

Since $\eta(x)$ is an arbitrary function.

$$\therefore \quad \frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) + \dots + (-1)^n \frac{d^n}{dx^n} \left(\frac{\partial F}{\partial y^n} \right) = 0$$

$$\text{or} \quad F_y - \frac{d}{dx} (F_{y'}) + \frac{d^2}{dx^2} (F_{y''}) + \dots + (-1)^n \frac{d^n}{dx^n} (F_{y^n}) = 0.$$

This is necessary condition for existence of extremal of higher derivatives.

This is known as **Euler-Poisson equation**.

Particular Case

$$1. \quad \text{If} \quad I[y(x)] = \int_{x_0}^{x_1} F(x, y, y', y'') dx.$$

Then necessary condition for existence of extremal is

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) = 0$$

2. If
$$I[y(x)] = \int_{x_0}^{x_1} F(x, y, y', y'', y''') dx.$$

Then necessary condition for existence of extremal is

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) - \frac{d^3}{dx^3} \left(\frac{\partial F}{\partial y'''} \right) = 0$$

Problem Set

Problem 1 : Find the extremals of the functional

$$I[y(x)] = \int_a^b [(y'')^2 - 2(y')^2 + y^2] dx.$$

[CSJM, Kanpur M. Sc. 2009]

Solution : Here $F = (y'')^2 - 2(y')^2 + y^2.$

The necessary condition for existence of extremal is

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) = 0$$

$$\Rightarrow 2y - \frac{d}{dx} (-4y') + \frac{d^2}{dx^2} (2y'') = 0$$

$$\text{or } y + 2 \frac{d}{dx} \left(\frac{dy}{dx} \right) + \frac{d^2}{dx^2} \left(\frac{d^2 y}{dx^2} \right) = 0$$

$$\text{or } y + 2 \frac{d^2 y}{dx^2} + \frac{d^4 y}{dx^4} = 0$$

$$\text{or } \frac{d^4 y}{dx^4} + 2 \frac{d^2 y}{dx^2} + y = 0. \quad \dots(1)$$

Auxiliary equation is

$$m^4 + 2m^2 + 1 = 0$$

$$\Rightarrow (m^2 + 1)^2 = 0$$

$$\Rightarrow m^2 + 1 = 0, \quad m^2 + 1 = 0$$

$$\Rightarrow m = \pm i, \pm i.$$

Hence solution is

$$y = (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x$$

This is required extremal.

Problem 2 : Find the extremal of the functional

$$I[y(x)] = \int_0^{\pi/2} [(y'')^2 - y^2 + x^2] dx$$

under the conditions $y(0) = 1$, $y'(0) = 0$, $y(\pi/2) = 0$ and $y'(\pi/2) = -1$.

[CSJM, Kanpur M. Sc. 2013]

Solution : Here

$$F = (y'')^2 - y^2 + x^2.$$

The necessary condition for existence of extremal is

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) = 0$$

$$\Rightarrow -2y - \frac{d}{dx} (0) + \frac{d^2}{dx^2} (2y'') = 0$$

$$i.e., -2y + 2 \frac{d^2}{dx^2} \left(\frac{d^2 y}{dx^2} \right) = 0$$

$$\Rightarrow -y + \frac{d^4 y}{dx^4} = 0$$

$$\Rightarrow \frac{d^4 y}{dx^4} - y = 0. \quad \dots(1)$$

Auxiliary equation is

$$m^4 - 1 = 0$$

$$\Rightarrow (m^2 - 1)(m^2 + 1) = 0$$

$$\Rightarrow m = \pm 1, \pm i$$



The solution of (1) is

$$y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x \quad \dots(2)$$

$$\therefore y'(x) = c_1 e^x - c_2 e^{-x} - c_3 \sin x + c_4 \cos x. \quad \dots(3)$$

Now $y(0) = 1 \Rightarrow c_1 + c_2 + c_3 = 1. \quad \dots(4)$

$$y'(0) = 0 \Rightarrow c_1 - c_2 + c_4 = 0 \quad \dots(5)$$

$$y(\pi/2) = 0 \Rightarrow c_1 e^{\pi/2} + c_2 e^{-\pi/2} + c_4 = 0 \quad \dots(6)$$

$$y'(\pi/2) = -1 \Rightarrow c_1 e^{\pi/2} - c_2 e^{-\pi/2} - c_3 = -1 \quad \dots(7)$$

Now, (5)–(6), we get

$$(1 - e^{\pi/2}) c_1 - (1 + e^{-\pi/2}) c_2 = 0 \quad \dots(8)$$

and adding (4) and (7), we get

$$(1 + e^{\pi/2}) c_1 + (1 - e^{-\pi/2}) c_2 = 0. \quad \dots(9)$$

Solving (8) and (9), we get

$$c_1 = 0, c_2 = 0$$

From (4) and (5)

$$c_3 = 1, c_4 = 0$$

Putting value of c_1, c_2, c_3 and c_4 in (2), we get

$$y = \cos x$$

This is required extremal.

Problem 3 : Find the extremal of

$$I[y(x)] = \int_{x_0}^{x_1} (16y^2 - y''^2 + x^2) dx$$

Solution : Here

$$F = 16y^2 - y''^2 + x^2.$$

The necessary condition for existence of extremal is

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) = 0$$

i.e.,
$$32y - \frac{d}{dx} (0) + \frac{d^2}{dx^2} (-2y'') = 0$$

or
$$32y - 2 \frac{d^2}{dx^2} \left(\frac{d^2 y}{dx^2} \right) = 0$$

or
$$\frac{d^4 y}{dx^4} - 16y = 0. \quad \dots(1)$$

Auxiliary equation is

$$m^4 - 16 = 0$$

or
$$(m^2 - 4)(m^2 + 4) = 0$$

$$\Rightarrow m = \pm 2, m = \pm 2i$$

$$\Rightarrow m = \pm 2, -2i.$$

Solution of (1) is

$$y = c_1 e^{2x} + c_2 e^{-2x} + c_3 \cos 2x + c_4 \sin 2x$$

This is required extremal.

Problem 4 : Find the extremals of the functional

$$I[y(x)] = \int_{x_0}^{x_1} (2xy + y'''^2) dx. \quad [\text{CSJM, Kanpur M. Sc. 2014}]$$

Solution : Here

$$F(x, y, y') = 2xy + y'''^2. \quad \dots(1)$$

The necessary condition of existence of extremal is

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) - \frac{d^3}{dx^3} \left(\frac{\partial F}{\partial y'''} \right) = 0.$$

By (1),

$$\Rightarrow 2x - \frac{d}{dx}(0) + \frac{d^2}{dx^2}(0) - \frac{d^3}{dx^3}(2y''') = 0$$

$$\text{or } 2x - 0 + 0 - 2 \frac{d^3}{dx^3} \left(\frac{d^3 y}{dx^3} \right) = 0$$

$$\text{i.e., } x - \frac{d^6 y}{dx^6} = 0$$

$$\Rightarrow \frac{d^6 y}{dx^6} = x.$$

Integrating, we get

$$y = \frac{x^7}{7!} + c_1 x^5 + c_2 x^4 + c_3 x^3 + c_4 x^2 + c_5 x + c_6$$

This is required extremals.

Problem 5 : Find the extremal of the functional

$$I[y(x)] = \int_{x_0}^{x_1} [(y'')^2 - 2(y')^2 + y^2 - 2y \sin x] dx$$

Solution : Here

$$F = y''^2 - 2y'^2 + y^2 - 2y \sin x. \quad \dots(1)$$

The necessary condition for existence of extremal is

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) = 0.$$

By (1),

$$\Rightarrow 2y - 2 \sin x - \frac{d}{dx}(-4y') + \frac{d^2}{dx^2}(2y'') = 0$$

$$\text{or } y - \sin x + 2 \frac{d}{dx} \left(\frac{dy}{dx} \right) + \frac{d^2}{dx^2} \left(\frac{d^2 y}{dx^2} \right) = 0$$

$$\text{or } y - \sin x + 2 \frac{d^2 y}{dx^2} + \frac{d^4 y}{dx^4} = 0$$

or
$$\frac{d^4 y}{dx^4} + 2 \frac{d^2 y}{dx^2} + y = \sin x$$

or
$$(D^4 + 2D^2 + 1) y = \sin x. \quad \dots(2)$$

Auxiliary equation is

$$m^4 + 2m^2 + 1 = 0$$

$$\Rightarrow (m^2 + 1)^2 = 0$$

$$\Rightarrow m^2 + 1 = 0, \quad m^2 + 1 = 0$$

$$\Rightarrow m = \pm i, \pm i$$

$$\therefore \text{C. F.} = (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x$$

Now
$$\text{P. I.} = \frac{1}{D^4 + 2D^2 + 1} \sin x$$

$$= x \frac{1}{4D^3 + 4D} \sin x$$

$$= x \cdot x \frac{1}{12D^2 + 4} \sin x$$

$$= x^2 \frac{1}{-12 + 4} \sin x$$

$$= -\frac{x^2}{8} \sin x.$$

Hence solution of (2) is

$$y = \text{C. F.} + \text{P. I.}$$

i.e.,

$$y = (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x - \frac{x^2}{8} \sin x$$

This is required extremal.

Problem 6 : Find the extremal of the functional $\int_0^{\pi/4} (y''^2 - y^2 + x^2) dx$

that satisfies the condition $y(0) = 0$, $y'(0) = 1$, $y(\pi/4) = y'(\pi/4) = \frac{1}{\sqrt{2}}$.

Solution : Here

$$F = y''^2 - y^2 + x^2. \quad \dots(1)$$

The necessary condition for existence of extremal is

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) = 0.$$

From (1),

$$\Rightarrow -2y - \frac{d}{dx} (0) + \frac{d^2}{dx^2} (2y'') = 0$$

$$\text{or} \quad -2y + 2 \frac{d^2}{dx^2} \left(\frac{d^2 y}{dx^2} \right) = 0$$

$$\text{or} \quad \frac{d^4 y}{dx^4} - y = 0$$

$$\text{or} \quad (D^4 - 1)y = 0 \quad \left[\therefore D \equiv \frac{d}{dx} \right] \quad \dots(2)$$

Auxiliary equation is

$$m^4 - 1 = 0$$

$$\Rightarrow (m^2 - 1)(m^2 + 1) = 0$$

$$\Rightarrow m = \pm 1, m = \pm i$$

$$\Rightarrow m = \pm 1, \pm i.$$

Hence solution of (2) is

$$y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x \quad \dots(3)$$

$$\therefore y'(x) = c_1 e^x - c_2 e^{-x} - c_3 \sin x + c_4 \cos x. \quad \dots(4)$$

$$\text{Now, } y(0) = 0 \Rightarrow c_1 + c_2 + c_3 = 0 \quad \dots(5)$$

$$y'(0) = 1 \Rightarrow c_1 - c_2 + c_4 = 1 \quad \dots(6)$$

$$y(\pi/4) = \frac{1}{\sqrt{2}} \Rightarrow c_1 e^{\pi/4} + c_2 e^{-\pi/4} + \frac{1}{\sqrt{2}} c_3 + \frac{1}{\sqrt{2}} c_4 = \frac{1}{\sqrt{2}}. \quad \dots(7)$$

$$y'(\pi/4) = \frac{1}{\sqrt{2}} \Rightarrow c_1 e^{\pi/4} - c_2 e^{-\pi/4} - \frac{1}{\sqrt{2}} c_3 + \frac{1}{\sqrt{2}} c_4 = \frac{1}{\sqrt{2}}. \quad \dots(8)$$

Solving (5), (6), (7) and (8), we get

$$c_1 = c_2 = c_3 = 0 \text{ and } c_4 = 1.$$

Therefore (3) becomes

$$y = \sin x$$

This is required extremal.

Problem 7 : Determine the extremal of the functional

$$I[y(x)] = \int_{-l}^l \left(\frac{1}{2} \mu y''^2 + \lambda y \right) dx$$

subject to $y(-l) = 0, y'(-l) = 0, y(l) = 0, y'(l) = 0.$

Solution : Here

$$F = \frac{1}{2} \mu y''^2 + \lambda y. \quad \dots(1)$$

The necessary condition of existence of extremal is

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) = 0$$

From (1),

$$\Rightarrow \lambda - \frac{d}{dx} (0) + \frac{d^2}{dx^2} \left(\frac{1}{2} \mu \cdot 2 y'' \right) = 0$$

$$\text{or } \mu \frac{d^2}{dx^2} \left(\frac{d^2 y}{dx^2} \right) + \lambda = 0$$

$$\text{or } \frac{d^4 y}{dx^4} = - \frac{\lambda}{\mu}.$$

Integrate w.r.t. (x), we get

$$\frac{d^3 y}{dx^3} = - \frac{\lambda}{\mu} x + c_1'.$$

Again integrate, we get

$$\frac{d^2 y}{dx^2} = - \frac{\lambda}{\mu} \frac{x^2}{2} + c_1' x + c_2'.$$

Again integrate, we get

$$\frac{dy}{dx} = -\frac{\lambda}{\mu} \cdot \frac{1}{2} \frac{x^3}{3} + c_1' \frac{x^2}{2} + c_2' x + c_3'.$$

Again integrate, we get

$$y = -\frac{\lambda}{\mu} \frac{x^4}{4 \cdot 3 \cdot 2} + \frac{c_1'}{3 \cdot 2} x^3 + \frac{c_2' x^2}{2} + c_3' x + c_4',$$

or

$$y = -\frac{\lambda}{\mu} \frac{x^4}{4!} + c_1 x^3 + c_2 x^2 + c_3 x + c_4 \quad \dots(2)$$

where

$$c_1 = \frac{c_1'}{6}, c_2 = \frac{c_2'}{2}, c_3 = c_1' \text{ and } c_4 = c_4'.$$

Now

$$y'(x) = -\frac{\lambda}{\mu} \cdot \frac{x^3}{3!} + 3c_1 x^2 + 2c_2 x + c_3. \quad \dots(3)$$

Now from (2) and (3), we have

$$y(-l) = 0 \Rightarrow -\frac{\lambda}{\mu} \frac{l^4}{4!} - c_1 l^3 + c_2 l^2 - c_3 l + c_4 = 0 \quad \dots(4)$$

$$y'(-l) = 0 \Rightarrow -\frac{\lambda}{\mu} \frac{l^3}{3!} + 3c_1 l^2 - 2c_2 l + c_3 = 0. \quad \dots(5)$$

$$y(l) = 0 \Rightarrow -\frac{\lambda}{\mu} \frac{l^4}{4!} + c_1 l^3 + c_2 l^2 + c_3 l + c_4 = 0 \quad \dots(6)$$

$$y'(l) = 0 \Rightarrow -\frac{\lambda}{\mu} \frac{l^3}{3!} + 3c_1 l^2 + 2c_2 l + c_3 = 0 \quad \dots(7)$$

Solving (4), (5), (6) and (7), we get

$$c_1 = c_3 = 0, c_2 = \frac{2}{4!} \frac{\lambda l^2}{\mu}$$

and

$$c_4 = -\frac{\lambda}{\mu} \frac{l^4}{4!}.$$

Putting these value in (2), we get

$$y = -\frac{\lambda}{4! \mu} (x^4 - 2l^2 x^2 + l^4)$$

or

$$y = -\frac{\lambda}{24\mu} (x^2 - l^2)^2$$

This is required extremal.

1.10 Functional for Several Dependent Variable

Theorem : A necessary condition for

$$I = \int_{x_0}^{x_1} F(x, y_1, y_2, \dots, y_n, y_1', y_2', \dots, y_n') dx$$

to be extremum is that

$$\frac{\partial F}{\partial y_i} - \frac{d}{dx} \left(\frac{\partial F}{\partial y_i'} \right) = 0; \quad i = 0, 1, 2, 3, \dots, n.$$

Proof : Consider the functional

$$I = \int_{x_0}^{x_1} F(x, y_1, y_2, \dots, y_n, y_1', y_2', \dots, y_n') dx \quad \dots(1)$$

where values of $x_0, x_1, y_1(x_0), y_1(x_1), y_2(x_0), y_2(x_1), \dots, y_n(x_0), y_n(x_1)$ are prescribed.

Let $y_1(x), y_2(x), \dots, y_n(x)$ are extremal of (1) and $\bar{y}_1(x), \bar{y}_2(x), \dots, \bar{y}_n(x)$ are neighbourhood of $y_1(x), y_2(x), \dots, y_n(x)$ such that

$$\left. \begin{aligned} \bar{y}_1(x) &= y_1(x) + \varepsilon_1 \eta_1(x) \\ \bar{y}_2(x) &= y_2(x) + \varepsilon_2 \eta_2(x) \\ &\vdots \\ \bar{y}_n(x) &= y_n(x) + \varepsilon_n \eta_n(x) \end{aligned} \right\} \quad \dots(2)$$

where $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n$ are small parameter and $\eta_1(x), \eta_2(x), \dots, \eta_n(x)$ are arbitrary function such that

$$\left. \begin{aligned} \eta_1(x_0) &= \eta_2(x_0) = \dots \eta_n(x_0) = 0 \\ \eta_1(x_1) &= \eta_2(x_1) = \dots \eta_n(x_1) = 0 \end{aligned} \right\} \quad \dots(3)$$

and



Now from (1)

$$I = \int_{x_0}^{x_1} F(x, \bar{y}_1, \bar{y}_2, \dots, \bar{y}_n, \bar{y}_1', \bar{y}_2' \dots \bar{y}_n') dx.$$

Using (2), we get

$$I = \int_{x_0}^{x_1} F(x, y_1 + \epsilon_1 \eta_1, y_2 + \epsilon_2 \eta_2 \dots y_n + \epsilon_n \eta_n,$$

$$y_1' + \epsilon_1 \eta_1' \dots y_n' + \epsilon_n \eta_n') dx$$

or
$$I(\epsilon_i) = \int_{x_0}^{x_1} F(x, y_i + \epsilon_i \eta_i, y_i' + \epsilon_i \eta_i') dx, \text{ where } i = 1, 2, 3, \dots, n$$

or
$$I(\epsilon_i) = \int_{x_0}^{x_1} \left[F(x, y_i, y_i') + \epsilon_i \eta_i \frac{\partial F}{\partial y_i} + \epsilon_i \eta_i' \frac{\partial F}{\partial y_i'} + 0(\epsilon_i^2) \right] dx \quad \dots(4)$$

[By Taylor's theorem]

The necessary condition for existence of extremal is

$$\left(\frac{\partial I}{\partial \epsilon_i} \right)_{\epsilon_i=0} = 0.$$

Therefore (4) becomes

$$\int_{x_0}^{x_1} \left[\eta_i \frac{\partial F}{\partial y_i} + \eta_i' \frac{\partial F}{\partial y_i'} \right] dx = 0$$

or
$$\int_{x_0}^{x_1} \eta_i \frac{\partial F}{\partial y_i} dx + \int_{x_0}^{x_1} \eta_i' \frac{\partial F}{\partial y_i'} dx = 0$$

or
$$\int_{x_0}^{x_1} \eta_i \frac{\partial F}{\partial y_i} dx + \left[\eta_i \frac{\partial F}{\partial y_i'} \right]_{x_0}^{x_1} - \int_{x_0}^{x_1} \eta_i \frac{d}{dx} \left(\frac{\partial F}{\partial y_i'} \right) dx = 0.$$

Using (3), we get

$$\int_{x_0}^{x_1} \eta_i(x) \frac{\partial F}{\partial y_i} dx + 0 - \int_{x_0}^{x_1} \eta_i(x) \frac{d}{dx} \left(\frac{\partial F}{\partial y_i'} \right) dx = 0$$

or
$$\int_{x_0}^{x_1} \eta_i(x) \left[\frac{\partial F}{\partial y_i} - \frac{d}{dx} \left(\frac{\partial F}{\partial y_i'} \right) \right] dx = 0.$$

Since $\eta_i(x)$ is an arbitrary function.

$$\therefore \quad \boxed{\frac{\partial F}{\partial y_i} - \frac{d}{dx} \left(\frac{\partial F}{\partial y_i'} \right) = 0} \quad ; i = 1, 2, 3, \dots, n$$

or

$$F_{y_i} - \frac{d}{dx} \left(\frac{\partial F}{\partial y_i'} \right) = 0.$$

This is required necessary condition for existence of extremal of functional (1).

Particular Case

1. If
$$I = \int_{x_0}^{x_1} F(x, y_1, y_2, y_1', y_2') dx.$$

Then necessary condition for existence of extremal is

$$\boxed{\begin{aligned} \frac{\partial F}{\partial y_1} - \frac{d}{dx} \left(\frac{\partial F}{\partial y_1'} \right) &= 0 \\ \frac{\partial F}{\partial y_2} - \frac{d}{dx} \left(\frac{\partial F}{\partial y_2'} \right) &= 0 \end{aligned}}$$

2. If
$$I = \int_{x_0}^{x_1} F(x, y_1, y_2, y_3, y_1', y_2', y_3') dx.$$

Then necessary condition for existence of extremal is

$$\boxed{\begin{aligned} \frac{\partial F}{\partial y_1} - \frac{d}{dx} \left(\frac{\partial F}{\partial y_1'} \right) &= 0 \\ \frac{\partial F}{\partial y_2} - \frac{d}{dx} \left(\frac{\partial F}{\partial y_2'} \right) &= 0 \\ \frac{\partial F}{\partial y_3} - \frac{d}{dx} \left(\frac{\partial F}{\partial y_3'} \right) &= 0 \end{aligned}}$$

and



Problem Set

Problem 1 : Find the extremal of the functional

$$I[y(x), z(x)] = \int_a^b (2zy - 2y^2 + y'^2 - z'^2) dx.$$

[CSJM, Kanpur M. Sc. 2009]

Solution : Here $F = F(x, y, z, y', z') = 2zy - 2y^2 + y'^2 - z'^2$... (1)

where y and z are two dependent variable.

The necessary condition for existence of extremal is

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

and
$$\frac{\partial F}{\partial z} - \frac{d}{dx} \left(\frac{\partial F}{\partial z'} \right) = 0.$$

From (1), it becomes

$$2z - 4y - \frac{d}{dx} (2y') = 0$$

and
$$2y - \frac{d}{dx} (-2z') = 0$$

or
$$\frac{d}{dx} \left(\frac{dy}{dx} \right) + 2y = z$$

and
$$\frac{d}{dx} \left(\frac{dz}{dx} \right) + y = 0$$

or
$$\frac{d^2 y}{dx^2} + 2y = z$$
 ... (2)

and
$$\frac{d^2 z}{dx^2} + y = 0$$
 ... (3)

Now differentiate (2) twice w.r.t. x , we get

$$\frac{d^4 y}{dx^4} + 2 \frac{d^2 y}{dx^2} = \frac{d^2 z}{dx^2}.$$

Using (3), we get

$$\frac{d^4 y}{dx^4} + 2 \frac{d^2 y}{dx^2} = -y$$

$$\text{i.e.,} \quad \frac{d^4 y}{dx^4} + 2 \frac{d^2 y}{dx^2} + y = 0$$

$$\text{or} \quad (D^4 + 2D^2 + 1) y = 0.$$

Auxiliary equation is

$$m^4 + 2m^2 + 1 = 0$$

$$\text{or} \quad (m^2 + 1)^2 = 0$$

$$\Rightarrow \quad m^2 + 1 = 0, \quad m^2 + 1 = 0$$

$$\Rightarrow \quad m = \pm i, \pm i.$$

Hence solution is

$$y = (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x$$

$$\text{Now} \quad \frac{dy}{dx} = -(c_1 + c_2 x) \sin x + c_2 \cos x + (c_3 + c_4 x) \cos x + c_4 \sin x$$

$$\begin{aligned} \frac{d^2 y}{dx^2} = & -(c_1 + c_2 x) \cos x - c_2 \sin x - c_2 \sin x - (c_3 + c_4 x) \sin x \\ & + c_4 \cos x + c_4 \cos x \end{aligned}$$

$$\begin{aligned} \text{or} \quad \frac{d^2 y}{dx^2} = & -(c_1 + c_2 x) \cos x - (c_3 + c_4 x) \sin x \\ & - c_2 (2 \sin x) + c_4 (2 \cos x). \end{aligned}$$

Therefore from (2), we have

$$z = \frac{d^2 y}{dx^2} + 2y$$

$$z = (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x - 2c_2 \sin x + 2c_4 \cos x.$$

Hence, the required extremals are

$$y = (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x$$

and

$$z = (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x - 2c_2 \sin x + 2c_4 \cos x.$$

Problem 2 : Extremize

$$I[y(x), z(x)] = \int_0^{\pi/2} (y'^2 + z'^2 + 2yz) dx$$

with $y(0) = 0$, $y(\pi/2) = 1$, $z(0) = 0$, $z(\pi/2) = -1$.

Solution : Here

$$F = y'^2 + z'^2 + 2yz. \quad \dots(1)$$

$\therefore$ The necessary condition for existence of extremal is

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

and

$$\frac{\partial F}{\partial z} - \frac{d}{dx} \left(\frac{\partial F}{\partial z'} \right) = 0.$$

From (1), it becomes

$$2z - \frac{d}{dx} (2y') = 0 \Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = z$$

and

$$2y - \frac{d}{dx} (2z') = 0 \Rightarrow \frac{d}{dx} \left(\frac{dz}{dx} \right) = y$$

i.e.,

$$\frac{d^2 y}{dx^2} = z \quad \dots(2)$$

and

$$\frac{d^2 z}{dx^2} = y \quad \dots(3)$$

Now, differentiate (2), w.r.t. x twice, we get

$$\frac{d^4 y}{dx^4} = \frac{d^2 z}{dx^2}.$$

Using (3), we get

$$\frac{d^4 y}{dx^4} = y$$

or

$$\frac{d^4 y}{dx^4} - y = 0$$

or

$$(D^4 - 1) y = 0.$$

Auxiliary equation is

$$m^4 - 1 = 0$$

$$\Rightarrow (m^2 - 1)(m^2 + 1) = 0$$

$$\Rightarrow m = \pm 1, \pm i.$$

$\therefore$ The solution is

$$y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x. \quad \dots(4)$$

Now
$$\frac{dy}{dx} = c_1 e^x - c_2 e^{-x} - c_3 \sin x + c_4 \cos x$$

$$\frac{d^2 y}{dx^2} = c_1 e^x + c_2 e^{-x} - c_3 \cos x - c_4 \sin x.$$

Therefore from (2), we get

$$z = \frac{d^2 y}{dx^2} = c_1 e^x + c_2 e^{-x} - c_3 \cos x - c_4 \sin x.$$

$$\therefore z = c_1 e^x + c_2 e^{-x} - c_3 \cos x - c_4 \sin x. \quad \dots(5)$$

Now from (4) and (5), we get

$$y(0) = 0 \Rightarrow c_1 + c_2 + c_3 = 0 \quad \dots(6)$$

$$y(\pi/2) = 1 \Rightarrow c_1 e^{\pi/2} + c_2 e^{-\pi/2} + c_4 = 1 \quad \dots(7)$$

and
$$z(0) = 0 \Rightarrow c_1 + c_2 - c_3 = 0 \quad \dots(8)$$

$$z(\pi/2) = -1 \Rightarrow c_1 e^{\pi/2} + c_2 e^{-\pi/2} - c_4 = -1. \quad \dots(9)$$

Solving (6), (7), (8) and (9), we get

$$c_1 = 0, \quad c_2 = 0, \quad c_3 = 0 \quad \text{and} \quad c_4 = 1.$$

Putting value of c_1, c_2, c_3 and c_4 in (4) and (5), we get required extremals are

$$y = \sin x, \quad z = -\sin x$$

Problem 3 : Find the extremals of the functional

$$\int_0^{\pi/2} \left[2xy + \left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 \right] dx$$

with conditions

$$x(0) = 0, \quad x(\pi/2) = -1$$

$$y(0) = 0, \quad y(\pi/2) = 1.$$

[CSJM, Kanpur M. Sc. 2011]

Solution : Here

$$\begin{aligned} F &= 2xy + \left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 \\ &= 2xy + \dot{x}^2 + \dot{y}^2 \end{aligned} \quad \dots(1)$$

$$\left[\text{where } \dot{x} = \frac{dx}{dt}, \dot{y} = \frac{dy}{dt} \right]$$

The necessary condition for existence of extremal is

$$\frac{\partial F}{\partial x} - \frac{d}{dt} \left(\frac{\partial F}{\partial \dot{x}} \right) = 0$$

and

$$\frac{\partial F}{\partial y} - \frac{d}{dt} \left(\frac{\partial F}{\partial \dot{y}} \right) = 0.$$

From (1), it becomes

$$2y - \frac{d}{dt} (2\dot{x}) = 0$$

and

$$2x - \frac{d}{dt} (2\dot{y}) = 0$$

or

$$\frac{d}{dt} \left(\frac{dx}{dt} \right) = y$$

and

$$\frac{d}{dt} \left(\frac{dy}{dt} \right) = x$$

or
$$\frac{d^2 x}{dt^2} = y \quad \dots(2)$$

$$\frac{d^2 y}{dt^2} = x. \quad \dots(3)$$

Now differentiating (2) w.r.t. t twice, we get

$$\frac{d^4 x}{dt^4} = \frac{d^2 y}{dt^2}$$

or
$$\frac{d^4 x}{dt^4} = x \quad [\text{By (3)}]$$

or
$$\frac{d^4 x}{dt^4} - x = 0$$

or
$$(D^4 - 1)x = 0; \quad D \equiv \frac{d}{dt}.$$

Auxiliary equation is

$$m^4 - 1 = 0$$

$$\Rightarrow (m^2 - 1)(m^2 + 1) = 0$$

$$\Rightarrow m = \pm 1, \pm i.$$

$\therefore$ The solution is

$$x = c_1 e^t + c_2 e^{-t} + c_3 \cos t + c_4 \sin t. \quad \dots(4)$$

Now
$$\frac{dx}{dt} = c_1 e^t - c_2 e^{-t} - c_3 \sin t + c_4 \cos t$$

$$\frac{d^2 x}{dt^2} = c_1 e^t + c_2 e^{-t} - c_3 \cos t - c_4 \sin t.$$

From (2),

$$y = \frac{d^2 x}{dt^2} = c_1 e^t + c_2 e^{-t} - c_3 \cos t - c_4 \sin t$$

i.e.,
$$y = c_1 e^t + c_2 e^{-t} - c_3 \cos t - c_4 \sin t. \quad \dots(5)$$

Now from (4) and (5), we have

$$x(0) = 0 \Rightarrow c_1 + c_2 + c_3 = 0 \quad \dots(6)$$

$$x(\pi/2) = -1 \Rightarrow c_1 e^{\pi/2} + c_2 e^{-\pi/2} + c_4 = -1 \quad \dots(7)$$

$$y(0) = 0 \Rightarrow c_1 + c_2 - c_3 = 0 \quad \dots(8)$$

$$y(\pi/2) = 1 \Rightarrow c_1 e^{\pi/2} + c_2 e^{-\pi/2} - c_4 = 1. \quad \dots(9)$$

Solving (6), (7), (8) and (9), we get

$$c_1 = 0, \quad c_2 = 0, \quad c_3 = 0 \quad \text{and} \quad c_4 = -1.$$

Putting these value in (4) and (5), we get

$$x = -\sin t, \quad y = \sin t$$

This is required extremal.

Problem 4 : Show that functional

$$\int_0^1 \left[2x + \left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 \right] dt$$

such that

$$x(0) = 0, \quad y(0) = 0, \quad x(1) = 1.5, \quad y(1) = 1$$

is stationary for $x = 1 + \frac{t^2}{2}, \quad y = t.$

[CSJM, Kanpur M. Sc. 2013]

Solution : Here
$$F = 2x + \left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2$$

or
$$F = 2x + \dot{x}^2 + \dot{y}^2 \quad \dots(1)$$

The necessary condition for existence of extremal is

$$\frac{\partial F}{\partial x} - \frac{d}{dt} \left(\frac{\partial F}{\partial \dot{x}} \right) = 0$$

and

$$\frac{\partial F}{\partial y} - \frac{d}{dt} \left(\frac{\partial F}{\partial \dot{y}} \right) = 0.$$

From (1), it becomes

$$2 - \frac{d}{dt}(2\dot{x}) = 0 \Rightarrow 1 - \frac{d}{dt}\left(\frac{dx}{dt}\right) = 0$$

$$0 - \frac{d}{dt}(2\dot{y}) = 0 \Rightarrow \frac{d}{dt}\left(\frac{dy}{dt}\right) = 0$$

or
$$\frac{d^2 x}{dt^2} = 1 \quad \dots(2)$$

$$\frac{d^2 y}{dt^2} = 0. \quad \dots(3)$$

Integrating w.r.t. t of (2) and (3), we get

$$x = \frac{t^2}{2} + a_1 t + a_2 \quad \dots(4)$$

$$y = b_1 t + b_2 \quad \dots(5)$$

where a_1, a_2, b_1, b_2 are constant.

Now $x(0) = 0 \Rightarrow a_2 = 0$

$$x(1) = 1.5 \Rightarrow \frac{1}{2} + a_1 + a_2 = 1.5$$

$$y(0) = 0 \Rightarrow b_2 = 0$$

$$y(1) = 1 \Rightarrow b_1 + b_2 = 1.$$

Solving these above equations, we get

$$a_1 = 1, a_2 = 0, b_1 = 1, b_2 = 0.$$

Putting these value in (4) and (5), we get

$$x = 1 + \frac{t^2}{2}, y = t$$

This is required extremal.

1.11 Functionals Dependent on Several Independent Variables

Theorem : *The necessary condition for existence of extremals for functional*

$$I = \iint_D F(x, y, u, u_x, u_y) dx dy \quad \dots(1)$$

is

$$\frac{\partial F}{\partial u} - \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial u_x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial u_y} \right) = 0$$

where $u(x, y)$ is continuous and has continuous derivatives upto the second order and is prescribed on the region of integration D . [Nagpur Univ. 2009]

Proof : Let $u(x, y)$ be extremal of (1) and $\bar{u}(x, y)$ be neighbourhood of $u(x, y)$ such that

$$\bar{u}(x, y) = u(x, y) + \epsilon \eta(x, y) \quad \dots(2)$$

where ϵ is small parameter and $\eta(x, y)$ is an arbitrary function such that

$$\eta(x, y) = 0 \quad \dots(3)$$

on the boundary C of D .

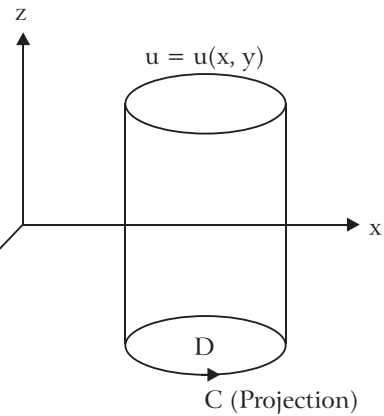


Fig. 1.9

Now from (1),

$$I [\bar{u}(x, y)] = \iint_D F(x, y, \bar{u}, \bar{u}_x, \bar{u}_y) dx dy$$

or

$$I(\epsilon) = \iint_D F(x, y, u + \epsilon \eta, u_x + \epsilon \eta_x, u_y + \epsilon \eta_y) dx dy$$

or

$$I(\epsilon) = \iint_D \left[F(x, y, u, u_x, u_y) + \epsilon \eta \frac{\partial F}{\partial u} + \epsilon \eta_x \frac{\partial F}{\partial u_x} + \epsilon \eta_y \frac{\partial F}{\partial u_y} + 0(\epsilon^2) \right] dx dy$$

[By Taylor's theorem]

The necessary condition for existence of extremal is

$$\left(\frac{\partial I}{\partial \varepsilon} \right)_{\varepsilon=0} = 0$$

$$\text{i.e.,} \quad \iint_D \left(\eta \frac{\partial F}{\partial u} + \eta_x \frac{\partial F}{\partial u_x} + \eta_y \frac{\partial F}{\partial u_y} \right) dx dy = 0$$

$$\text{or} \quad \iint_D \eta \frac{\partial F}{\partial u} dx dy + \iint_D \left(\eta_x \frac{\partial F}{\partial u_x} + \eta_y \frac{\partial F}{\partial u_y} \right) dx dy = 0. \quad \dots(4)$$

Now consider

$$\begin{aligned} \eta_x \frac{\partial F}{\partial u_x} + \eta_y \frac{\partial F}{\partial u_y} &= \frac{\partial}{\partial x} \left(\eta \frac{\partial F}{\partial u_x} \right) + \frac{\partial}{\partial y} \left(\eta \frac{\partial F}{\partial u_y} \right) \\ &\quad - \eta \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial u_x} \right) - \eta \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial u_y} \right). \end{aligned}$$

$$\begin{aligned} \therefore \quad \iint_D \left(\eta_x \frac{\partial F}{\partial u_x} + \eta_y \frac{\partial F}{\partial u_y} \right) dx dy &= \iint_D \left[\frac{\partial}{\partial x} (\eta F_{u_x}) - \frac{\partial}{\partial y} (-\eta F_{u_y}) \right] dx dy \\ &\quad - \iint_D \eta \left[\frac{\partial}{\partial x} \left(\frac{\partial F}{\partial u_x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial u_y} \right) \right] dx dy \\ &= \int_C (\eta F_{u_x} dy - \eta F_{u_y} dx) \\ &\quad - \iint_D \eta \left[\frac{\partial}{\partial x} \left(\frac{\partial F}{\partial u_x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial u_y} \right) \right] dx dy \end{aligned}$$

[Using Green's theorem]

$$= 0 - \iint_D \eta \left[\frac{\partial}{\partial x} \left(\frac{\partial F}{\partial u_x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial u_y} \right) \right] dx dy$$

[By (3) $\eta(x, y) = 0$ on C]

$$= - \iint_D \eta \left[\frac{\partial}{\partial x} \left(\frac{\partial F}{\partial u_x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial u_y} \right) \right] dx dy.$$

Therefore (4) becomes

$$\iint_D \eta \frac{\partial F}{\partial u} dx dy - \iint_D \eta \left[\frac{\partial}{\partial x} \left(\frac{\partial F}{\partial u_x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial u_y} \right) \right] dx dy = 0$$

$$\Rightarrow \iint_D \eta(x, y) \left[\frac{\partial F}{\partial u} - \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial u_x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial u_y} \right) \right] dx dy = 0.$$

Since $\eta(x, y)$ is an arbitrary function. Therefore, we have

$$\frac{\partial F}{\partial u} - \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial u_x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial u_y} \right) = 0$$

or
$$F_u - \frac{\partial}{\partial x} (F_{u_x}) - \frac{\partial}{\partial y} (F_{u_y}) = 0.$$

This is required necessary condition for existence of extremals.

This equation is known as **Euler-Ostrogradsky Equation**.

Remark : If functional $I [u(x, y)]$ contains derivative of higher order two i.e.,

$$I [u(x, y)] = \iint_D F(x, y, u, u_x, u_y, u_{xx}, u_{xy}, u_{yy}) dx dy$$

Then Euler's Ostrogradsky equation is

$$F_u - \frac{\partial}{\partial x} (F_{u_x}) - \frac{\partial}{\partial y} (F_{u_y}) + \frac{\partial^2}{\partial x^2} F_{u_{xx}} + \frac{\partial^2}{\partial x \partial y} F_{u_{xy}} + \frac{\partial^2}{\partial y^2} F_{u_{yy}} = 0$$

Problem Set

Problem 1 : Dirichlet's Problem : Find the Euler-Ostrogradsky equation for

$$I[u(x, y)] = \iint_D \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right] dx dy$$

where the values of u are prescribed on the boundary C of the domain D .

Solution : Here

$$F = F(x, y, u_x, u_y) = \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 = u_x^2 + u_y^2 \quad \dots(1)$$

By Euler's-Ostrogradsky equation

$$\frac{\partial F}{\partial u} - \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial u_x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial u_y} \right) = 0.$$

From (1)

$$\Rightarrow 0 - \frac{\partial}{\partial x} (2u_x) - \frac{\partial}{\partial y} (2u_y) = 0$$

$$\Rightarrow \frac{\partial}{\partial x} (u_x) + \frac{\partial}{\partial y} (u_y) = 0$$

or
$$\frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) = 0$$

or

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

or

$$\nabla^2 u = 0.$$

This is Laplace equation and its solution gives required extremal $u(x, y)$.

Problem 2 : Find the extremizing function for

$$I[z(x, y)] = \iint_D \left[\left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 + 2z f(x, y) \right] dx dy$$

where $f(x, y)$ is known function.

Solution : Here

$$\begin{aligned} F &= F(x, y, z, z_x, z_y) = \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 2z f(x, y) \\ &= z_x^2 + z_y^2 + 2z f(x, y). \end{aligned} \quad \dots(1)$$

By Euler-Ostrogradsky equation,

$$\frac{\partial F}{\partial z} - \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial z_x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial z_y} \right) = 0.$$

From (1),

$$\Rightarrow 2f(x, y) - \frac{\partial}{\partial x} (2z_x) - \frac{\partial}{\partial y} (2z_y) = 0$$

$$\text{or } f(x, y) - \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = 0$$

or

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = f(x, y)$$

This gives required extremal $z(x, y)$.

Problem 3 : Find the Euler's-Ostrogradsky equation for

$$I[u(x, y, z)] = \iiint_D \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial u}{\partial z} \right)^2 + 2uf(x, y, z) \right] dx dy dz$$

where $f(x, y, z)$ is known function.

Solution : Here

$$\begin{aligned} F &= F(x, y, z, u, u_x, u_y, u_z) \\ &= \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial u}{\partial z} \right)^2 + 2u f(x, y, z) \\ &= u_x^2 + u_y^2 + u_z^2 + 2u f(x, y, z) \end{aligned} \quad \dots(1)$$

By Euler-Ostrogradsky equation

$$\frac{\partial F}{\partial u} - \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial u_x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial u_y} \right) - \frac{\partial}{\partial z} \left(\frac{\partial F}{\partial u_z} \right) = 0.$$

From (1)

$$\Rightarrow 2f(x, y, z) - \frac{\partial}{\partial x} (2u_x) - \frac{\partial}{\partial y} (2u_y) - \frac{\partial}{\partial z} (2u_z) = 0$$

$$\Rightarrow f(x, y, z) - \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) - \frac{\partial}{\partial z} \left(\frac{\partial u}{\partial z} \right) = 0$$

or

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = f(x, y, z)$$

This gives required extremal $u(x, y, z)$.

Problem 4 : Find the extremizing function for

$$I[z(x, y)] = \iint_D \left[\left(\frac{\partial^2 z}{\partial x^2} \right)^2 + \left(\frac{\partial^2 z}{\partial y^2} \right)^2 + 2 \left(\frac{\partial^2 z}{\partial x \partial y} \right) - 2z f(x, y) \right] dx dy$$

where $f(x, y)$ is known function.

Solution : Here

$$\begin{aligned} F &= F(x, y, z, z_x, z_y, z_{xx}, z_{xy}, z_{yy}) \\ &= \left(\frac{\partial^2 z}{\partial x^2} \right)^2 + \left(\frac{\partial^2 z}{\partial y^2} \right)^2 + 2 \left(\frac{\partial^2 z}{\partial x \partial y} \right) - 2z f(x, y) \end{aligned}$$

$$i.e., \quad F = z_{xx}^2 + z_{yy}^2 + 2z_{xy}^2 - 2z f(x, y). \quad \dots(1)$$

By Euler-Ostrogradsky equation

$$F_z - \frac{\partial}{\partial x} (F_{z_x}) - \frac{\partial}{\partial y} (F_{z_y}) + \frac{\partial^2}{\partial x^2} (F_{z_{xx}}) + \frac{\partial^2}{\partial x \partial y} (F_{z_{xy}}) + \frac{\partial^2}{\partial y^2} (F_{z_{yy}}) = 0.$$

From (1)

$$-2f(x, y) - 0 - 0 + \frac{\partial^2}{\partial x^2} (2z_{xx}) + \frac{\partial^2}{\partial x \partial y} (2z_{xy}) + \frac{\partial^2}{\partial y^2} (2z_{yy}) = 0$$

or

$$-f(x, y) + \frac{\partial^2}{\partial x^2} \left(\frac{\partial^2 z}{\partial x^2} \right) + \frac{\partial^2}{\partial x \partial y} \left(\frac{\partial^2 z}{\partial x \partial y} \right) + \frac{\partial^2}{\partial y^2} \left(\frac{\partial^2 z}{\partial y^2} \right) = 0$$

or

$$\frac{\partial^4 z}{\partial x^4} + \frac{\partial^4 z}{\partial x^2 \partial y^2} + \frac{\partial^4 z}{\partial y^4} = f(x, y)$$

or

$$\nabla^2 \nabla^2 z = 0$$

This is **bi-harmonic** equation which gives required extremal.

1.12 Isoperimetric Problems

(Conditional Extremum)

There are some problems in which one has to find geometric figure *i.e.*, extremum under the given condition, such problems are called isoperimetric problem. Such problem is solved by Lagrange's multiplier's method.

To find the extremals of functional

$$I[y(x)] = \int_{x_0}^{x_1} f(x, y, y') dx \quad \dots(1)$$

subject to the condition (constraint)

$$J[y(x)] = \int_{x_0}^{x_1} g(x, y, y') dx = \text{constant} \quad \dots(2)$$

Consider

$$F = f + \lambda g$$

where λ is called Lagrange multiplier.

Then by Euler equation, the necessary condition for existence of extremal is

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

This gives required extremal of functional (1) under the condition (2).

Problem Set

Problem 1 : Find the extremal of functional

$$I = \int_0^{\pi} (y'^2 - y^2) dx$$

under the conditions $y(0) = 0$, $y(\pi) = 1$ and subject to constraint

$$\int_0^{\pi} y dx = 1. \quad [\text{CSJM, Kanpur M. Sc. 2014}]$$

Solution : Let
$$I = \int_0^{\pi} (y'^2 - y^2) dx$$

and
$$J = \int_0^{\pi} y dx = 1.$$

Here
$$f = y'^2 - y^2, g = y.$$

Consider
$$F = f + \lambda g$$

i.e.,
$$F = (y'^2 - y^2) + \lambda y \quad \dots(1)$$

where λ is Lagrange multiplier.

Now, by Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

From (1),

$$(-2y + \lambda) - \frac{d}{dx} (2y') = 0$$

or
$$-2y + \lambda - 2 \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

or
$$2 \frac{d^2 y}{dx^2} = \lambda - 2y$$

or
$$\frac{d^2 y}{dx^2} + y = \frac{\lambda}{2}$$

or
$$(D^2 + 1) y = \frac{\lambda}{2}. \quad \dots(2)$$



Auxiliary equation is

$$m^2 + 1 = 0$$

$$\Rightarrow m = \pm i.$$

$$\therefore \text{C. F.} = c_1 \cos x + c_2 \sin x$$

Now

$$\begin{aligned} \text{P. I.} &= \frac{1}{D^2 + 1} \frac{\lambda}{2} \\ &= \frac{\lambda}{2} \frac{1}{D^2 + 1} e^{ax} \\ &= \frac{\lambda}{2} \frac{1}{0 + 1} e^{ax} \\ &= \frac{\lambda}{2}. \end{aligned}$$

Hence solution of (2) is

$$y = \text{C. F.} + \text{P. I.}$$

$$i.e., \quad y = c_1 \cos x + c_2 \sin x + \frac{\lambda}{2}. \quad \dots(3)$$

Now

$$\begin{aligned} y(0) = 0 &\Rightarrow c_1 + \frac{\lambda}{2} = 0 \\ &\Rightarrow c_1 = -\frac{\lambda}{2} \quad \dots(4) \end{aligned}$$

$$\begin{aligned} y(\pi) = 1 &\Rightarrow -c_1 + \frac{\lambda}{2} = 1 \\ &\Rightarrow c_1 = \frac{\lambda}{2} - 1 \quad \dots(5) \end{aligned}$$

Solving (4) and (5), we get

$$c_1 = -\frac{1}{2}, \lambda = 1. \quad \dots(6)$$

Therefore (3) becomes

$$y = -\frac{1}{2} \cos x + c_2 \sin x + \frac{1}{2} \quad \dots(7)$$

Now from (2), we have

$$\int_0^{\pi} y \, dx = 1.$$

Using (7), we get

$$\int_0^{\pi} \left(-\frac{1}{2} \cos x + c_2 \sin x + \frac{1}{2} \right) dx = 1$$

$$\Rightarrow \left[-\frac{1}{2} \sin x - c_2 \cos x + \frac{1}{2} x \right]_0^{\pi} = 1$$

$$\Rightarrow 2c_2 + \frac{\pi}{2} = 1$$

$$\text{or } c_2 = \frac{1}{2} - \frac{\pi}{4}.$$

Putting value of c_2 in (7), we get

$$y = -\frac{1}{2} \cos x + \left(\frac{1}{2} - \frac{\pi}{4} \right) \sin x + \frac{1}{2}$$

or

$$y = \frac{1}{2} (1 - \cos x) + \frac{1}{4} (2 - \pi) \sin x$$

This is required extremal.

Problem 2 : Find the extremal of the functional $\int_0^2 y'^2 \, dx$ under the constraint $\int_0^2 y \, dx = 1$ given $y(0) = 0$ and $y(2) = 1$.

$$\text{Solution : Let } I = \int_0^2 y'^2 \, dx \quad \dots(1)$$

$$\text{and } J = \int_0^2 y \, dx = 1 \quad \dots(2)$$

$$\text{Here } f = y'^2, \quad g = y.$$

$$\text{Consider } F = f + \lambda g$$

$$\text{i.e., } F = y'^2 + \lambda y. \quad \dots(3)$$



By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

From (3),

$$\Rightarrow \lambda - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow \lambda - 2 \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

$$i.e., \quad 2 \frac{d^2 y}{dx^2} = \lambda$$

$$\text{or} \quad \frac{d^2 y}{dx^2} = \frac{\lambda}{2}.$$

On integrating twice, we get

$$y = \frac{\lambda}{4} x^2 + c_1 x + c_2 \quad \dots(4)$$

where c_1 and c_2 are arbitrary constant.

$$\text{Now} \quad y(0) = 0 \Rightarrow c_2 = 0$$

$$y(2) = 1 \Rightarrow \lambda + 2c_1 = 1.$$

$$\therefore c_1 = \frac{1 - \lambda}{2}, c_2 = 0. \quad \dots(5)$$

Therefore (4) becomes

$$y = \frac{\lambda}{4} x^2 + \frac{1 - \lambda}{2} x. \quad \dots(6)$$

Now from (2),

$$\int_0^2 y \, dx = 1$$

$$\Rightarrow \int_0^2 \left(\frac{\lambda}{4} x^2 + \frac{1 - \lambda}{2} x \right) dx = 1$$

$$\Rightarrow \left[\frac{\lambda}{4} \frac{x^3}{3} + \frac{1-\lambda}{2} \cdot \frac{x^2}{2} \right]_0^2 = 1$$

$$\Rightarrow \frac{8\lambda}{12} + 1 - \lambda = 1$$

$$\Rightarrow \lambda = 0.$$

Therefore (5) becomes

$$y = \frac{1}{2} x$$

This is required extremal.

Problem 3 : Find the extremals of the isoperimetric problem

$$I[y(x)] = \int_{x_0}^{x_1} y'^2 dx \text{ given that } \int_{x_0}^{x_1} y dx = c, \text{ a constant.}$$

Solution : Here $f = y'^2, g = y.$

Consider $F = f + \lambda g$

i.e., $F = y'^2 + \lambda y.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

i.e., $\lambda - \frac{d}{dx} (2y') = 0$

$$\Rightarrow \lambda - 2 \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

$$\Rightarrow \lambda - 2 \frac{d^2 y}{dx^2} = 0$$

or $\frac{d^2 y}{dx^2} = \frac{\lambda}{2}.$

On integrating twice, we get

$$y = \frac{\lambda}{4} x^2 + c_1 x + c_2 \quad \text{where } c_1 \text{ and } c_2 \text{ are constant.}$$

This is required extremal.

Problem 4 : Find the shape of the curve of the given parimeter enclosing maximum area.

or

Find the plane curve of fixed perimeter and maximum area.

[CSJM Kanpur M. Sc. 2010, 2013]

Solution : Let P be perimeter of plane curve $y = y(x)$ passing through (x_1, y_1) and (x_2, y_2) .

$$\therefore P = \int_{x_1}^{x_2} ds$$

$$i.e., \quad P = \int_{x_1}^{x_2} \sqrt{1 + y'^2} dx. \quad \dots(1)$$

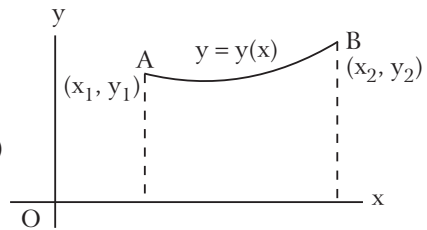


Fig. 1.10

Also, the area enclosed by the curve $y = y(x)$, x -axis and two perpendicular lines

$$A = \int_{x_1}^{x_2} y dx \quad \dots(2)$$

We have to find maximum value of (2) under the condition (1).

Here $f = y, \quad g = \sqrt{1 + y'^2}$.

By Lagrange's multiplier method

Consider $F = f + \lambda g$

$$i.e., \quad F = y + \lambda \sqrt{1 + y'^2}. \quad \dots(3)$$

∴ By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

From (3),

$$1 - \lambda \frac{d}{dx} \frac{y'}{\sqrt{1 + y'^2}} = 0.$$

On integrating, we get

$$x - \frac{\lambda y'}{\sqrt{1 + y'^2}} = \text{constant} = a$$

or

$$\frac{\lambda y'}{\sqrt{1 + y'^2}} = x - a$$

⇒

$$\lambda y' = (x - a) \sqrt{1 + y'^2}.$$

Squaring, we get

$$\lambda^2 y'^2 = (x - a)^2 (1 + y'^2)$$

or

$$[\lambda^2 - (x - a)^2] y'^2 = (x - a)^2$$

or

$$y'^2 = \frac{(x - a)^2}{[\lambda^2 - (x - a)^2]}$$

i.e.,

$$y' = \frac{(x - a)}{\sqrt{\lambda^2 - (x - a)^2}}$$

i.e.,

$$\frac{dy}{dx} = \frac{(x - a)}{\sqrt{\lambda^2 - (x - a)^2}}.$$

On integrating, we get

$$y = -\sqrt{\lambda^2 - (x - a)^2} + b, \text{ where } b \text{ is constant.}$$

or

$$y - b = -\sqrt{\lambda^2 - (x - a)^2}.$$

Squaring, we get

$$(y - b)^2 = \lambda^2 - (x - a)^2$$

i.e.,

$$(x - a)^2 + (y - b)^2 = \lambda^2$$

This is required extremal which represents a circle with centre (a, b) and radius λ .

Problem 5 : Find the solid of maximum volume formed by the revolution of a given surface.

Solution : Let the curve OA passes through the origin and it rotated about x -axis.

$\therefore$ The surface area enclosed by the curve $y = y(x)$ is

$$S = \int_0^a 2\pi y \, ds$$

$$S = \int_0^a 2\pi y \sqrt{1 + y'^2} \, dx \quad \dots(1)$$

and volume V is

$$V = \int_0^a \pi y^2 \, dx. \quad \dots(2)$$

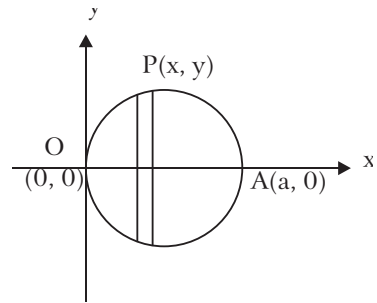


Fig. 1.11

Here we have to extremize V under given S .

Let $f = \pi y^2$, $g = 2\pi y \sqrt{1 + y'^2}$.

By Lagrange multiplier's method.

Consider $F = f + \lambda g$.

$$\therefore F = F(y, y') = \pi y^2 + \lambda 2\pi y \sqrt{1 + y'^2}. \quad \dots(3)$$

Therefore, the necessary condition for existence of extremal is

$$F - y' F_{y'} = \text{constant}.$$

From (3),

$$\Rightarrow \pi y^2 + \lambda \cdot 2\pi y \sqrt{1 + y'^2} - y' \times \frac{2\pi y y'}{\sqrt{1 + y'^2}} = \text{constant}$$

$$\Rightarrow \pi y^2 + 2\pi y \lambda \sqrt{1 + y'^2} - \frac{2\pi \lambda y y'^2}{\sqrt{1 + y'^2}} = c$$

$$\text{or } \pi y^2 + \frac{2\pi y \lambda (1 + y'^2) - 2\pi y \lambda y'^2}{\sqrt{1 + y'^2}} = c$$

$$\text{or } \pi y^2 + \frac{2\pi y \lambda}{\sqrt{1 + y'^2}} = c. \quad \dots(4)$$

Since curve passes through origin $(0, 0)$.

$$\therefore c = 0.$$

Then (4) becomes

$$\pi y^2 + \frac{2\pi y \lambda}{\sqrt{1 + y'^2}} = 0$$

$$\text{or } y + \frac{2\lambda}{\sqrt{1 + y'^2}} = 0$$

$$\text{or } y \sqrt{1 + y'^2} = -2\lambda$$

$$\text{or } \sqrt{1 + y'^2} = -\frac{2\lambda}{y}.$$

Squaring, we get

$$1 + y'^2 = \frac{4\lambda^2}{y^2}$$

$$\text{or } y'^2 = \frac{4\lambda^2}{y^2} - 1 = \frac{4\lambda^2 - y^2}{y^2}$$

$$\text{or } \left(\frac{dy}{dx}\right)^2 = \frac{4\lambda^2 - y^2}{y^2}$$

$$\text{or } \frac{dy}{dx} = \frac{\sqrt{4\lambda^2 - y^2}}{y}$$

$$\text{or } \frac{y \, dy}{\sqrt{4\lambda^2 - y^2}} = dx.$$

Integrating, we get

$$\int \frac{y \, dy}{\sqrt{4\lambda^2 - y^2}} = x + c_1$$

$$\Rightarrow -\sqrt{4\lambda^2 - y^2} = x + c_1. \quad \dots(5)$$

Since curve passes through $(0, 0)$

$$\therefore -2\lambda = c_1$$

$$i.e., \quad c_1 = -2\lambda.$$

Therefore (5) becomes

$$-\sqrt{4\lambda^2 - y^2} = x - 2\lambda.$$

Squaring, we get

$$4\lambda^2 - y^2 = (x - 2\lambda)^2$$

or

$$(x - 2\lambda)^2 + y^2 = 4\lambda^2$$

This is required extremal which represents a circle.

Hence on revolving the circle about x -axis, the solid formed is a sphere.

Problem 6 : Find the extremal of the functional

$$A = \int_{t_1}^{t_2} \frac{1}{2} (x\dot{y} - y\dot{x}) dt$$

subject to integral constraint

$$\int_{t_1}^{t_2} \sqrt{\dot{x}^2 + \dot{y}^2} = l.$$

Solution : Here

$$f = \frac{1}{2} (x\dot{y} - y\dot{x}), \quad g = \sqrt{\dot{x}^2 + \dot{y}^2}.$$

Consider

$$F = f + \lambda g$$

$$i.e., \quad F = \frac{1}{2} (x\dot{y} - y\dot{x}) + \lambda \sqrt{\dot{x}^2 + \dot{y}^2} \quad \dots(1)$$

Here x, y are dependent variable and t is independent.

Therefore, the necessary condition for existence of extremal is

$$\frac{\partial F}{\partial x} - \frac{d}{dt} \left(\frac{\partial F}{\partial \dot{x}} \right) = 0$$

$$\frac{\partial F}{\partial y} - \frac{d}{dt} \left(\frac{\partial F}{\partial \dot{y}} \right) = 0.$$

From (1), it becomes

$$\frac{1}{2} \dot{y} - \frac{d}{dt} \left(-\frac{y}{2} + \frac{\lambda 2\dot{x}}{2\sqrt{\dot{x}^2 + \dot{y}^2}} \right) = 0$$

or
$$\frac{d}{dt} \left[y - \frac{\lambda \dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \right] = 0 \quad \dots(2)$$

and
$$-\frac{1}{2} \dot{x} - \frac{d}{dt} \left[\frac{x}{2} + \frac{\lambda \dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \right] = 0$$

or
$$\frac{d}{dt} \left[x - \frac{\lambda \dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \right] = 0. \quad \dots(3)$$

Integrating (2) and (3), we get

$$y - \frac{\lambda \dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} = c_1 \quad \text{or} \quad y - c_1 = -\frac{\lambda \dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \quad \dots(4)$$

and
$$x - \frac{\lambda \dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} = c_2 \quad \text{or} \quad x - c_2 = \frac{\lambda \dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \quad \dots(5)$$

Squaring (4) and (5), we get

$$(x - c_2)^2 + (y - c_1)^2 = \lambda^2 \left(\frac{\dot{x}^2 + \dot{y}^2}{\dot{x}^2 + \dot{y}^2} \right)$$

or

$$(x - c_2)^2 + (y - c_1)^2 = \lambda^2$$

This is required extremal.

Problem 7 : Find a function $y(x)$ for which $\int_0^1 (x^2 - y'^2) dx$ is stationary given that

$$\int_0^1 y'^2 dx = 2, \quad y(0) = 0, \quad y(1) = 0.$$

Solution : Here $f = x^2 - y'^2$, $g = y^2$.

Consider $F = f + \lambda g$

$$\text{i.e.,} \quad F = (x^2 - y'^2) + \lambda y^2. \quad \dots(1)$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

From (1),

$$\Rightarrow \quad 2\lambda y - \frac{d}{dx} (-2y') = 0$$

$$\text{or} \quad \lambda y + \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

$$\text{or} \quad \frac{d^2 y}{dx^2} + \lambda y = 0$$

$$\text{or} \quad (D^2 + \lambda) y = 0.$$

Auxiliary equation is

$$m^2 + \lambda = 0$$

$$\text{i.e.,} \quad m = \pm \sqrt{\lambda} \, i.$$

$\therefore$ The solution is

$$y = c_1 \cos \sqrt{\lambda} \, x + c_2 \sin \sqrt{\lambda} \, x. \quad \dots(2)$$

$$\text{Now} \quad y(0) = 0 \Rightarrow c_1 = 0$$

$$y(1) = 1 \Rightarrow c_1 \cos \sqrt{\lambda} + c_2 \sin \sqrt{\lambda} = 0$$

$$\Rightarrow c_2 \sin \sqrt{\lambda} = 0 \quad [\because c_1 = 0]$$

$$\Rightarrow \sin \sqrt{\lambda} = 0 \text{ when } c_2 \neq 0$$

$$\Rightarrow \sin \sqrt{\lambda} = \sin n\pi$$

$$\Rightarrow \sqrt{\lambda} = n\pi.$$

Therefore (2) becomes

$$y = c_2 \sin n\pi x. \quad \dots(3)$$

Now from the constraint

$$\int_0^1 y^2 dx = 2$$

$$\Rightarrow \int_0^1 c_2^2 \sin^2 n\pi x dx = 2 \quad [\text{By (2)}]$$

$$\Rightarrow \frac{c_2^2}{2} \int_0^1 (1 - \cos 2n\pi x) dx = 2$$

$$\text{or} \quad \frac{c_2^2}{2} \left[x - \frac{\sin 2n\pi x}{2n\pi} \right]_0^1 = 2$$

$$\text{or} \quad \frac{c_2^2}{2} [1 - 0] = 2$$

$$\Rightarrow c_2^2 = 4$$

$$\text{or} \quad c_2 = \pm 2$$

Putting value of c_2 in (3), we get

$$y = \pm 2 \sin n\pi x$$

This is required extremal.

Problem 8 : Prove that the extremal of the isoperimetric problem

$I[y(x)] = \int_1^4 y'^2 dx$ with $y(1) = 3$, $y(4) = 24$ subject to condition

$\int_1^4 y dx = 36$ is a parabola.

Solution : Here $f = y'^2$, $g = y$.

Consider $F = f + \lambda g$

$$\text{i.e.,} \quad F = y'^2 + \lambda y. \quad \dots(1)$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

From (1),

$$\Rightarrow \lambda - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow \lambda - 2 \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

$$i.e., \quad \frac{d^2 y}{dx^2} = \frac{\lambda}{2}.$$

On integrating, we get

$$y = \frac{\lambda}{4} x^2 + c_1 x + c_2. \quad \dots(2)$$

$$\text{Now } y(1) = 3 \Rightarrow \frac{\lambda}{4} + c_1 + c_2 = 3 \quad \dots(3)$$

$$y(4) = 24 \Rightarrow 4\lambda + 4c_1 + c_2 = 24. \quad \dots(4)$$

Solving (3) and (4), we get

$$c_1 = 7 - \frac{5\lambda}{4}, \quad c_2 = \lambda - 4.$$

Therefore (2) becomes

$$y = \frac{\lambda}{4} x^2 + \left(7 - \frac{5\lambda}{4} \right) x + \lambda - 4. \quad \dots(5)$$

Now since

$$\int_1^4 y \, dx = 36$$

$$i.e., \quad \int_1^4 \left[\frac{\lambda}{4} x^2 + \left(7 - \frac{5\lambda}{4} \right) x + (\lambda - 4) \right] dx = 36$$

$$\text{or } \left[\frac{\lambda}{4} \frac{x^3}{3} + \left(7 - \frac{5\lambda}{4} \right) \frac{x^2}{2} + (\lambda - 4) x \right]_1^4 = 36$$

or
$$-\frac{27}{24} \lambda = -\frac{9}{2}$$

or
$$\lambda = 4.$$

Putting value of λ in (5), we get

$$y = x^2 + 2x.$$

This is required extremal which represents a parabola.

Theorem 1 : (Sturm-Liouville's Theorem). *The extremals of*

$$I = \int_{x_1}^{x_2} [p(x) y'^2 - q(x) y^2] dx$$

subject to constraint

$$J = \int_{x_1}^{x_2} r(x) y^2 dx = 1$$

are the solution of the Sturm-Liouville equation

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [q(x) + \lambda r(x)] y = 0.$$

Proof : Consider the functional

$$H = I - \lambda J$$

where λ is the Lagrange's multiplier.

Therefore, I will have extremals provided H will have extremals.

Now
$$\begin{aligned} H &= \int_{x_1}^{x_2} [p(x) y'^2 - q(x) y^2] dx - \lambda \int_{x_1}^{x_2} r(x) y^2 dx \\ &= \int_{x_1}^{x_2} [p(x) y'^2 - q(x) y^2 - \lambda r(x) y^2] dx. \end{aligned}$$

Here
$$F = p(x) y'^2 - q(x) y^2 - \lambda r(x) y^2.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

$$\therefore -2q(x) y - 2\lambda r(x) y - \frac{d}{dx} [2p(x) y'] = 0$$

$$\text{or } \frac{d}{dx} [p(x) y'] + [q(x) + \lambda r(x)] y = 0$$

which is Sturm-Liouville's equation.

Theorem 2 : *Extremals of*

$$\lambda = \frac{\int_{x_1}^{x_2} [p(x) y'^2 - q(x) y^2] dx}{\int_{x_1}^{x_2} r(x) y^2 dx} \quad \dots(1)$$

where the denominator need not be equal to unity, are solution of Sturm-Liouville's system.

Proof : Let $I = \int_{x_1}^{x_2} [p(x) y'^2 - q(x) y^2] dx$

and $J = \int_{x_1}^{x_2} r(x) y^2 dx.$

Then by (1),

$$\lambda = \frac{I}{J}.$$

$$\begin{aligned} \therefore \delta\lambda &= \frac{J \delta I - I \delta J}{J^2} \\ &= \frac{\delta I - \lambda \delta J}{J} \end{aligned}$$

$$\text{or } \delta\lambda = \frac{1}{J} \delta (I - \lambda J).$$

If λ be an extremal, then by above theorem

$$\delta\lambda = \frac{-2 \int_{x_1}^{x_2} [py' + qy + \lambda ry] \delta y dx}{\int_{x_1}^{x_2} ry^2 dx} = 0$$

$$\Rightarrow 2 \int_{x_1}^{x_2} [py' + qy + \lambda ry] \delta y dx = 0.$$

Since $\delta y = 0$ at x_1 and x_2 and δy is arbitrary.

Therefore we get Sturm-Liouville equation

$$py' + qy + \lambda ry = 0.$$

1.13 Invariance of Euler's Equation under Co-ordinate Transformation

Consider the functional

$$I[y(x)] = \int_{x_0}^{x_1} f(x, y, y') dx \quad \dots(1)$$

and suppose the two variables x and y are transformed to two new variable u and v by following rule :

$$x = x(u, v); \quad y = y(u, v). \quad \dots(2)$$

Then the Jacobian of transformation is

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} \neq 0.$$

Now from (2),

$$x = x(u, v).$$

$$\therefore \frac{dx}{du} = \frac{\partial x}{\partial u} \frac{du}{du} + \frac{\partial x}{\partial v} \frac{dv}{du}$$

$$\text{or} \quad \frac{dx}{du} = x_u + x_v v_u'.$$

$$\text{Similarly} \quad \frac{dy}{du} = y_u + y_v v_u'.$$

$$\begin{aligned} \therefore y' &= \frac{dy}{dx} = \frac{dy / du}{dx / du} \\ &= \frac{y_u + y_v v_u'}{x_u + x_v v_u'}. \end{aligned}$$

Then

$$\begin{aligned}\int_{x_0}^{x_1} f(x, y, y') dx &= \int_{x_0}^x f \left[x(u, v), y(u, v), \frac{y_u + y_v v_u'}{x_u + x_v v_u'} \right] (x_u + x_v v_u') du \\ &= \int_{x_0}^{x_1} F(u, v, v_u') du\end{aligned}$$

which satisfy Euler's equation

$$\frac{\partial F}{\partial v} - \frac{d}{du} \left(\frac{\partial F}{\partial v'} \right) = 0.$$

This principle is known as the principal of invariance of Euler's equation.

Hence, Euler's equation is invariance under co-ordinate transformation.

Problem Set

Problem 1 : Find the extremal of functional

$$I[y(x)] = \int_0^{\log 2} (e^{-x} y'^2 - e^x y^2) dx.$$

Solution : Here

$$f = e^{-x} y'^2 - e^x y^2. \quad \dots(1)$$

By Euler's equation

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$$

$$\Rightarrow -2ye^x - \frac{d}{dx} (2e^{-x} y') = 0$$

$$\text{or} \quad -2ye^x - 2y''e^{-x} + 2y'e^{-x} = 0$$

$$\text{or} \quad y'' - y' + e^{2x} y = 0$$

$$\text{i.e.,} \quad \frac{d^2 y}{dx^2} - \frac{dy}{dx} + e^{2x} y = 0. \quad \dots(2)$$

The solution of (2) possesses some problem.

Now, we introduce two new variable u and v defined as

$$x = \log u, \quad y = v.$$

Then given functional reduces to

$$\begin{aligned} I_1 [v(u)] &= \int_1^2 [u^{-1} v'^2 - uv^2] \frac{1}{u} du \\ &= \int_1^2 (v'^2 - v^2) du. \end{aligned}$$

Here $F = v'^2 - v^2$.

By Euler's equation

$$\frac{\partial F}{\partial v} - \frac{d}{du} \left(\frac{\partial F}{\partial v'} \right) = 0$$

$$\Rightarrow -2v - \frac{d}{du} (2v') = 0$$

$$\text{or } v + \frac{d}{dx} \left(\frac{dv}{dx} \right) = 0$$

$$\text{or } \frac{d^2 v}{dx^2} + v = 0$$

$$\text{or } (D^2 + 1) v = 0; \quad D \equiv \frac{d}{dv}.$$

Auxiliary equation is

$$m^2 + 1 = 0$$

$$\Rightarrow m = \pm i.$$

$\therefore$ The solution is

$$v = c_1 \cos u + c_2 \sin u.$$

Hence solution of original functional is

$$y = c_1 \cos(e^x) + c_2 \sin(e^x) \quad [\because \log u = x \Rightarrow u = e^x]$$

Hence it varies the invariance of Euler's equation under co-ordinate transformation.

Exercise 1.2

1. Find the extremals of the functional

$$I[y(x)] = \int_0^1 (1 + y''^2) dx$$

with $y(0) = 0$; $y'(0) = 1$; $y(1) = 1$ and $y'(1) = 1$.

2. Find the extremals of the functional

$$I[y(x)] = \int_{-1}^1 \left(\frac{1}{2} y''^2 + y \right) dx$$

when $y(\pm 1) = 0$; $y'(\pm 1) = 0$.

3. Find the extremals of the functional

$$I[y(x)] = \int_{x_1}^{x_2} \left(\frac{1}{2} myy'' + \frac{1}{2} ky^2 \right) dx.$$

4. Find the extremals of the functional

$$I = \int_0^\pi (y'^2 - z'^2 + 2yz - 2y^2) dx$$

with conditions

$$y(0) = 0; \quad y(\pi) = 1; \quad z(0) = 0 \quad \text{and} \quad z(\pi) = 1.$$

5. Find the extremals of the functional

$$I = \int_{x_1}^{x_2} \sqrt{1 + y'^2 + z'^2} dx.$$

6. Find the extremal of

$$\int_0^\pi y'^2 dx, \quad y(0) = 0, \quad y(\pi) = 0$$

with the constraint

$$\int_0^\pi y^2 dx = 1.$$

7. Find the extremal of $\int_0^\pi y''^2$ with the constraint $\int_0^\pi y^2 dx = 1$ with boundary conditions

$$y(0) = y''(0) = y(\pi) = y''(\pi) = 0.$$

8. Find the curve of length l which passes through the point $(0,0)$ and $(1,0)$ for which the area between the curve and the x -axis is maximum.

9. Find function $y(x)$ for which $\int_0^1 (x^2 + y'^2) dx$ is stationary given that

$$\int_0^1 y^2 dx = 2, \quad y(0) = 0.$$

Answers

- | | |
|----|--|
| 1. | $y = x$ |
| 2. | $y = -\frac{(x^2 - 1)^2}{24}$ |
| 3. | $y = c_1 \cos \omega x + c_2 \sin \omega x$, where $\omega = \sqrt{\frac{k}{m}}$ |
| 4. | $y = -\frac{1}{\pi} x \cos x + c_3 \sin x$
$z = -\frac{1}{\pi} (-x \cos x + 2 \sin x) + c_3 \sin x$ |
| 5. | $y = ax + b$ and $z = cx + d$, where a, b, c, d are constants |
| 6. | $y = \sqrt{\frac{2}{\pi}} \sin nx$ |
| 7. | $y = \sqrt{\frac{2}{\pi}} \sin n\pi x$ |
| 8. | $(x + c)^2 + (y - c')^2 = \lambda^2$, where c and c' are constant |
| 9. | $y = \pm 2 \sin m\pi x$ |

HIGHLIGHTS

- * The object of calculus of variation is to discuss extrema of functional.
- * Functional is a function of function and mathematically, the integral $\int_{x_1}^{x_2} F(x, y, y') dx$ is called functional.
- * The length between two point in xy -plane is an example of functional. The length is given by

$$I [y(x)] = \int_{x_1}^{x_2} (1 + y'^2)^{1/2} dx.$$

- * A function $y = y(x)$ which extremize the functional

$$I [y(x)] = \int_{x_1}^{x_2} F(x, y, y') dx$$

is called extremal.

- * The necessary condition for existence of extremal for functional

$$I [y(x)] = \int_{x_1}^{x_2} F(x, y, y') dx$$

is
$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

This equation is called Euler's equation or Euler's Lagrange equation.

- * The solution of Euler's equation which satisfy given boundary condition is extremal of functional.
- * The Euler's equation for functional

$$I [y(x)] = \int_{x_1}^{x_2} F(y, y') dx$$

is
$$F - y' F_{y'} = \text{constant}.$$

- * The famous Brachistochrone problem is developed by **J. Bernoulli** in 1696.
- * The solution of Brachistochrone problem is **cycloid**.
i.e., the cycloid extremize the Brachistochrone problem.

- * The necessary condition for existence of extremal of functional of higher derivative

$$I[y(x)] = \int_{x_0}^{x_1} F(x, y, y', y'', \dots, y^{(n)}) dx$$

is
$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) + \dots (-1)^n \frac{d^n}{dx^n} \left(\frac{\partial F}{\partial y^{(n)}} \right) = 0.$$

This equation is called Euler-Poisson's equation.

- * The necessary condition for existence of extremal of functional of several dependent variables

$$I = \int_{x_0}^{x_2} F(x, y_1, y_2, \dots, y_n, y_1', y_2' \dots y_n') dx$$

is
$$\frac{\partial F}{\partial y_i} - \frac{d}{dx} \left(\frac{\partial F}{\partial y_i'} \right) = 0; \quad i = 1, 2, \dots, n.$$

- * The necessary condition for existence of functional of several independent variable

$$I = \iint_D F(x, y, u, u_x, u_y) dx dy$$

is
$$\frac{\partial F}{\partial u} - \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial u_x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial u_y} \right) = 0.$$

- * In order to finding the extremal of given functional under the given constraint is isoperimetric problem.
- * Isoperimetric problem is solved by Lagrange multiplier method.
- * Euler's equation is invariant under co-ordinate transformation.

Chapter 2



Variational Problems with Moving Boundaries

2.1 Introduction

Consider the functional

$$I[y(x)] = \int_{x_1}^{x_2} F(x, y, y') dx$$

In chapter 1, we have discussed that if the boundary point (x_1, y_1) and (x_2, y_2) are fixed.

Then the necessary condition for existence of extremal is given by Euler's equation.

i.e.,

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

or

$$F_y - \frac{d}{dx} (F_{y'}) = 0$$

In this chapter, we consider the case when one or both the boundary points can move along the curve.

i.e., if the boundary point (x_1, y_1) moves along the curve $y = \psi(x)$ and boundary point (x_2, y_2) move along the curve $y = \phi(x)$.

Then, such problem is known as variational problem with moving or free boundaries.

Then our aim is to find necessary condition for existence of extremal of such problem.

2.2 Transversality Conditions

Consider the functional

$$I[y(x)] = \int_{x_1}^{x_2} F(x, y, y') dx \quad \dots(1)$$

For the sake of simplicity, let us assume that one of the boundary point (x_1, y_1) is fixed while the other boundary point (x_2, y_2) can move and varies from (x_2, y_2) to $(x_2 + \delta x_2, y_2 + \delta y_2)$ on the curve $y = f(x)$.

And let $\delta y(a)$ be variation in y as right end points varies.

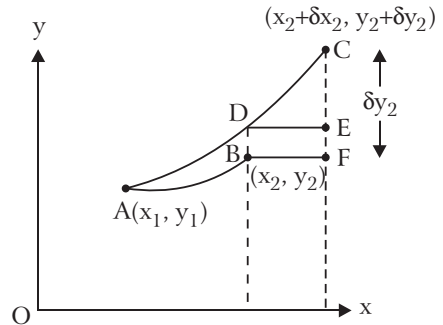


Fig. 2.1

Then total variation in I is given by

$$\begin{aligned} \Delta I &= I[y(x) + \delta y(x)] - I[y(x)] \\ &= \int_{x_1}^{x_2 + \delta x_2} F(x, y + \delta y, y' + \delta y') dx - \int_{x_1}^{x_2} F(x, y, y') dx \\ &= \int_{x_1}^{x_2} F(x, y + \delta y, y' + \delta y') dx + \int_{x_2}^{x_2 + \delta x_2} F(x, y + \delta y, y' + \delta y') dx \\ &\quad - \int_{x_1}^{x_2} F(x, y, y') dx \end{aligned}$$

$$\Delta I = \int_{x_1}^{x_2} [F(x, y + \delta y, y' + \delta y') - F(x, y, y')] dx + \int_{x_2}^{x_2 + \delta x_2} F(x, y + \delta y, y' + \delta y') dx \quad \dots(2)$$

Now by mean value theorem of integral calculus, we have

$$\int_{x_2}^{x_2 + \delta x_2} F(x, y + \delta y, y' + \delta y') dx = [F(x, y, y')]_{x_2 + \theta \delta x_2}^{x_2} \delta x_2 \dots (3)$$

where $0 < \theta < 1$.

By the virtue of continuity of F , we may write

$$[F]_{x_2 + \theta \delta x_2} = [F]_{x_2} + \varepsilon \quad \dots (4)$$

where $\varepsilon \rightarrow 0$ as $\delta x_2 \rightarrow 0$ and $\delta y_2 \rightarrow 0$.

Then by (3) and (4), we have

$$\int_{x_2}^{x_2 + \delta x_2} F(x, y + \delta y, y' + \delta y') dx = [F]_{x=x_2} \delta x_2. \quad \dots(5)$$

Now consider,

$$\begin{aligned}
& \int_{x_1}^{x_2} [F(x, y + \delta y, y' + \delta y') - F(x, y, y')] dx \\
&= \int_{x_1}^{x_2} \left[\left\{ F(x, y, y') + \delta y \frac{\partial F}{\partial y} + \delta y' \frac{\partial F}{\partial y'} + \dots \right\} - F(x, y, y') \right] dx \\
&\hspace{15em} [\text{By Taylor's theorem}] \\
&= \int_{x_1}^{x_2} \left[\frac{\partial F}{\partial y} \delta y + \frac{\partial F}{\partial y'} \delta y' \right] dx \hspace{2em} [\text{Neglecting higher term}] \\
&= \int_{x_1}^{x_1} \left(\frac{\partial F}{\partial y} \delta y \right) dx + \int_{x_1}^{x_1} \left(\frac{\partial F}{\partial y'} \delta y' \right) dx \\
&= \int_{x_1}^{x_1} \left(\frac{\partial F}{\partial y} \delta y \right) dx + \left[\frac{\partial F}{\partial y'} \delta y \right]_{x_1}^{x_2} - \int_{x_1}^{x_1} \left\{ \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \delta y \right\} dx \\
&= \int_{x_1}^{x_2} \left[\left\{ \frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right\} \delta y \right] dx + \left[\frac{\partial F}{\partial y'} \delta y \right]_{x_1}^{x_2} \hspace{2em} \dots(6)
\end{aligned}$$

Combining (5) and (6), we have

$$\Delta I = \int_{x_1}^{x_2} \left[\left\{ \frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right\} \delta y \right] dx + \left[\frac{\partial F}{\partial y'} \delta y \right]_{x_1}^{x_2} + [F]_{x=x_2} \delta x_2 \quad \dots(7)$$

For extremum, we have

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

Therefore, (7) becomes

$$\Delta I = \left[\frac{\partial F}{\partial y'} \delta y \right]_{x_1}^{x_2} + [F]_{x=x_2} \delta x_2 \quad \dots(8)$$

Since point (x_1, y_1) is fixed.

$$\therefore \delta y(x_1) = 0$$

and it is clear from figure 2.1 that

$$BD = (\delta y)_{x_2} \text{ and } FC = \delta y_2$$

$$\text{Further } EC = y'(x_2) \delta x_2$$

and hence $BD = FC - EC$ gives

$$(\delta y)_{x_2} = \delta y_2 - y'(x_2) \delta x_2 \quad \dots(9)$$

Therefore, (8) becomes

$$\begin{aligned} \Delta I &= \left[\frac{\partial F}{\partial y'} \delta y \right]_{x=x_2} + [F]_{x=x_2} \delta x_2 \\ &= [F_{y'}]_{x=x_2} [\delta y]_{x=x_2} + [F]_{x=x_2} \delta x_2 \\ &= [F_{y'}]_{x=x_2} \{ \delta y_2 - y'(x_2) \delta x_2 \} + [F]_{x=x_2} \delta x_2 \quad [\text{By (9)}] \\ &= [F_{y'}]_{x=x_2} \delta y_2 - [F_{y'} y']_{x=x_2} \delta x_2 + [F]_{x=x_2} \delta x_2 \\ \Delta I &= [F - y' F_{y'}]_{x=x_2} \delta x_2 + [F_{y'}]_{x=x_2} \delta y_2 \quad \dots(10) \end{aligned}$$

The necessary condition for extremum is

$$\Delta I = 0$$

$$\Rightarrow [F - y' F_{y'}]_{x=x_2} \delta x_2 + [F_{y'}]_{x=x_2} \delta y_2 = 0$$

Since δx_2 and δy_2 are independent.

$$\Rightarrow \left[\begin{array}{l} [F - y' F_{y'}]_{x=x_2} = 0 \\ [F_{y'}]_{x=x_2} = 0 \end{array} \right] \quad \dots(11)$$

If δx_2 and δy_2 are not independent.

For example, if the boundary point (x_2, y_2) moves along the curve $y = \phi(x)$

$$\therefore y_2 = \phi(x_2) \delta x_2 \quad \dots(12)$$

$$\text{Then } \delta y_2 = \phi'(x_2) \delta x_2$$

Thus, from (10), we get

$$\begin{aligned} \Delta I &= [F - y' F_{y'}]_{x=x_2} \delta x_2 + [F_{y'}]_{x=x_2} \phi'(x_2) \delta x_2 \\ &= [F + (\phi' - y') F_{y'}]_{x=x_2} \delta x_2. \end{aligned}$$

$\therefore$ The necessary condition for extremal is

$$\Delta I = 0$$

$$\Rightarrow [F + (\phi' - y') F_{y'}]_{x=x_2} \delta x_2 = 0$$

Since δx_2 is arbitrary.

$$\therefore [F + (\phi' - y') F_{y'}]_{x=x_2} = 0 \quad \dots(13)$$

which is the required condition at the free boundary.

This is known as **transversality condition**.

Particular Case : If boundary point (x_1, y_1) moves along the curve $y = \psi(x)$ and boundary point (x_2, y_2) move along the curve $y = \phi(x)$.

Then transversality condition is

$$\begin{array}{l} [F + (\psi' - y') F_{y'}]_{x=x_1} = 0 \\ [F + (\phi' - y') F_{y'}]_{x=x_2} = 0 \end{array}$$

This gives required extremal of functional.

2.3 Orthogonality Conditions

When F in (1) is given by

$$A(x, y) (1 + y'^2)^{1/2}$$

i.e.,

$$F = A(x, y) (1 + y'^2)^{1/2}$$

where $A(x, y)$ does not vanish at the movable point x_2 .

In this case (13) reduces to

$$A(x, y) \cdot \frac{(1 + \phi' y')}{\sqrt{1 + y'^2}} = 0 \text{ at } x = x_2$$

Since $A(x, y) \neq 0$ at $x = x_2$, we have

$$\frac{1 + \phi' y'}{\sqrt{1 + y'^2}} = 0 \text{ at } x = x_2$$

or

$$y' = -\frac{1}{\phi'} \text{ at } x = x_2 \quad \text{i.e., } \phi' y' = -1$$

which is the **orthogonality condition**.

Problem Set

Problem 1 : Find the shortest distance between the parabola $y = x^2$ and the straight line $x - y = 5$.

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Solution : The problem is to find extremal of the function.

$$I[y(x)] = \int_{x_1}^{x_2} ds$$

or
$$I[y(x)] = \int_{x_1}^{x_2} \sqrt{1 + y'^2} dx \quad \dots(1)$$

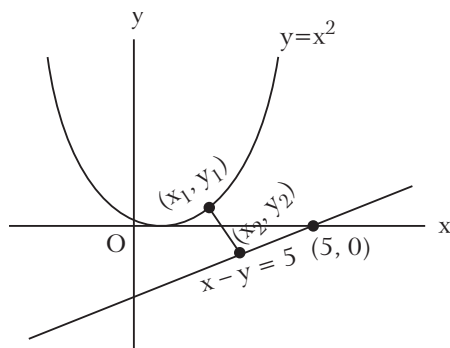


Fig. 2.2

Here $F = \sqrt{1 + y'^2}$.

Subject to condition that point (x_1, y_1) moves along the curve

$$y = x^2$$

and point (x_2, y_2) moves along the curve

$$x - y = 5 \text{ i.e., } y = x - 5$$

Let $\Psi(x) = x^2, \phi(x) = x - 5$

By transversality condition, we have

$$[F + (\Psi' - y') F_{y'}]_{x=x_1} = 0$$

$$[F + (\phi' - y') F_{y'}]_{x=x_2} = 0$$

Since $F = \sqrt{1 + y'^2}, \Psi(x) = x^2, \phi(x) = x - 5$

$$\therefore \left[\sqrt{1 + y'^2} + (2x - y') \frac{y'}{\sqrt{1 + y'^2}} \right]_{x=x_1} = 0 \quad \dots(2)$$

$$\text{and} \quad \left[\sqrt{1 + y'^2} + (1 - y') \frac{y'}{\sqrt{1 + y'^2}} \right]_{x=x_2} = 0 \quad \dots(3)$$

Now, since $F = \sqrt{1 + y'^2}$

Then by Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 0 - \frac{d}{dx} \left(\frac{y'}{\sqrt{1 + y'^2}} \right) = 0$$

$$\text{or} \quad \frac{d}{dx} \left(\frac{y'}{\sqrt{1 + y'^2}} \right) = 0$$

Integrating, we get

$$\frac{y'}{\sqrt{1+y'^2}} = \text{constant} = c_1'$$

or

$$y' = c_1' \sqrt{1+y'^2}$$

Squaring, we get

$$y'^2 = c_1'^2 (1+y'^2)$$

or

$$(1-c_1'^2) y'^2 = c_1'^2$$

or

$$y'^2 = \frac{c_1'^2}{(1-c_1'^2)}$$

or

$$y' = \frac{c_1'}{\sqrt{1-c_1'^2}} = c_1$$

$\therefore$

$$\frac{dy}{dx} = c_1$$

On integrating, we get required extremal is

$$y = c_1 x + c_2 \quad \dots(4)$$

$\therefore$

$$y' = c_1. \quad \dots(5)$$

Since both end point (x_1, y_1) and (x_2, y_2) lies on the extremal (4).

$\therefore$

$$c_1 x_1 + c_2 = y_1$$

$\Rightarrow$

$$c_1 x_1 + c_2 = x_1^2 \quad [\because y = x^2] \quad \dots(6)$$

and

$$c_1 x_2 + c_2 = y_2$$

$\Rightarrow$

$$c_1 x_2 + c_2 = x_2 - 5 \quad [\because y = x - 5] \quad \dots(7)$$

Now, put $y' = c_1$ in (2), we get

$$\sqrt{(1+c_1^2)} + (2x_1 - c_1) \frac{c_1}{\sqrt{1+c_1^2}} = 0$$

or

$$(1+c_1^2) + 2x_1 c_1 - c_1^2 = 0$$

or

$$1 + 2x_1 c_1 = 0 \quad \dots(8)$$



Similarly, put $y' = c_1$ in (3), we get

$$\sqrt{1 + c_1^2} + (1 - c_1) \frac{c_1}{\sqrt{1 + c_1^2}} = 0$$

or $(1 + c_1^2) + (c_1 - c_1^2) = 0$

or $1 + c_1 = 0$

$\Rightarrow$

$$c_1 = -1$$

Put $c_1 = -1$ in (8), we get

$$1 - 2x_1 = 0$$

$\Rightarrow$

$$x_1 = \frac{1}{2}$$

Putting value of c_1 and x_1 in (6), we get

$$-\frac{1}{2} + c_2 = \frac{1}{4}$$

$\Rightarrow$

$$c_2 = \frac{3}{4}$$

Putting value of c_1 and c_2 in (7), we get

$$-x_2 + \frac{3}{4} = x_2 - 5$$

$\Rightarrow$

$$2x_2 = \frac{3}{4} + 5 = \frac{23}{4}$$

$\Rightarrow$

$$x_2 = \frac{23}{8}$$

Hence, we get

$$c_1 = -1, c_2 = \frac{3}{4}, x_1 = \frac{1}{2}, x_2 = \frac{23}{8}$$

Therefore, from (4), required extremal is

$$y = -x + \frac{3}{4}$$

and shortest distance between parabola and straight line is

$$\begin{aligned} I &= \int_{x_1}^{x_2} \sqrt{1 + y'^2} \, dx \\ &= \int_{1/2}^{23/8} \sqrt{1+1} \, dx \quad [\because y' = c_1 = -1] \\ &= \sqrt{2} [x]_{1/2}^{23/8} \\ &= \sqrt{2} \left[\frac{23}{8} - \frac{1}{2} \right] = \frac{19}{8} \sqrt{2} \end{aligned}$$

Problem 2 : Using only the basic necessary condition $\delta I = 0$. Find the curve on which an extremum of the functional

$$I[y(x)] = \int_0^{x_1} \frac{(1 + y'^2)^{1/2}}{y} \, dx, \quad y(0) = 0$$

can be achieved if the second boundary point (x_1, y_1) can move along the circumference

$$(x - 9)^2 + y^2 = 9.$$

Solution : Given functional is

$$I[y(x)] = \int_0^{x_1} \frac{(1 + y'^2)^{1/2}}{y} \, dx \quad \dots(1)$$

$$y(0) = 0$$

Here,

$$F = F(y, y') = \frac{(1 + y'^2)^{1/2}}{y}$$

Therefore, the necessary condition for existence of extremal is

$$F - y' F_{y'} = \text{constant}$$

$$\frac{\sqrt{1+y'^2}}{y} - y' \frac{y'}{y \sqrt{1+y'^2}} = \text{constant}$$

or
$$\frac{1+y'^2 - y'^2}{y \sqrt{1+y'^2}} = \text{constant} = c'_1$$

or
$$\frac{1}{y \sqrt{1+y'^2}} = c'_1$$

Put $y' = \tan t$(2)

$\therefore \frac{1}{y \sec t} = c'_1$

$\Rightarrow y = \frac{1}{c'_1} \cos t$

$\Rightarrow y = c_1 \cos t$, where $c_1 = \frac{1}{c'_1}$...(3)

Now,
$$\frac{dx}{dt} = \frac{dx}{dy} \frac{dy}{dt}$$

= $\cot t (-c_1 \sin t)$ [By (2) and (3)]

$$\frac{dx}{dt} = -c_1 \cos t$$

On integrating, we get

$$x = -c_1 \sin t + c_2$$

or $x - c_2 = -c_1 \sin t$...(4)

Now, squaring and adding (3) and (4), we get

$$(x - c_2)^2 + y^2 = c_1^2. \quad \dots(5)$$

Now, since $y(0) = 0$.

Then from (5), we get

$$c_2^2 = c_1^2$$

$\Rightarrow c_1 = c_2$

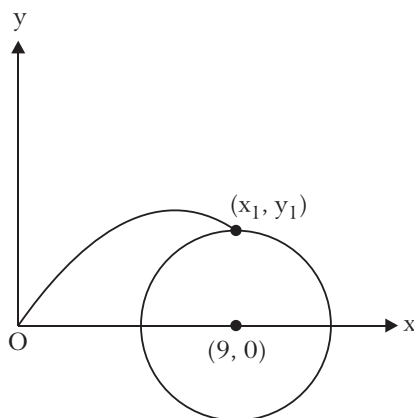


Fig. 2.3

Therefore, equation (5) becomes

$$(x - c_1)^2 + y^2 = c_1^2 \quad \dots(6)$$

or
$$x^2 + c_1^2 - 2c_1x + y^2 = c_1^2$$

or
$$x^2 + y^2 - 2c_1x = 0 \quad \dots(7)$$

Now, since (x_1, y_1) lies on extremal (7) and given curve

$$(x - 9)^2 + y^2 = 9.$$

Therefore, we have

$$x_1^2 + y_1^2 - 2c_1x_1 = 0 \quad \dots(8)$$

and
$$(x_1 - 9)^2 + y_1^2 = 9$$

or
$$x_1^2 + y_1^2 - 18x_1 = -72 \quad \dots(9)$$

Now subtracting (8) and (9), we get

$$-2c_1x_1 + 18x_1 = 72$$

or
$$x_1 (9 - c_1) = 36$$

or
$$x_1 (c_1 - 9) = -36 \quad \dots(10)$$

Now tangent at (x_1, y_1) to given circle $(x - 9)^2 + y^2 = 9$ and extremal circle (6) are orthogonal to each other.

$$\therefore m_1 \cdot m_2 = -1$$

i.e.,
$$\left(\frac{x_1 - c_1}{y_1} \right) \left(\frac{x_1 - 9}{y_1} \right) = -1$$

or
$$(x_1 - c_1)(x_1 - 9) = -y_1^2$$

or
$$x_1^2 - 9x_1 - c_1x_1 + 9c_1 = -y_1^2$$

or
$$x_1^2 + y_1^2 - 9x_1 - c_1x_1 + 9c_1 = 0$$

or
$$2c_1x_1 - 9x_1 - c_1x_1 + 9c_1 = 0 \quad \text{[From (8)]}$$

or
$$c_1x_1 - 9x_1 + 9c_1 = 0$$

$$\text{or} \quad x_1 (c_1 - 9) + 9c_1 = 0$$

$$\text{or} \quad -36 + 9c_1 = 0 \quad [\text{From (10)}]$$

$$9c_1 = 36$$

$$\therefore \quad c_1 = 4$$

Putting value of c_1 in (7), we get

$$x^2 + y^2 - 8x = 0.$$

This is required extremal.

Problem 3 : Find the shortest distance between the point $(1, 0)$ and the ellipse $4x^2 + 9y^2 = 36$.

Solution : We have to find shortest distance between $A(1, 0)$ and $B(x_2, y_2)$ where B lies on the ellipse.

$$4x^2 + 9y^2 = 36 \quad \dots(1)$$

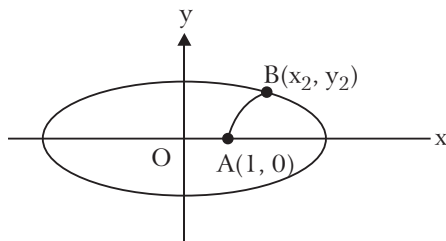


Fig. 2.4

The arc length AB of the minimizing curve $y = f(x)$ is given by

$$\begin{aligned} I[y(x)] &= \int_1^{x_2} ds \\ &= \int_1^{x_2} \sqrt{1 + y'^2} \, dx \end{aligned} \quad \dots(2)$$

where the end point $A(1, 0)$ is fixed and other end $B(x_2, y_2)$ lies on (1).

Here $F = \sqrt{1 + y'^2}$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\text{i.e.,} \quad 0 - \frac{d}{dx} \left(\frac{y'}{\sqrt{1 + y'^2}} \right) = 0$$

$$\text{or} \quad \frac{d}{dx} \left(\frac{y'}{\sqrt{1 + y'^2}} \right) = 0$$

Integrating, we get

$$\frac{y'}{\sqrt{1+y'^2}} = \text{constant} = c_1'$$

or
$$y'^2 = c_1'^2 (1 + y'^2)$$

or
$$(1 - c_1'^2) y'^2 = c_1'^2$$

or
$$y'^2 = \frac{c_1'^2}{1 - c_1'^2}$$

or
$$y' = \frac{c_1'}{\sqrt{1 - c_1'^2}} = c_1 \text{ (say)}$$

i.e.,
$$\frac{dy}{dx} = c_1$$

Integrating, we get

$$y = c_1 x + c_2. \quad \dots(3)$$

which is straight line along which the required shortest distance is attained.

Now since (3) passes through $A(1, 0)$

$$\therefore c_1 + c_2 = 0 \Rightarrow c_2 = -c_1$$

Then (3) becomes

$$y = c_1 x - c_1$$

or
$$y = c_1 (x - 1) \quad \dots(4)$$

Also it passes through (x_2, y_2)

$$\therefore y_2 = c_1 (x_2 - 1) \quad \dots(5)$$

Now from equation (1)

$$y = \frac{2}{3} \sqrt{9 - x^2} = \psi(x) \text{ (say)} \quad \dots(6)$$



$\therefore$ By transversality condition for $\psi(x)$, we get

$$[F + (\psi' - y') F_{y'}]_{x=x_2} = 0$$

$$\Rightarrow \left[\sqrt{1 + y'^2} + \left(\frac{2}{3} \cdot \frac{1}{2} \frac{-2x}{\sqrt{9 - x^2}} - y' \right) \frac{y'}{\sqrt{1 + y'^2}} \right]_{x=x_2} = 0$$

$$\Rightarrow \sqrt{1 + c_1^2} - \frac{2}{3} \cdot \frac{x_2}{\sqrt{9 - x_2^2}} \frac{c_1}{\sqrt{1 + c_1^2}} - \frac{c_1^2}{\sqrt{1 + c_1^2}} = 0$$

$$\Rightarrow 1 + c_1^2 - \frac{2}{3} \frac{x_2 c_1}{\sqrt{9 - x_2^2}} - c_1^2 = 0$$

$$\Rightarrow 1 - \frac{2}{3} \frac{x_2 c_1}{\sqrt{9 - x_2^2}} = 0$$

$$\Rightarrow 3 \sqrt{9 - x_2^2} = 2x_2 c_1$$

Squaring both sides, we get

$$9(9 - x_2^2) = 4x_2^2 c_1^2 \quad \dots(7)$$

Now since the point (x_2, y_2) lies on ellipse (1).

$$\therefore 4x_2^2 + 9y_2^2 = 36 \quad \dots(8)$$

Now, from (5) and (8), we get

$$4x_2^2 + 9c_1^2 (x_2 - 1)^2 = 36$$

$$\Rightarrow 4(9 - x_2^2) = 9c_1^2 (x_2 - 1)^2$$

From (7),

$$\Rightarrow 4 \cdot \frac{4x_2^2 c_1^2}{9} = 9c_1^2 (x_2 - 1)^2$$

$$\Rightarrow 16x_2^2 = 81(x_2 - 1)^2$$

$$\Rightarrow \left(\frac{x_2}{x_2 - 1} \right)^2 = \frac{81}{16}$$

$$\Rightarrow \frac{x_2}{x_2 - 1} = \frac{9}{4}$$

$$\Rightarrow 9x_2 - 9 = 4x_2$$

$$\Rightarrow 5x_2 = 9$$

$$\therefore x_2 = \frac{9}{5}$$

Now from equation (7),

$$9 \left(9 - \frac{81}{25} \right) = 4 \times \frac{81}{25} c_1^2$$

$$\Rightarrow 9 \times \frac{144}{25} = \frac{4 \times 81}{25} c_1^2$$

$$\Rightarrow c_1^2 = 4$$

$$\therefore c_1 = 2$$

Now putting value of c_1 and x_2 in (5), we get

$$y_2 = 2 \left(\frac{9}{5} - 1 \right)$$

$$\Rightarrow y_2 = \frac{8}{5}$$

Hence, we get point

$$B(x_2, y_2) = B\left(\frac{9}{5}, \frac{8}{5}\right)$$

$\therefore$ The required shortest distance AB i.e., the distance between $A(1, 0)$ and $B\left(\frac{9}{5}, \frac{8}{5}\right)$ is

$$= \sqrt{\left(\frac{9}{5} - 1\right)^2 + \left(\frac{8}{5} - 0\right)^2}$$

$$= \sqrt{\frac{16}{25} + \frac{64}{25}}$$

$$= \sqrt{\frac{80}{25}} = \frac{4\sqrt{5}}{5}.$$

Problem 4 : Find the minimum distance between circle $x^2 + y^2 = 1$ and straight line $x + y = 4$.
 [CSJM Kanpur M. Sc. 2011]

Solution : We have to extremize functional.

$$\begin{aligned} I[y(x)] &= \int_{x_1}^{x_2} ds \\ &= \int_{x_1}^{x_2} \sqrt{1 + y'^2} dx \quad \dots(1) \end{aligned}$$

Subject to condition that end point $A(x_1, y_1)$ and $B(x_2, y_2)$ varies along the circle $x^2 + y^2 = 1$ and straight line $x + y = 4$ respectively.

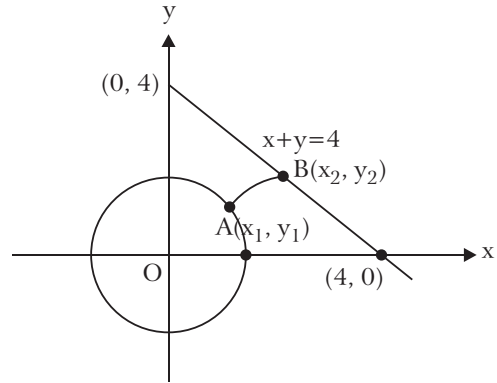


Fig. 2.5

Here $F = \sqrt{1 + y'^2}$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 0 - \frac{d}{dx} \left(\frac{y'}{\sqrt{1 + y'^2}} \right) = 0$$

$$\text{or} \quad \frac{d}{dx} \left(\frac{y'}{\sqrt{1 + y'^2}} \right) = 0$$

On integrating, we get

$$\frac{y'}{\sqrt{1 + y'^2}} = c_1'$$

This gives

$$y = c_1 x + c_2 \quad [\text{By above problem}] \quad \dots(2)$$

$$\therefore y' = c_1$$

which is straight line along which the required shortest distance is attained.

Now, since

$$x^2 + y^2 = 1 \Rightarrow y = \sqrt{1 - x^2}$$

and

$$x + y = 4 \Rightarrow y = 4 - x$$

$\therefore$ Let

$$\Psi(x) = \sqrt{1 - x^2}, \quad \phi(x) = 4 - x$$

$\therefore$ By transversality condition

$$[F + (\Psi' - y') F_{y'}]_{x=x_1} = 0$$

and

$$[F + (\phi' - y') F_{y'}]_{x=x_2} = 0$$

$\Rightarrow$

$$\left[\sqrt{1 + y'^2} + \left(-\frac{x}{\sqrt{1 - x^2}} - y' \right) \frac{y'}{\sqrt{1 + y'^2}} \right]_{x=x_1} = 0$$

and

$$\left[\sqrt{1 + y'^2} + (-1 - y') \frac{y'}{\sqrt{1 + y'^2}} \right]_{x=x_2} = 0$$

$\Rightarrow$

$$\sqrt{1 + c_1^2} + \left(-\frac{x_1}{\sqrt{1 - x_1^2}} - c_1 \right) \frac{c_1}{\sqrt{1 + c_1^2}} = 0 \quad [\because y' = c_1]$$

and

$$\sqrt{1 + c_1^2} + (-1 - c_1) \frac{c_1}{\sqrt{1 + c_1^2}} = 0$$

$\Rightarrow$

$$(1 + c_1^2) + \left(-\frac{x_1}{\sqrt{1 - x_1^2}} - c_1 \right) c_1 = 0$$

and

$$(1 + c_1^2) + (-1 - c_1) c_1 = 0$$

$\Rightarrow$

$$1 + c_1^2 - \frac{x_1 c_1}{\sqrt{1 - x_1^2}} - c_1^2 = 0$$

and

$$1 + c_1^2 - c_1 - c_1^2 = 0$$

$\Rightarrow$

$$1 - \frac{x_1 c_1}{\sqrt{1 - x_1^2}} = 0$$

and

$$1 - c_1 = 0$$

$$\Rightarrow \quad 1 = \frac{x_1 c_1}{\sqrt{1 - x_1^2}} \quad \text{and } c_1 = 1$$

$$\therefore \quad c_1 = 1 \quad \text{and} \quad 1 = \frac{x_1}{\sqrt{1 - x_1^2}}$$

$$\Rightarrow \quad c_1 = 1 \quad \text{and} \quad 1 - x_1^2 = x_1^2$$

$$\Rightarrow \quad c_1 = 1 \quad \text{and} \quad 2x_1^2 = 1$$

$$\Rightarrow \quad c_1 = 1 \quad \text{and} \quad x_1 = \frac{1}{\sqrt{2}} \quad \dots(3)$$

Now since both point (x_1, y_1) and (x_2, y_2) lies on extremal (2).

$$i.e., \quad y = c_1 x + c_2$$

$$\therefore \quad y_1 = c_1 x_1 + c_2 \Rightarrow c_1 x_1 + c_2 = \sqrt{1 - x_1^2} \quad \dots(4)$$

$$y_2 = c_1 x_2 + c_2 \Rightarrow c_1 x_2 + c_2 = 4 - x_2 \quad \dots(5)$$

$$[\because y = \sqrt{1 - x^2} \quad \text{and} \quad y = 4 - x^2]$$

Putting value of c_1 and x_1 in (4) and (5), we get

$$1 \cdot \frac{1}{\sqrt{2}} + c_2 = \sqrt{1 - \frac{1}{2}}$$

$$\Rightarrow \quad \frac{1}{\sqrt{2}} + c_2 = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \quad c_2 = 0$$

$$\text{and} \quad 1 \cdot x_2 + 0 = 4 - x_2$$

$$\Rightarrow \quad 2x_2 = 4$$

$$\Rightarrow \quad x_2 = 2$$

$$\begin{aligned} \text{and} \quad y_2 &= c_1 x_2 + c_2 \\ &= 1 \cdot 2 + 0 \\ &= 2 \end{aligned}$$

Hence, we get

$$c_1 = 1, c_2 = 0$$

and

$$(x_1, y_1) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

$$(x_2, y_2) = (2, 2)$$

Therefore, from (2) required extremal is

$$y = x$$

and required shortest distance

$$\begin{aligned} &= \int_{1/\sqrt{2}}^2 \sqrt{1+1} \, dx && [\because y' = c_1 = 1] \\ &= \sqrt{2} [x]_{1/\sqrt{2}}^2 \\ &= \sqrt{2} \left(2 - \frac{1}{\sqrt{2}} \right) \\ &= 2\sqrt{2} - 1. \end{aligned}$$

Problem 5 : Find the shortest distance between circle $x^2 + y^2 = 4$ and straight line $2x + y = 6$.

Solution : We have to extremize the functional

$$I[y(x)] = \int_{x_1}^{x_2} \sqrt{1 + y'^2} \, dx \quad \dots(1)$$

subject to condition that

end point $A(x_1, y_1)$ and $B(x_2, y_2)$ varies along the circle $x^2 + y^2 = 4$ and straight line $2x + y = 6$ respectively.

Now here

$$F = \sqrt{1 + y'^2}$$

Proceed as above example, we get

$$y = c_1 x + c_2 \quad \dots(2)$$

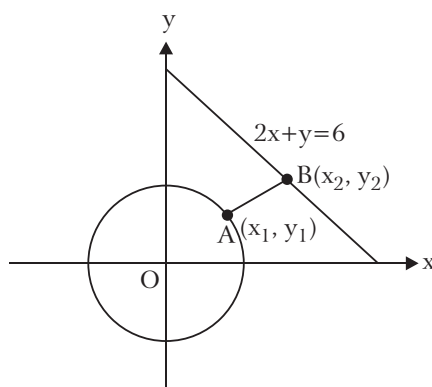


Fig. 2.6

$$\therefore y' = c_1$$

$$\text{Now, since } x^2 + y^2 = 4 \Rightarrow y = \sqrt{4 - x^2}$$

$$\text{and } 2x + y = 6 \Rightarrow y = 6 - 2x$$

$$\text{Let } \Psi(x) = \sqrt{4 - x^2}, \phi(x) = 6 - 2x$$

$\therefore$ By transversality condition

$$[F + (\Psi' - y') F_{y'}]_{x=x_1} = 0$$

$$\text{and } [F + (\phi' - y') F_{y'}]_{x=x_2} = 0$$

$$\Rightarrow \left[\sqrt{1 + y'^2} + \left(-\frac{x}{\sqrt{4 - x^2}} - y' \right) \frac{y'}{\sqrt{1 + y'^2}} \right]_{x=x_1} = 0$$

$$\text{and } \left[\sqrt{1 + y'^2} + (-2 - y') \frac{y'}{\sqrt{1 + y'^2}} \right]_{x=x_2} = 0$$

$$\Rightarrow \left[\sqrt{1 + c_1^2} + \left(\frac{-x_1}{\sqrt{4 - x_1^2}} - c_1 \right) \frac{c_1}{\sqrt{1 + c_1^2}} \right] = 0 \quad [\because y' = c_1]$$

$$\text{and } \sqrt{1 + c_1^2} + (-2 + c_1) \frac{c_1}{\sqrt{1 + c_1^2}} = 0$$

$$\Rightarrow (1 + c_1^2) - \frac{x_1 c_1}{\sqrt{4 - x_1^2}} - c_1^2 = 0$$

$$\text{i.e., } x_1 c_1 = \sqrt{4 - x_1^2} \quad \dots(3)$$

$$\text{and } 1 + c_1^2 - 2c_1 + c_1^2 = 0$$

i.e.,

$$c_1 = 1/2$$

$\dots(4)$

Putting value of c_1 in (3), we get

$$x_1 \times \frac{1}{2} = \sqrt{4 - x_1^2}$$

$$\Rightarrow \frac{x_1}{2} = \sqrt{4 - x_1^2}$$

Squaring both sides,

$$\frac{x_1^2}{4} = 4 - x_1^2$$

$$\Rightarrow \frac{5x_1^2}{4} = 4$$

$$\Rightarrow x_1^2 = \frac{16}{5}$$

$$\Rightarrow \boxed{x_1 = \frac{4}{\sqrt{5}}}$$

Now since both end point (x_1, y_1) and (x_2, y_2) lies on the extremal (2).

$$\therefore y_1 = c_1 x_1 + c_2 \Rightarrow c_1 x_1 + c_2 = \sqrt{4 - x_1^2} \quad \dots(4)$$

$$y_2 = c_1 x_2 + c_2 \Rightarrow c_1 x_1 + c_2 = 6 - 2x_2 \quad \dots(5)$$

Putting value of c_1 and x_1 in (4) and (5), we get

$$c_2 = 0, x_2 = \frac{12}{5}$$

$$\text{and } y_1 = \frac{2}{\sqrt{5}}, y_2 = \frac{6}{5}$$

Hence, we get

$$c_1 = \frac{1}{2}, c_2 = 0$$

$$A(x_1, y_1) = \left(\frac{4}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right)$$

$$\text{and } B(x_2, y_2) = \left(\frac{12}{5}, \frac{6}{5} \right).$$

Therefore, from (2), required extremal AB is

$$y = \frac{1}{2} x$$

and shortest distance between A and B

$$\begin{aligned}
 &= \int_{x_1}^{x_2} \sqrt{1 + y'^2} \, dx \\
 &= \int_{4/\sqrt{5}}^{12/5} \sqrt{1 + \left(\frac{1}{2}\right)^2} \, dx \quad \left[\because y = \frac{x}{2} \Rightarrow y' = \frac{1}{2} \right] \\
 &= \frac{\sqrt{5}}{2} [x]_{4/\sqrt{5}}^{12/5} \\
 &= \frac{\sqrt{5}}{2} \left[\frac{12}{5} - \frac{4}{\sqrt{5}} \right] \\
 &= \sqrt{5} \left(\frac{6}{5} - \frac{2}{\sqrt{5}} \right).
 \end{aligned}$$

Problem 6 : Find the shortest distance between $y^2 = 4x$ and $(x - 9)^2 + y^2 = 4$.

Solution : We have to extremize the functional

$$I[y(x)] = \int_{x_1}^{x_2} \sqrt{1 + y'^2} \, dx \quad \dots(1)$$

Subject to condition that end point $A(x_1, y_1)$, and $B(x_2, y_2)$ moves along the parabola $y^2 = 4x$ and circle $(x - 9)^2 + y^2 = 4$ respectively.

Here $F = \sqrt{1 + y'^2}$

Then proceeding as above example, we get extremal curve

$$y = c_1 x + c_2 \quad \dots(2)$$

$$\therefore y' = c_1$$

Now, since $y^2 = 4x \Rightarrow y = 2\sqrt{x}$

and $(x - 9)^2 + y^2 = 4 \Rightarrow y = 2\sqrt{4 - (x - 9)^2}$

Let $\psi(x) = 2\sqrt{x}$, $\phi(x) = 2\sqrt{4 - (x - 9)^2}$

∴ By transversality condition

$$[F + (\Psi' - y') F_{y'}]_{x=x_1} = 0$$

and $[F + (\Phi' - y') F_{y'}]_{x=x_2} = 0$

$$\Rightarrow \left[\sqrt{1 + y'^2} + \left(\frac{1}{\sqrt{x}} - y' \right) \frac{y'}{\sqrt{1 + y'^2}} \right]_{x=x_1} = 0$$

and $\left[\sqrt{1 + y'^2} + \left(-\frac{(x-9)}{\sqrt{4 - (x-9)^2}} - y' \right) \frac{y'}{\sqrt{1 + y'^2}} \right]_{x=x_2} = 0$

$$\Rightarrow \sqrt{1 + c_1^2} + \left(\frac{1}{\sqrt{x_1}} - c_1 \right) \frac{c_1}{\sqrt{1 + c_1^2}} = 0$$

and $\sqrt{1 + c_1^2} + \left(-\frac{(x_2 - 9)}{\sqrt{4 - (x_2 - 9)^2}} - c_1 \right) \frac{c_1}{\sqrt{1 + c_1^2}} = 0$

$$\Rightarrow 1 + c_1^2 + \frac{c_1}{\sqrt{x_1}} - c_1^2 = 0$$

and $1 + c_1^2 - \frac{(x_2 - 9) c_1}{\sqrt{4 - (x_2 - 9)^2}} - c_1^2 = 0$

$$\Rightarrow 1 + \frac{c_1}{\sqrt{x_1}} = 0$$

and $\frac{(x_2 - 9) c_1}{\sqrt{4 - (x_2 - 9)^2}} = 1$

$$\Rightarrow c_1 = -\sqrt{x_1} \text{ and } c_1 (x_2 - 9) = \sqrt{4 - (x_2 - 9)^2} = c_1 x_2 + c_2$$

$$\Rightarrow c_1 = -\sqrt{x_1} \text{ and } c_2 = -9c_1 \quad \dots(3)$$

Now, since both end point (x_1, y_1) and (x_2, y_2) lies on extremal (2), we get

$$y_1 = c_1 x_1 + c_2 \Rightarrow c_1 x_1 + c_2 = 2\sqrt{x_1} \quad \dots(4)$$

$$y_2 = c_1 x_2 + c_2 \Rightarrow c_1 x_2 + c_2 = \sqrt{4 - (x_2 - 9)^2} \quad \dots(5)$$

Now, from (3), (4) becomes

$$-x_1 \sqrt{x_1} + 9 \sqrt{x_1} = 2\sqrt{x_1}$$

$$\Rightarrow -x_1 + 9 = 2$$

$$\Rightarrow x_1 = 7$$

Therefore, from (3)

$$c_1 = -\sqrt{7}, c_2 = 9\sqrt{7}$$

Now, from (4) and (5)

$$y_1 = 2\sqrt{7}, x_2 = 9 \pm \frac{1}{\sqrt{2}}, y_2 = \frac{\sqrt{7}}{2}$$

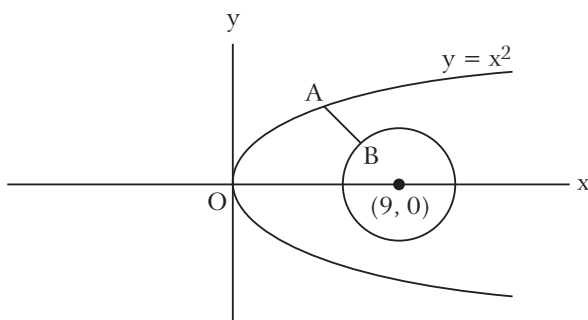


Fig. 2.7

Hence, we get

$$A(x_1, y_1) = (7, 2\sqrt{7})$$

$$B(x_2, y_2) = \left(9 - \frac{1}{\sqrt{2}}, \frac{\sqrt{7}}{2}\right) \text{ or } \left(9 + \frac{1}{\sqrt{2}}, \frac{\sqrt{7}}{2}\right),$$

Therefore, the distance between $A(7, 2\sqrt{7})$ and $B\left(9 - \frac{1}{\sqrt{2}}, \frac{\sqrt{7}}{2}\right)$ gives shortest distance.

$$\text{Hence, required shortest distance} = \int_7^{9 - \frac{1}{\sqrt{2}}} \sqrt{1 + c_1^2} \, dx$$

$$= \int_7^{9 - \frac{1}{\sqrt{2}}} \sqrt{1 + 7} \, dx$$

$$\begin{aligned}
 &= \sqrt{8} [x]_7^{9 - \frac{1}{\sqrt{2}}} \\
 &= \sqrt{8} \left(9 - \frac{1}{\sqrt{2}} - 7 \right) = \sqrt{8} \left(2 - \frac{1}{\sqrt{2}} \right)
 \end{aligned}$$

2.4 Variational Problem with a Moving Boundary for a Functional Dependent on Two Functions

Consider the functional

$$I[y(x), z(x)] = \int_{x_1}^{x_2} F(x, y(x), z(x), y'(x), z'(x)) dx \quad \dots(1)$$

where the lower point $A(x_1, y_1, z_1)$ be fixed and upper point $B(x_2, y_2, z_2)$ move in an arbitrary manner, or along a given curve or surface.

It is clear that extremum of (1) can be obtained by Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

and

$$\frac{\partial F}{\partial z} - \frac{d}{dx} \left(\frac{\partial F}{\partial z'} \right) = 0$$

The general solution of these equations contains four arbitrary constant.

Since the boundary point $A(x_1, y_1, z_1)$ is fixed, it is possible to eliminate two arbitrary constant. The other two constant can be determined from the necessary condition $\delta I = 0$ for extremum, where δI is the variation of I

Hence, $\delta I = 0$ gives

$$\Rightarrow (F - y' F_{y'} - z' F_{z'})_{x=x_2} \cdot \delta x_2 + (F_{y'})_{x=x_2} \cdot \delta y_2 + (F_{z'})_{x=x_2} \cdot \delta z_2 = 0. \quad \dots(2)$$

If $\delta x_2, \delta y_2$ and δz_2 are independent.

Then,

$$\begin{aligned}
 [F - y' F_{y'} - z' F_{z'}]_{x=x_2} &= 0 \\
 [F_{y'}]_{x=x_2} &= 0 \text{ and } [F_{z'}]_{x=x_2} = 0.
 \end{aligned} \quad \dots(3)$$

If the boundary point (x_2, y_2, z_2) moves along some curve $y_2 = \phi(x_2)$, $z_2 = \psi(x_2)$

then

$$\delta y_2 = \phi'(x_2) \delta x_2$$

and

$$\delta z_2 = \psi'(x_2) \delta x_2$$



Then, we get

$$[F + (\phi' - y') F_{y'} + (\psi' - z') F_{z'}]_{x=x_2} \delta x_2 = 0$$

Since δx_2 is arbitrary, we have

$$[F + (\phi' - y') F_{y'} + (\psi' - z') F_{z'}]_{x=x_2} = 0. \quad \dots(4)$$

This is **transversality** condition in the problem of extremum of (1).

Along with the equation $y_2 = \phi(x_2)$, $z_2 = \psi(x_2)$ the condition (4) gives the equations necessary for determining the two arbitrary constants in the general solution of Euler's equation.

Note : If the boundary point $B(x_2, y_2, z_2)$ moves along a given surface $z_2 = \phi(x_2, y_2)$ then $\delta z_2 = \frac{\partial \phi}{\partial x_2} \delta x_2 + \frac{\partial \phi}{\partial y_2} \delta y_2$ such that the variation δx_2 and δy_2 are arbitrary.

In this case (4) reduces to

$$[F - y' F_{y'} + (\phi_x - z') F_{z'}]_{x=x_2} \cdot \delta x_2 + [F_{y'} + \phi_y F_{z'}]_{x=x_2} \delta y_2 = 0$$

Since δx_2 and δy_2 are independent, we get

$$[F - y' F_{y'} + (\phi_x - z') F_{z'}]_{x=x_2} = 0$$

$$[F_{y'} + \phi_y F_{z'}]_{x=x_2} = 0$$

These two conditions together with $z_2 = \phi(x_2, y_2)$ enable us to determine two arbitrary constants in the general solution of Euler's equation.

SOLVED EXAMPLES

Example 1 : Find the extremum of the functional

$$I = \int_{x_1}^{x_2} (y'^2 + z'^2 + 2yz) dx$$

with $y(0) = 0$, $z(0) = 0$ and the point (x_2, y_2, z_2) moves over the fixed plane $x = x_2$.

Solution : Here,

$$F = y'^2 + z'^2 + 2yz \quad \dots(1)$$

By Euler's equation,

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\frac{\partial F}{\partial z} - \frac{d}{dx} \left(\frac{\partial F}{\partial z'} \right) = 0$$

From (1),

$$\Rightarrow 2z - \frac{d}{dx} (2y') = 0$$

$$\text{and } 2y - \frac{d}{dx} (2z') = 0$$

$$\Rightarrow \frac{d^2 y}{dx^2} - z = 0 \quad \dots(2)$$

$$\text{and } \frac{d^2 z}{dx^2} - y = 0 \quad \dots(3)$$

Now differentiating (2) w.r.t. x , we get

$$\frac{d^4 y}{dx^4} - \frac{d^2 z}{dx^2} = 0$$

$$\text{or } \frac{d^4 y}{dx^4} - y = 0 \quad [\text{By (3)}]$$

$$\text{or } (D^4 - 1) y = 0$$

Auxiliary equation

$$m^4 - 1 = 0$$

$$\Rightarrow (m^2 - 1)(m^2 + 1) = 0$$

$$\Rightarrow m = \pm 1, \pm i$$

Therefore, solution is

$$y = c_1 \cosh x + c_2 \sinh x + c_3 \cos x + c_4 \sin x \quad \dots(4)$$

From (3),

$$z = \frac{d^2 y}{dx^2}$$

$$\text{i.e.,} \quad z = c_1 \cosh x + c_2 \sinh x - c_3 \cos x - c_4 \sin x \quad \dots(5)$$

$$\text{Now,} \quad y(0) = 0, z(0) = 0$$

$$\Rightarrow \quad c_1 = c_3 = 0$$

Now since x_2 is fixed therefore, by condition of moving boundary point (x_2, y_2, z_2) .

$$[F_{y'}]_{x=x_2} = 0, [F_{z'}]_{x=x_2} = 0$$

$$\Rightarrow \quad y'(x_2) = 0, z'(x_2) = 0.$$

Then equation (4) and (5) gives

$$c_2 \cosh x_2 + c_4 \cos x_2 = 0$$

$$c_2 \cosh x_2 - c_4 \cos x_2 = 0.$$

If $\cosh x_2 \neq 0$, then $c_2 = c_4 = 0$

and therefore, an extremum is attained on $y = 0, z = 0$.

But if $\cos x_2 = 0$

then $c_2 = 0$ and c_4 remains arbitrary.

Hence, in this case extremum is

$$y = c_4 \sin x$$

$$z = -c_4 \sin x$$

2.5 One Sided Variations

Consider the functional

$$I[y(x)] = \int_{x_1}^{x_2} F(x, y, y') dx \quad \dots(1)$$

Earlier, we have discussed that the extremal curve passes through end point (x_1, y_1) and (x_2, y_2) .

But in this case, suppose that a restriction is imposed on the class of permissible curves in such a way that the curve cannot pass through the point of certain R bounded by the curve $\psi(x, y) = 0$.

In such a problem that extremizing curve C either passes through a region which is completely outside R or C consists of arcs lying outside R and also consists of parts of the boundary of the region R .

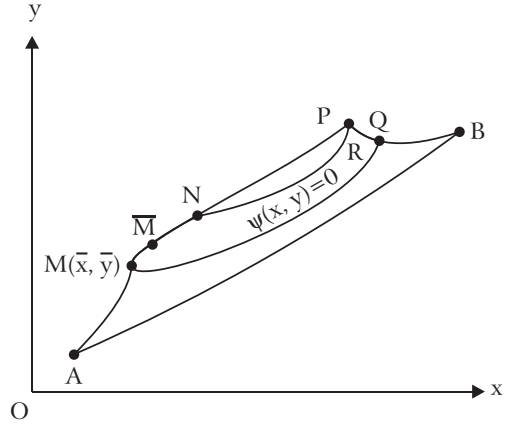


Fig. 2.8

Since on these parts two sided variation (unaffected by the region R) are possible. We now derive conditions at the points of transition M, N, P and Q .

Now, if co-ordinate of M be $(\bar{x}, \bar{y})$ then the functional can be written as

$$\begin{aligned} I &= \int_{x_1}^{x_2} F(x, y, y') dx \\ &= \int_{x_1}^{\bar{x}} F(x, y, y') dx + \int_{\bar{x}}^{x_2} F(x, y, y') dx \\ I &= I_1 + I_2 \end{aligned} \quad \dots(2)$$

where

$$I_1 = \int_{x_1}^{\bar{x}} F(x, y, y') dx$$

and

$$I_2 = \int_{\bar{x}}^{x_2} F(x, y, \bar{y}) dx.$$

If the point $M(\bar{x}, \bar{y})$ moves to neighbouring point $\bar{M}(\bar{x} + \delta\bar{x}, \bar{y} + \delta\bar{y})$ on the boundary of region R and if $y = \phi(x)$ be equation of boundary then

$$\delta I_1 = [F + (\phi' - y') F_{y'}]_{x=\bar{x}} \delta\bar{x} \quad \dots(3)$$

Now,

$$\begin{aligned} \delta I_2 &= I_2(\bar{y} + \delta\bar{y}) - I_2(\bar{y}) \\ &= \int_{\bar{x} + \delta\bar{x}}^{x_2} F(x, y, y') dx - \int_{\bar{x}}^{x_2} F(x, y, y') dx \end{aligned}$$

$$\Rightarrow \Delta I_2 = - \int_{\bar{x}}^{\bar{x} + \delta \bar{x}} F(x, y, y') dx$$

$$\text{or} \quad \Delta I_2 = - \int_{\bar{x}}^{\bar{x} + \delta \bar{x}} F(x, \phi(x), \phi'(x)) dx \quad [\because y = \phi(x)] \quad \dots(4)$$

Using mean value theorem of integral calculus, we have

$$\Delta I_2 = - [F(x, \phi, \phi')]_{x=\bar{x}} \cdot \Delta \bar{x} + \alpha \Delta \bar{x}$$

where $\alpha \rightarrow 0$ as $\Delta \bar{x} \rightarrow 0$.

Therefore, this gives

$$\delta I_2 = - [F(x, \phi, \phi')]_{x=\bar{x}} \cdot \Delta \bar{x}. \quad \dots(5)$$

Combining (3) and (5), we find that

$$\delta I = \delta I_1 + \delta I_2$$

$$\delta I = [F(x, y, y') - F(x, y, \phi') - (y' - \phi') F_{y'}(x, y, y')]_{x=\bar{x}} \cdot \delta \bar{x}$$

with $y(\bar{x}) = \phi(\bar{x})$

Since $\delta \bar{x}$ is arbitrary.

Then the necessary condition $\delta I = 0$ for an extremum reduces to

$$[F(x, y, y') - F(x, y, \phi') - (y' - \phi') F_{y'}(x, y, y')]_{x=\bar{x}} = 0 \quad \dots(6)$$

Applying the mean value theorem to this equation, we get

$$(y' - \phi') [F_{y'}(x, y, q) - F_{y'}(x, y, y')]_{x=\bar{x}} = 0 \quad \dots(7)$$

where q lies between $y'(\bar{x})$ and $\phi'(\bar{x})$.

Again, applying mean value theorem to (7), we get

$$[(y' - \phi')(q - y') F_{yy'}(x, y, \bar{q})]_{x=\bar{x}} = 0$$

where $\bar{q}$ lies between q and $y'(\bar{x})$.

Assume $F_{y'y'}(x, y, \bar{q}) \neq 0$

In this case $y'(\bar{x}) = \phi'(\bar{x})$ because $q = y'$ only when $y'(\bar{x}) = \phi'(\bar{x})$.

Hence, we conclude that at the point M , the extremal AM meets the boundary curve MN tangentially.

Example 2 : Find the shortest path from the point $A(-2, 3)$ to the point $B(2, 3)$ located in the region $y \leq x^2$.

Solution : Here, we find the extremum of the functional.

$$I[y] = \int_{-2}^2 \sqrt{1 + y'^2} (x) dx \quad \dots(1)$$

Subject to condition that

$$y \leq x^2, y(-2) = 3, y(2) = 3$$

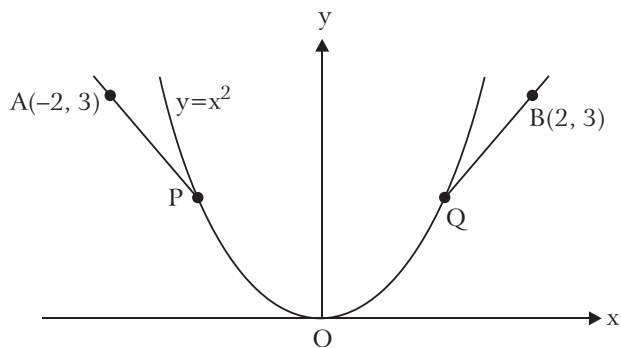


Fig. 2.9

Now, here $F = \sqrt{1 + y'^2}.$... (2)

$\therefore$ By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 0 - \frac{d}{dx} \left(\frac{y'}{\sqrt{1 + y'^2}} \right) = 0$$

$$\Rightarrow \frac{d}{dx} \left(\frac{y'}{\sqrt{1 + y'^2}} \right) = 0$$

Integrating, we get

$$\frac{y'}{\sqrt{1 + y'^2}} = \text{constant} = c_1'$$

$$\Rightarrow y'^2 = c_1'^2 (1 + y'^2)$$

$$\Rightarrow (1 - c_1'^2) y'^2 = c_1'^2$$

$$\Rightarrow y'^2 = \frac{c_1'^2}{1 - c_1'^2}$$

$$\text{or} \quad y' = \frac{c_1'}{\sqrt{1 - c_1'^2}} = c_2$$

$$\text{or} \quad \frac{dy}{dx} = c_2$$

Integrating, we get

$$y = c_1 + c_2 x. \quad \dots(3)$$

This is required extremal curve where c_1 and c_2 are constant.

$$\text{Now,} \quad F_{y'y'} = [1 + y'(x)]^{3/2} \neq 0$$

Thus, the required extremal will consist of portion of the straight line AP and QB both tangent to the parabola $y = x^2$ and the position POQ at the parabola.

Let $-\bar{x}$ and $\bar{x}$ be the abscissae of P and Q respectively.

Then, the condition of tangent of AP and BQ at P and Q are

$$\left. \begin{aligned} c_1 + c_2 \bar{x} &= \bar{x}^2 \\ c_2 &= 2\bar{x} \end{aligned} \right\}. \quad \dots(4)$$

Since tangent QB passes through $(2, 3)$.

$$\therefore \quad c_1 + 2c_2 = 3. \quad \dots(5)$$

Solving (4) and (5), we get two value of $\bar{x}$

$$\text{i.e.,} \quad \bar{x} = 1 \text{ and } \bar{x} = 3$$

The second value is clearly not possible.

$$\therefore \quad \bar{x} = 1$$

Therefore, (4) becomes

$$c_1 + c_2 = 1, c_2 = 2$$

$$\Rightarrow \quad c_1 = -1, c_2 = 2$$

Hence, the required extremal to

$$y = \begin{cases} -2x - 1, & \text{if } -2 \leq x \leq -1 \\ x^2, & \text{if } -1 \leq x \leq 1 \\ 2x - 1, & \text{if } 1 \leq x \leq 2 \end{cases}$$

This, clearly minimize the functional.

Exercise 2.1

1. Use the calculus of variation to find the shortest distance between the line $y = x$ and the parabola $y^2 = x - 1$.
2. If l is not prescribed show that the extremals corresponding to the problem $\delta \int_0^l y'^2 dx = 0$, $y(0) = 2$, $y(l) = \sin l$ are of the form $y(x) = 2 + 2x \cos l$ where l satisfies the transcendental equation $2 + 2l \cos l - \sin l = 0$.
3. If l is not prescribed show that the extremals of the problem

$$\delta \int_0^l [y'^2 + 4(y - l)] dx = 0$$

$$y(0) = 2, y(l) = l^2$$

are of the form $y(x) = x^2 + 2 - \frac{2x}{l}$.

where l is root of equation $2l^4 - 2l^3 + 1 = 0$.

4. Find the shortest distance between the point $A(-1, 5)$ and the parabola $y^2 = x$.
5. Find the shortest distance between the point $A(-1, 3)$ and the straight line $y = 1 - 3x$.

Answers

1.	$\frac{3\sqrt{2}}{8}$
4.	$\sqrt{20}$
5.	$\frac{1}{\sqrt{10}}$

HIGHLIGHTS

- * If the functional

$$I[y(x)] = \int_{x_1}^{x_2} F(x, y, y') dx$$

such that boundary point (x_1, y_1) is fixed and other boundary point (x_2, y_2) is moving along curve $y = \phi(x)$.

Then Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

and transversality condition.

$$[F + (\phi' - y') F_{y'}]_{x=x_2} = 0$$

gives extremal.

- * If the functional

$$I[y(x)] = \int_{x_1}^{x_2} F(x, y, y') dx$$

such that boundary point (x_1, y_1) moves along curve $y = \psi(x)$ and (x_2, y_2) moves along the curve $y = \phi(x)$.

Then Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

and transversality condition

$$[F + (\psi' - y') F_{y'}]_{x=x_1} = 0$$

and

$$[F + (\phi' - y') F_{y'}]_{x=x_2} = 0$$

gives the extremal.

- * If the functional

$$I[y(x), z(x)] = \int_{x_1}^{x_2} F(x, y(x), z(x), y'(x), z'(x)) dx$$

such that the point (x_1, y_1, z_1) is fixed and (x_2, y_2, z_2) moves in an arbitrary manner along a given curve

$$y = \phi(x), \quad z = \psi(x)$$

Then Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

and

$$\frac{\partial F}{\partial z} - \frac{d}{dx} \left(\frac{\partial F}{\partial z'} \right) = 0$$

and transversality condition

$$[F + (\phi' - y') F_{y'} + (\psi' - z') F_{z'}]_{x=x_2} = 0$$

gives extremal.

○○○



Chapter 3



Sufficient Conditions for an Extremum

3.1 Definitions

3.1.1 Proper Field

A family of curves $y = y(x, c)$ where c is parameter is said to form a **proper field** in a given region D of the xy -plane if one and only one curve of the family passes through every point of the region D .

Examples :

1. Inside the circle $x^2 + y^2 = 1$ the family of parallel lines $y = x + c$ (c being an arbitrary constant) forms a proper field; because through any point at given circle there passes one and only one straight line of the family.

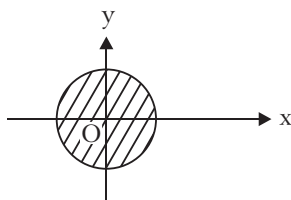


Fig. 3.1

2. The family of parabola $y = (x + a)^2$ inside the circle $x^2 + y^2 = 1$ does not form a proper field, because parabola of this family intersect inside the circle as shown in figure 3.2.

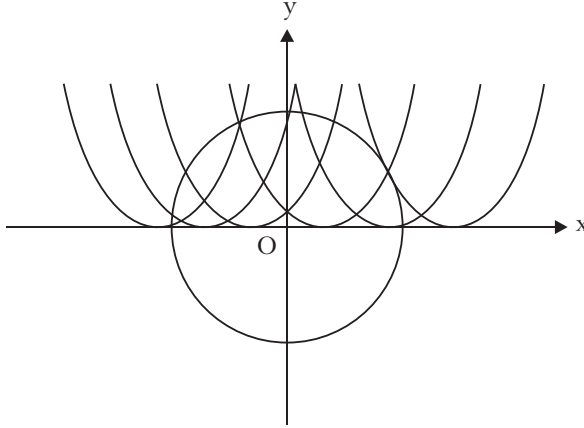


Fig. 3.2

3.1.2 Central Field

If all the curves of family $y = y(x, c)$ passes through a single point (x_0, y_0) of the domain D , then such a family is said to form a **central field** over the domain D , if these curves cover D without self intersection and the point (x_0, y_0) lies outside D .

Note : If family of curves $y = y(x, c)$ passes through a single point (x_0, y_0) which is not in domain D , then the point (x_0, y_0) is called the centre at **pencil of curves**.

Example : Let D is the domain $x > 0$, then $(0, 0) \notin D$.

Since family of straight line $y = cx$ passes through $(0, 0)$.

$\Rightarrow (0, 0)$ is the centre at pencil of straight lines.

Hence, family of straight line $y = cx$ forms a **central field** in the domain $x > 0$.

3.1.3 Extremal Field (Field of Extremal)

If a proper field or a central field is formed by a family of extremals of given variational problem, then it is called an **extremal field** or **field of extremals**.

Definition (Embedding in a Field of Extremals) :

Let $y = y(x)$ be extremal of functional

$$I[y(x)] = \int_{x_1}^{x_2} F(x, y, y') dx$$

with (x_1, y_1) and (x_2, y_2) as boundary point.

If it is possible to find a family of extremals $y = y(x, c)$ in such a way that the family forms a fixed and $y = y(x)$ is a member of this family for some value at c and extremal does not lie on the boundary of domain in which the family forms a field then $y = y(x)$ is said to be **embedded in a extremal field**.

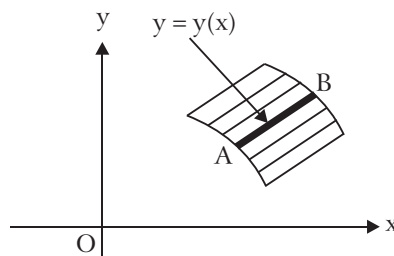


Fig. 3.3

3.1.4 Definition (Embedding in a Central Field)

If a pencil at extremals originating from the point (x_1, y_1) form a central field which includes the extremal $y = y(x)$, then extremal curve $y = y(x)$ is said to be **embedded in central field**.

In this case, the parameter of the family of curves can be taken as the slope of **tangent line** to the curve at A .

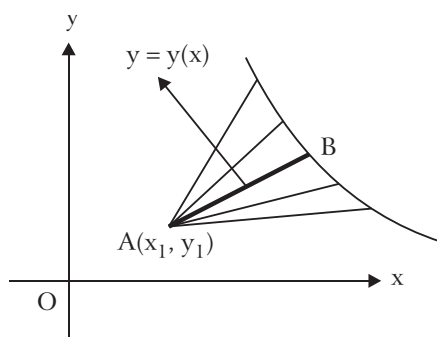


Fig. 3.4

Problem Set

Problem 1 : Find the proper and central fields of extremals for the functional

$$\int_0^{\pi/4} (y'^2 - y^2 + 2x^2 + 4) dx.$$

Solution : $I[y(x)] = \int_0^{\pi/4} (y'^2 - y^2 + 2x^2 + 4) dx$

Here $F = y'^2 - y^2 + 2x^2 + 4$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow -2y - \frac{d}{dx} (2y') = 0$$

or $y + \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$

or $\frac{d^2 y}{dx^2} + y = 0$

or $(D^2 + 1) y = 0. \quad \dots(1)$

Auxiliary equation is

$$m^2 + 1 = 0$$

$$\Rightarrow m = \pm i$$

$\therefore$ The solution is

$$y = c_1 \cos x + c_2 \sin x \quad \dots(2)$$

where c_1 and c_2 are arbitrary constant.

This is required extremal of the functional.

Now for $c_2 = 0$ the equation (2) becomes

$$y = c_1 \cos x$$

which is a proper field of extremals in a domain $0 \leq x \leq \frac{\pi}{4}$.

Again for $c_1 = 0$, the equation (2) becomes

$$y = c_2 \sin x$$

which is a central field of extremals in domain $0 \leq x \leq \frac{\pi}{4}$.

Problem 2 : Show that the extremal of the variational problem

$$\int_0^2 (y'^2 + x^2) dx$$

$$y(0) = 1, y(2) = 3$$

is included in a proper field of extremal of the given functional.

Solution : $I[y(x)] = \int_0^2 (y'^2 + x^2) dx$

Here $F = y'^2 + x^2$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 0 - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

$$\Rightarrow \frac{d^2 y}{dx^2} = 0$$

On integrating twice, we get

$$y = c_1 x + c_2 \quad \dots(1)$$

Now, $y(0) = 1 \Rightarrow c_2 = 1$

and $y(2) = 3 \Rightarrow 2c_1 + c_2 = 3$

$$\Rightarrow 2c_1 + 1 = 3$$

$$\therefore \Rightarrow c_1 = 1$$

Hence, we get $c_1 = c_2 = 1$

Therefore, (1) becomes

$$y = x + 1$$

This is required extremal of functional.

Now, for $c_1 = 1$, equation (1) becomes

$$y = x + c_2$$

which is proper field of extremals in the domain $0 \leq x \leq 2$.

For $c_2 = 1$, $y = x + c_2$ gives $y = x + 1$.

which shows that the extremal $y = x + 1$ is included in the proper field of extremals $y = x + c_2$.

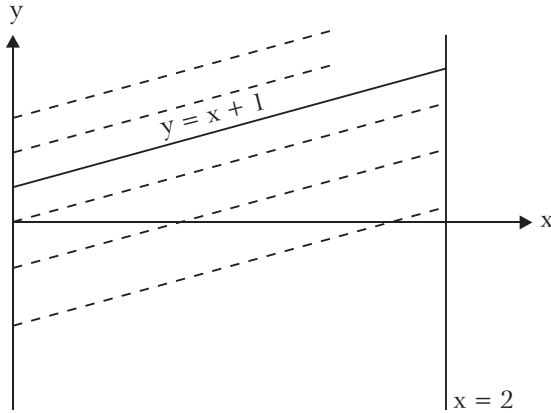


Fig. 3.5

Problem 3 : Discuss the extremal field for the functional as

$$I[y(x)] = \int_0^a (y'^2 - y^2) dx$$

$$y(0) = 0, y(a) = 0$$

Solution : Here,

$$F = y'^2 - y^2$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow -2y - \frac{d}{dx} (2y') = 0$$

$$\text{or } y + \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

or
$$\frac{d^2 y}{dx^2} + y = 0 \quad \dots(1)$$

Auxiliary equation is

$$m^2 + 1 = 0$$

$$\Rightarrow m = \pm i.$$

$\therefore$ The general solution of (1) is

$$y = c_1 \cos x + c_2 \sin x \quad \dots(2)$$

Now, $y(0) = 0 \Rightarrow c_1 = 0.$

Then (2) becomes

$$y = c_2 \sin x \quad \dots(3)$$

which pencil curve form a central field on $0 \leq x \leq a$; $a < \pi$ including for $c_2 = 0$ the extremal is $y = 0$.

For $a \geq \pi$, the family at extremals equation (3) does not form an extremal field.

3.2 Jacobi Condition

Consider one parameter family of plane curves

$$\phi(x, y, c) = 0$$

where c is parameter.

For this family c -discriminant is the locus of point of intersection of

$$\phi(x, y, c) = 0 \text{ and } \frac{\partial \phi}{\partial c} = 0$$

which include envelope of the family, the locus of cuts and locus of nodal points too.

If we have a pencil of curves with centre at $A(x_1, y_1)$ then $A(x_1, y_1)$ also belongs to this locus.

Suppose a pencil of extremals passing from $A(x_1, y_1)$ such that the curve $\phi(x, y) = 0$ is the c -discriminant.

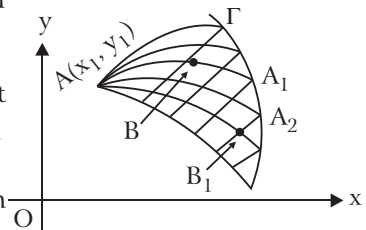


Fig. 3.6

Then the envelope Γ of this pencil of extremals will belongs to $\phi(x, y) = 0$.

Every extremals of the family will touch this envelope Γ . The point A_1 where the extremal $y = y(x)$ touches the envelope is called the conjugate point of A . If $B(x_2, y_2)$ be a point which lies in between A and A_1 then the extremals of the pencil close to AB do not intersect. Hence, it follows that extremal close to AB form a central field including the arc AB .

Now for the extremal AB_1 it follows that the conjugate point A_2 of A lies in between A and B_1 and the curves of the pencil closed to AB_1 intersect.

Therefore, the extremal AB_1 cannot be embedded in a central field.

Hence, **to embed an arc AB of the extremal in a central field of extremals it is sufficient that the conjugate point of A does not lie on the curve.**

This is known as **Jacobi condition**.

3.2.1 Mathematical Definition

Let $y = y(x, c)$ be equation of pencil of extremals with c is parameter and A centre. The parameter c is regarded as slope $y' = \frac{dy}{dx}$ of the extremals at A .

The c -discriminant is given by $y = y(x, c)$ and $\frac{\partial f}{\partial c} = 0$.

Let $u = \frac{\partial y(x, c)}{\partial c}$ which is a function of x along for every fixed curve of family.

For the extremum of

$$I[y(x)] = \int_{x_1}^{x_2} F(x, y, y') dx \quad \dots(1)$$

$y = y(x, c)$ is solution of Euler's equation.

$$\therefore F_y[x, y(x, c); y'_x(x, c)] - \frac{d}{dx} F_{y'}[x, y(x, c), y'_x(x, c)] = 0$$

Differentiating w.r.t. c , we get

$$\left(F_{yy} - \frac{d}{dx} F_{yy'} \right) u - \frac{d}{dx} (F_{y'y'} u') = 0 \quad \dots(2)$$

This is **Jacobi** equation.

Let $y(x)$ is a solution of Euler's equation with $c = c_0$ for the extremal AB .

Further, if the solution $u = \frac{\partial y}{\partial c}$ vanishes at $A(x_1, y_1)$ then the centre of the pencil belongs to the c -discriminant curve, also vanishes at some point of the interval $x_1 < x < x_2$, then the point conjugate to A given by

$$y = y(x, c_0) \text{ and } \left(\frac{\partial y}{\partial c} \right)_{c=c_0} = 0$$

lies on the arc AB of the extremal with B at the point (x_2, y_2) .

If there exists a solution of (2) which vanishes for $x = x_1$ and does not vanish at any point in $x_1 \leq x \leq x_2$ then there are no points conjugate to A lying on the arc AB .

Thus, the Jacobi condition is satisfied and the arc at the extremal can be embedded in a central field of extremals with centre at A .

3.3 Sufficient Condition for Extremum (Legendre Condition)

Legendre condition is sufficient condition to find out the nature of extremal.

Consider the functional

$$I[y(x)] = \int_{x_1}^{x_2} F(x, y, y') dx \quad \dots(1)$$

with $y(x_1) = y_1, y(x_2) = y_2$

Let C be extremal curve of functional (1) and $\bar{C}$ be neighbouring curve of C .

Therefore, consider

$$I_1 = \int_C F(x, y, y') dx$$

$$I_2 = \int_{\bar{C}} F(x, y, y') dx$$

Let $p = \frac{dy}{dx}$ on C .

Consider the auxiliary functional

$$\int_{\bar{C}} \left[F(x, y, p) + \left(\frac{dy}{dx} - p \right) F_p(x, y, p) \right] dx$$

The integral in this integration is an exact differentiation of function. Therefore, it is independent of path.

$$\therefore \int_{\bar{C}} [F(x, y, p) + (y' - p) F_p(x, y, p)] dx = \int_C F(x, y, y') dx \quad \dots(2)$$

$$\begin{aligned} \text{Now, } \Delta I &= \int_{\bar{C}} F(x, y, y') dx - \int_C F(x, y, y') dx \\ &= \int_{\bar{C}} F(x, y, y') dx - \int_{\bar{C}} [F(x, y, p) + (y' - p) F_p(x, y, p)] dx \quad [\text{By (2)}] \\ &= \int_{\bar{C}} [F(x, y, y') - F(x, y, p) - (y' - p) F_p(x, y, p)] dx \end{aligned}$$

$$\text{or } \Delta I = \int_{\bar{C}} E(x, y, p, y') dx$$

where

$$E(x, y, p, y') = F(x, y, y') - F(x, y, p) - (y' - p) F_p(x, y, p)$$

This $E(x, y, p, y')$ is called **Weirstrass function**.

Now,

if $E \leq 0$, then extremal is maximum.
and if $E \geq 0$, then extremal is maximum.

This is required **Legendre condition**.

SOLVED EXAMPLES

Example 1 : Find Weirstrass function and test the extremal of the functional

$$I[y(x)] = \int_0^a y'^2 dx$$

$$y(0) = 0, y(a) = b$$

where $a > 0, b > 0$.

Solution : Here,

$$F = y'^2$$

...(1)

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 0 - \frac{d}{dx} (y'^2) = 0$$

$$\Rightarrow \frac{d}{dx} (y'^2) = 0$$

On integrating, we get

$$y'^2 = \text{constant}$$

$$\text{or} \quad \left(\frac{dy}{dx} \right)^2 = \text{constant}$$

$$\text{or} \quad \frac{dy}{dx} = \text{constant} = c_1.$$

Again integrating, we get

$$y = c_1 x + c_2 \quad \dots(2)$$

$$\text{Now,} \quad y(0) = 0 \Rightarrow c_2 = 0$$

$$y(a) = b \Rightarrow c_1 a = b \Rightarrow c_1 = \frac{b}{a}$$

Putting value of c_1 and c_2 in (2), we get

$$y = \frac{b}{a} x \quad \dots(3)$$

This is required extremal.

Weirstrass Function

The Weirstrass function is

$$\begin{aligned} E(x, y, y', p) &= F(x, y, y') - F(x, y, p) - (y' - p) F_p(x, y, p) \\ &= y'^3 - p^3 - (y' - p) \cdot 3p^2 \quad [\because F(x, y, y') = y'^3 \text{ By (1)}] \end{aligned}$$

$$\begin{aligned}
 &= y'^3 - p^3 - 3p^2 y' + 3p^3 \\
 &= y'^3 + 2p^3 - 2p^2 y'
 \end{aligned}$$

$$E(x, y, y', p) = (y' - p)^2 (y' + 2p)$$

This is required Weirstrass function.

Now, since $E(x, y, y', p) = (y' - p)^2 (y' + 2p) \geq 0$

Therefore, extremal is maxima.

3.4 Weak and Strong Extremum

Consider the functional

$$I[y(x)] = \int_{x_1}^{x_2} F(x, y, y') dx$$

with $y(x_1) = x_1, y(x_2) = y_2$

Let C be the curve of the extremal of the given functional.

Also, assume that the extremal of curve C is included in a field of extremals

Then Legendre condition for weak extremum and strong extremum are :

3.4.1 Weak Extremum

1. The curve C is extremal satisfying the boundary condition.
2. Jacobi condition must be satisfied *i.e.*, the extremal C must be embedded in the field of extremals.
3. The Weirstrass function E does not change sign at any point (x, y) close to the curve C and for arbitrary values of y' close to $p(x, y)$ on the extremals.
4. For weak minimum $E \geq 0$ or $F_{y'y'} > 0$ on C
and for weak maximum $E \leq 0$ or $F_{y'y'} < 0$ on C .

3.4.2 Strong Extremum

1. The curve C is an extremal which satisfy given boundary condition.
2. The extremal C is embedded in a field of extremals *i.e.*, Jacobi conditions are satisfied.
3. At a point (x, y) closed to the curve C and for arbitrary value of y' , the Weirstrass function E does not change sign.
4. For strong minimum $E \geq 0$ or $F_{y'y'} \geq 0$ at point close to C and also arbitrary value of y' and for strong maximum $E \leq 0$ or $F_{y'y'} < 0$ at points closed curve C and also for arbitrary value of y' .

Problem Set

Problem 1 : Test for an extremal the functional

$$I[y(x)] = \int_0^2 (e^{y'} + 3) dx$$

$$y(0) = 0, y(2) = 1$$

Solution : Here

$$F(x, y, y') = e^{y'} + 3 \quad \dots(1)$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\text{By (1)} \Rightarrow 0 - \frac{d}{dx} (e^{y'}) = 0$$

$$\Rightarrow \frac{d}{dx} (e^{y'}) = 0$$

$$\Rightarrow e^{y'} y'' = 0$$

$$\Rightarrow y'' = 0 \quad [\because e^{y'} \neq 0]$$

$$\text{i.e.,} \quad \frac{d^2 y}{dx^2} = 0$$

On integrating, we get

$$y = c_1 x + c_2 \quad \dots(2)$$

Hence, extremal of the given functional is attained only on the straight line.

Now, from (2)

$$y(0) = 0 \Rightarrow c_2 = 0$$

$$y(2) = 1 \Rightarrow 2c_1 = 0$$

$$\therefore c_1 = \frac{1}{2}, c_2 = 0$$

Then (1) becomes

$$y = \frac{1}{2} x$$

Hence, the extremal satisfying the boundary condition is $y = \frac{x}{2}$ which is included in the central field of extremals $y = c_1 x$.

Now,
$$F(x, y, y') = e^{y'} + 3$$

$$\therefore F_{y'} = e^{y'}$$

and
$$F_{y'y'} = e^{y'} > 0 \text{ for any value of } y'.$$

Therefore, by Legendre condition, the given functional is strong minimum on extremal $y = \frac{x}{2}$.

Problem 2 : Test for an extremum the functional

$$I[y(x)] = \int_0^a (y'^2 - y^2) dx ;$$

$$y(0) = 0, y(a) = 0, a > 0.$$

Solution : Here
$$F = y'^2 - y^2 \quad \dots(1)$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

From (1),

$$\Rightarrow -2y - \frac{d}{dx}(2y') = 0$$

$$\Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) + y = 0$$

$$\text{i.e.,} \quad \frac{d^2 y}{dx^2} + y = 0$$

$$\text{or} \quad (D^2 + 1) y = 0$$

Auxiliary equation

$$m^2 + 1 = 0.$$

$$\Rightarrow m = \pm i.$$

$\therefore$ The general solution is

$$y = c_1 \cos x + c_2 \sin x. \quad \dots(2)$$

$$\text{Now,} \quad y(0) = 0 \Rightarrow c_1 = 0$$

$$y(a) = 0 \Rightarrow c_2 \sin a = 0$$

Hence, we get

$$c_1 = 0, \quad c_2 \sin a = 0$$

Now, if $a \neq n\pi$ i.e., $\sin a \neq 0$

$$\text{Then} \quad c_1 = 0, \quad c_2 = 0$$

Hence, if $a \neq n\pi$, the extremum is attained only on the straight line $y = 0$.

Now, if $a < \pi$, then pencil at extremals $y = c_2 \sin x$ with centre $(0, 0)$ for a central field.

$$\text{and now since} \quad F = y'^2 - y^2$$

$$\therefore F_{y'} = 2y'$$

$$\text{and} \quad F_{y'y'} = 2 > 0; \quad \forall y'$$

Therefore, a strong minimum is attained on $y = 0$ for $a < \pi$.

For $a > \pi$, extremals $y = c_2 \sin x$ neither form a proper field nor form a central field.

Hence, for $a > \pi$, minimum is not attained on $y = 0$.

Problem 3 : Investigate for an extremal the functional

$$I[y(x)] = \int_0^1 \left(x + 2y - \frac{1}{2} y'^2 \right) dx$$

$$y(0) = 0 ; y(1) = 0.$$

Solution : Here, $F = x + 2y - \frac{1}{2} y'^2$(1)

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

From (1),

$$\Rightarrow 2 - \frac{d}{dx} \left(\frac{1}{2} \cdot 2y' \right) = 0$$

$$\Rightarrow 2 - \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

i.e.,
$$\frac{d^2 y}{dx^2} = 2$$

On integrating, we get

$$y = x^2 + c_1 x + c_2 \quad \dots(2)$$

Now,

$$y(0) = 0 \Rightarrow c_2 = 0$$

$$y(1) = 0 \Rightarrow 1 + c_1 + c_2 = 0$$

$$\Rightarrow c_1 = -1 \quad [\because c_2 = 0]$$

$$\therefore c_1 = -1, c_2 = 0$$

Therefore (2) becomes

$$y = x^2 - x$$

Hence extremal satisfying the boundary condition is $y = x^2 - x$ which is included in the central field of extremals $y = x_1^2 + c_1 x$. Whose centre at $(0, 0)$.

Now, since
$$F = x + 2y + \frac{1}{2} y'^2$$

$$\therefore F_{y'} = \frac{1}{2} \cdot 2y' = y'$$

$$F_{y'y'} = 1 > 0$$

Therefore, by Legendre condition the given functional is strong minimum on extremal $y = x^2 - x$.

Problem 4 : Show that the Jacobi condition satisfied for the extremal of the functional.

$$I[y(x)] = \int_0^a (y'^2 + y^2 + x^2) dx$$

which passes through $(0, 0)$ and $(a, 0)$.

Solution : Here,

$$F = y'^2 + y^2 + x^2$$

Therefore,
$$F_y = \frac{\partial F}{\partial y} = 2y, \quad F_{yy} = \frac{\partial^2 F}{\partial y^2} = 2$$

$$F_{y'} = \frac{\partial F}{\partial y'} = 2y', \quad F_{yy'} = \frac{\partial^2 F}{\partial y \partial y'} = 0$$

$$F_{y'y'} = \frac{\partial^2 F}{\partial y'^2} = 2$$

By Jacobi equation,

$$\left[F_{yy} - \frac{d}{dx} F_{yy'} \right] u - \frac{d}{dx} [F_{y'y'} u'] = 0$$

$$\Rightarrow \left[2 - \frac{d}{dx} (0) \right] u - \frac{d}{dx} [2u'] = 0$$

$$\Rightarrow 2u - 2 \frac{d}{dx} \left(\frac{du}{dx} \right) = 0$$

$$\text{or} \quad \frac{d^2 u}{dx^2} - u = 0$$

$$\text{or} \quad (D^2 - 1) y = 0 \quad \dots(1)$$

Auxiliary equation is

$$m^2 - 1 = 0$$

$$\Rightarrow m = \pm 1$$

Therefore, general solution of (1) is

$$u = c_1 \sinh x + c_2 \cosh x \quad \dots(2)$$

$$\text{Now,} \quad u(0) = 0 \Rightarrow c_2 = 0$$

Then (2) becomes

$$u = c_1 \sinh x.$$

Hence, pencil curve $u = c_1 \sinh x$ is zero at $x = x_1 = 0$ and does not vanish at any other point $0 < x < a$ (i.e., $x_1 < x < x_2$).

Hence, Jacobi condition is satisfied for all values of a .

Problem 5 : Show that the Jacobi condition for the central field of extremals for

$$I[y(x)] = \int_0^{\pi/2} \left(xyy' + y^2 - \frac{y'^2}{2} \right) dx$$

$$u(0) = 0$$

where $u = \delta y$ is fulfilled.

Also, show that for extremals the functional is maximum.

Solution : Here

$$F = x + y' + y^2 - \frac{y'^2}{2}$$

$$\therefore F_y = \frac{\partial F}{\partial y} = xy' + 2y$$

$$F_{yy} = \frac{\partial^2 F}{\partial y^2} = 2$$

$$F_{y'} = \frac{\partial F}{\partial y'} = xy - y'$$

$$F_{y'y'} = \frac{\partial^2 F}{\partial y'^2} = -1$$

Therefore, Jacobi equation

$$\left[F_{yy} - \frac{d}{dx} F_{yy'} \right] u - \frac{d}{dx} [F_{y'y'} u'] = 0$$

becomes

$$\left[2 - \frac{d}{dx} (x) \right] u - \frac{d}{dx} [-u'] = 0$$

$$\Rightarrow 2u - u + \frac{d}{dx} \left(\frac{du}{dx} \right) = 0$$

$$\Rightarrow \frac{d^2 u}{dx^2} + u = 0$$

$$\text{or} \quad (D^2 + 1) u = 0 \quad \dots(1)$$

Auxiliary equation is

$$m^2 + 1 = 0$$

$$\Rightarrow m = \pm i$$

$\therefore$ The solution of (1) is

$$u = c_1 \cos x + c_2 \sin x \quad \dots(2)$$

$$\text{Now,} \quad u(0) = 0 \Rightarrow c_1 = 0$$

Then (2) becomes

$$u = c_2 \sin x. \quad \dots(3)$$

This is required central field of extremal.

Now, $u = 0$ at $x = 0$

and u does not vanish in the internal $0 \leq x \leq \frac{\pi}{2}$.

$\therefore$ Jacobi condition is satisfied.

Now, $F_{y'y'} = -1 < 0$

Therefore, by Legendre condition for $u = c_2 \sin x$ the given functional is maximum.

Problem 6 : Discuss the Jacobi and Legendre conditions for extremum for the functional

$$I[y(x)] = \int_0^1 \left(\frac{x^2 y'^2}{2} - 2xyy' + y \right) dx$$

$$u(0) = 0, \text{ where } u = \delta y.$$

Further, derive the extremal satisfying $u(1) = \frac{1}{2}$ and emanating from $(0, 1)$.

Solution : Here, $F = \frac{x^2 y'^2}{2} - 2xyy' + y$

$$\therefore F_y = \frac{\partial F}{\partial y} = -2xy' + 1$$

$$F_{yy} = \frac{\partial^2 F}{\partial y^2} = 0$$

$$F_{y'} = \frac{\partial F}{\partial y'} = x^2 y' - 2xy$$

$$F_{y'y'} = \frac{\partial^2 F}{\partial y'^2} = x^2$$

$$F_{yy'} = \frac{\partial^2 F}{\partial y y'} = -2x$$



Therefore, Jacobi equation

$$\left[F_{yy} - \frac{d}{dx} F_{yy'} \right] u - \frac{d}{dx} [F_{y'y'} u'] = 0$$

becomes

$$\left[0 - \frac{d}{dx} (-2x) \right] u - \frac{d}{dx} [x^2 u'] = 0$$

or $\frac{d}{dx} (x^2 u') - 2u = 0$

or $x^2 u'' + 2xu' - 2u = 0$

or $x^2 \frac{d^2 u}{dx^2} + 2x \frac{du}{dx} - 2u = 0 \quad \dots(1)$

Put $z = \log x$ i.e., $x = e^z$

Then homogeneous differential equation (1) becomes

$$[D(D-1) + 2D - 2] u = 0, \text{ where } D \equiv \frac{d}{dz}$$

$\Rightarrow (D^2 + D - 2) u = 0 \quad \dots(2)$

Now, auxiliary equation is

$$m^2 + m - 2 = 0$$

$\Rightarrow (m-1)(m+2) = 0$

$\Rightarrow m = 1, -2$

$\therefore$ The solution of (2) is

$$u = c_1 e^z + c_2 e^{-2z}$$

$\Rightarrow u = c_1 x + \frac{c_2}{x^2} \quad [\because x = e^z] \quad \dots(3)$

Now, $u(0) = 0 \Rightarrow c_2 = 0$

Then (3) becomes

$$u = c_1 x$$

Now,

$$u(1) = \frac{1}{2} \Rightarrow \frac{1}{2} = c_1$$

$\therefore$

$$u = \frac{1}{2} x.$$

This is required extremal satisfying the boundary condition which is included in the central field of extremal.

$$u = c_1 x$$

whose centre is at $(0, 0)$.

Also,

$$u = 0 \text{ at } x = 0$$

But never vanishes

$$0 < x \leq 1.$$

Therefore, Jacobi condition is satisfied.

Now,

$$F = \frac{x^2 y'^2}{2} - 2xyy' + y$$

$\therefore$

$$F_{y'} = x^2 y' - 2xy$$

$$F_{y'y'} = x^2 > 0, \forall x$$

Then by Legendre condition for the extremal $u = \frac{x}{2}$ the given functional is minimum.

Also,

$$u = \delta y$$

$\Rightarrow$

$$\delta y = \frac{x}{2}$$

or

$$y = \frac{x^2}{4} + c_3. \quad \dots(4)$$

Since extremal emanating from $(0, 1)$.

i.e.,

$$y(0) = 1$$

$\Rightarrow$

$$1 = c_3$$



Then (4) becomes

$$y = \frac{x^2}{4} + 1$$

i.e.,
$$x^2 - 4y + 1 = 0$$

This is required extremal emanating from (0, 1).

3.5 Application of the Calculus of Variation

The calculus of variation is widely applied in mechanics, mechanical engineering control theory etc. and is used in solving some important problem in economics.

In mechanics, Hamilton's principle and Lagrange's equation can be derived very easily with the help of calculus of variation.

3.6 Hamilton's Principle

A particle moves in a conservative field in such a way that

$$\int_{t_1}^{t_2} (T - V) dt$$

is extremum, actually a minimum, where T is kinetic energy and V is potential energy of the system.

Proof : Let there be n particles of mass m_i ; $i = 1, 2, \dots, n$ and their position vectors are r_i ; $i = 1, 2, \dots, n$ relative to co-ordinate system. Let F_i ; $i = 1, 2, \dots, n$ be applied force acting on the i^{th} particle.

Then, the equation of motion of particles are

$$m_i \frac{d^2 r_i}{dt^2} = F_i ; i = 1, 2, \dots, n \quad \dots(1)$$

and from these equations, we can determine the path c_i ; $i = 1, 2, \dots, n$ transversed by n particles.

Next, assume that the path of i^{th} particle has been varied without changing the end points.

If the variation of path be δr_i , sometimes called virtual displacement.

Then from (1), we get

$$\left(m_i \frac{d^2 r_i}{dt^2} \right) \delta r_i = F_i \delta r_i \quad \dots(2)$$

Summing for all particles, we get

$$\sum_{i=1}^n m_i \frac{d^2 r_i}{dt^2} \delta r_i = \sum_{i=1}^n F_i \delta r_i \quad \dots(3)$$

where the right hand side indicates the total work done δW under the displacement of the path, then

$$\delta W = \sum_{i=1}^n F_i \delta r_i$$

or

$$\delta W = \sum_{i=1}^n m_i \frac{d^2 r_i}{dt^2} \delta r_i \quad \dots(4)$$

Now, the kinetic energy of the system is

$$T = \frac{1}{2} \sum_{i=1}^n m_i \left(\frac{dr_i}{dt} \right)^2 \quad \dots(5)$$

$\therefore$

$$\begin{aligned} \delta T &= \sum_{i=1}^n m_i \frac{dr_i}{dt} \delta \left(\frac{dr_i}{dt} \right) \\ &= \sum_{i=1}^n m_i \frac{dr_i}{dt} \cdot \frac{d}{dt} (\delta r_i) \end{aligned} \quad \dots(6)$$

But
$$\frac{d}{dt} \left(\frac{dr_i}{dt} \cdot \delta r_i \right) = \frac{d^2 r_i}{dt^2} \cdot \delta r_i + \frac{dr_i}{dt} \cdot \frac{d}{dt} (\delta r_i).$$

Multiplying both sides by m_i and summing from $i = 1$ to n , we get

$$\begin{aligned} \sum_{i=1}^n m_i \frac{d}{dt} \left[\frac{dr_i}{dt} \delta r_i \right] &= \sum_{i=1}^n m_i \frac{d^2 r_i}{dt^2} \delta r_i + \sum_{i=1}^n m_i \frac{dr_i}{dt} \frac{d}{dt} [\delta r_i] \\ &= \delta T + \delta W \end{aligned} \quad [\text{By (4) and (6)}]$$

Hence,
$$\delta T + \delta W = \sum_{i=1}^n m_i \frac{d}{dt} \left[\frac{dr_i}{dt} \delta r_i \right].$$

Integrate w.r.t. t from t_1 to t_2 , we get

$$\int_{t_1}^{t_2} (\delta T + \delta W) dt = \left[\sum_{i=1}^n m_i \frac{dr_i}{dt} \delta r_i \right]_{t_1}^{t_2} = 0 \text{ because } \delta r_i = 0 \text{ at } t_1 \text{ and } t_2$$

$$\therefore \int_{t_1}^{t_2} (\delta T + \delta W) dt = 0. \quad \dots(7)$$

Now, if force is conservative then $W = -V$

where $V =$ potential function.

Therefore, (7) becomes

$$\int_{t_1}^{t_2} [\delta T + \delta(-V)] dt = 0$$

or
$$\delta \int_{t_1}^{t_2} (T - V) dt = 0$$

From this it can be stated that in a conservative field, a system moves from t_1 to t_2 in such a way that

$$\int_{t_1}^{t_2} (T - V) dt$$

is an extremum, actually a minimum.

Hence, we conclude that in a conservative field, a system moves from t_1 to t_2 in such a way that $\int_{t_1}^{t_2} (T - V) dt$ is minimum.

3.7 Lagrangian of a System

The expression

$$L = T - V$$

where T is kinetic energy and V is potential energy of the system is called Lagrangian.

Note : Lagrangian L is function of the co-ordinate of particle *i.e.*, generalized co-ordinate (q_i), their velocities ($\dot{q}_i$) and time (t).

i.e.,
$$L = L(q_1, q_2, \dots, q_n, \dot{q}_1, \dot{q}_2, \dots, \dot{q}_n, t)$$

Hence, Hamilton's principle state that for a conservative system

$$\int_{t_1}^{t_2} \delta L \, dt = 0$$

3.8 Lagrange's Equation

Consider Lagrangian

$$L = L(q_1, q_2, \dots, q_n, \dot{q}_1, \dot{q}_2, \dots, \dot{q}_n) \quad \dots(1)$$

where $q_1, q_2, \dots, q_n$ are generalized co-ordinate of system of particles.

Now, from (1)

$$\delta L = \sum_{j=1}^n \frac{\partial L}{\partial q_j} \delta q_j + \sum_{j=1}^n \frac{\partial L}{\partial \dot{q}_j} \delta \dot{q}_j \quad \dots(2)$$

By Hamilton's principle

$$\int_{t_1}^{t_2} \delta L \, dt = 0$$

Using (2), we get

$$\int_{t_1}^{t_2} \left[\sum_{j=1}^n \frac{\partial L}{\partial q_j} \delta q_j + \sum_{j=1}^n \frac{\partial L}{\partial \dot{q}_j} \delta \dot{q}_j \right] dt = 0$$

$$\Rightarrow \int_{t_1}^{t_2} \sum_{j=1}^n \frac{\partial L}{\partial q_j} \delta q_j \, dt + \int_{t_1}^{t_2} \sum_{j=1}^n \frac{\partial L}{\partial \dot{q}_j} \delta \dot{q}_j \, dt = 0$$

$$\Rightarrow \int_{t_1}^{t_2} \sum_{j=1}^n \frac{\partial L}{\partial q_j} \delta q_j \, dt + \int_{t_1}^{t_2} \sum_{j=1}^n \frac{\partial L}{\partial \dot{q}_j} \frac{d}{dt} (\delta \dot{q}_j) \, dt = 0$$

$$\Rightarrow \int_{t_1}^{t_2} \sum_{j=1}^n \frac{\partial L}{\partial q_j} \delta q_j \, dt + \left[\sum_{j=1}^n \frac{\partial L}{\partial \dot{q}_j} \delta \dot{q}_j \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \sum_{j=1}^n \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) \delta q_j \, dt = 0$$

Since $\delta q_j = 0$ at t_1 and t_2

$$\Rightarrow \int_{t_1}^{t_2} \sum_{j=1}^n \frac{\partial L}{\partial q_j} \delta q_j dt - \int_{t_1}^{t_2} \sum_{j=1}^n \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) \delta q_j dt = 0$$

$$\Rightarrow \int_{t_1}^{t_2} \left[\left\{ \sum_{j=1}^n \frac{\partial L}{\partial q_j} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) \right\} \delta q_j \right] dt = 0$$

$$\Rightarrow \sum_{j=1}^n \left\{ \frac{\partial L}{\partial q_j} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) \right\} \delta q_j = 0$$

Since δq_j 's are arbitrary and independent to each other.

$\therefore$

$$\boxed{\frac{\partial L}{\partial q_j} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) = 0}$$

This is Lagrange's equation.

3.9 Hamiltonian

Consider the Lagrangian

$$L = L(q_1, q_2, \dots, q_n, \dot{q}_1, \dot{q}_2, \dots, \dot{q}_n) \quad \dots(1)$$

where $q_1, q_2, \dots, q_n$ are generalized co-ordinate and $\dot{q}_1, \dot{q}_2, \dots, \dot{q}_n$ are generalized velocity.

Since,

$$L = T - V = \text{Kinetic Energy} - \text{Potential Energy}$$

$$= \frac{1}{2} m \dot{q}_j^2 - f(q)$$

$\therefore$

$$\frac{\partial L}{\partial \dot{q}_j} = m \dot{q}_j = \text{momentum of particle}$$

$$= p_j \text{ (say)}$$

∴ Generalized momentum is defined as

$$p_j = \frac{\partial L}{\partial \dot{q}_j} \quad \dots(2)$$

Now, by Lagrange equation

$$\frac{\partial L}{\partial q_j} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) = 0$$

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) = \frac{\partial L}{\partial q_j}$$

$$\Rightarrow \frac{d}{dt} (p_j) = \frac{\partial L}{\partial q_j} \quad [\text{By (2)}]$$

$$\text{or} \quad \dot{p}_j = \frac{\partial L}{\partial q_j} \quad \dots(3)$$

Now, from (1),

$$\begin{aligned} \frac{dL}{dt} &= \sum_{j=1}^n \frac{\partial L}{\partial q_j} \frac{dq_j}{dt} + \sum_{j=1}^n \frac{\partial L}{\partial \dot{q}_j} \frac{d\dot{q}_j}{dt} \\ &= \sum_{j=1}^n \dot{p}_j \dot{q}_j + \sum_{j=1}^n p_j \ddot{q}_j \quad [\text{by (2) and (3)}] \end{aligned}$$

$$\therefore \frac{dL}{dt} = \sum_{j=1}^n \frac{d}{dt} (p_j \dot{q}_j)$$

$$\text{or} \quad \frac{d}{dt} \left[\sum_{j=1}^n (p_j \dot{q}_j - L) \right] = 0$$

Integrating, we get

$$\sum_{j=1}^n p_j \dot{q}_j - L = \text{constant}$$

$$= H \text{ (say)}$$

Hence,

$$H = \sum_{j=1}^n p_j \dot{q}_j - L$$

This H is called **Hamiltonian**.

Note : Hamiltonian

$$H = H(q_1, q_2, \dots, q_n; p_1, p_2, \dots, p_n, t)$$

i.e., H is function of generalized co-ordinate, generalized momentum and time t .

3.10 Hamilton's Canonical Equation of Motion

Consider the Hamiltonian

$$H = H(q_1, q_2, \dots, q_n, p_1, p_2, \dots, p_n, t) \quad \dots(1)$$

where $q_1, q_2, \dots, q_n$ are generalized co-ordinate and $p_1, p_2, \dots, p_n$ are generalized momentum.

Consider Lagrangian

$$L = L(q_1, q_2, \dots, q_n, \dot{q}_1, \dot{q}_2, \dots, \dot{q}_n, t) \quad \dots(2)$$

$\therefore$ Generalized momentum is

$$\left. \begin{aligned} p_j &= \frac{\partial L}{\partial \dot{q}_j} \\ \dot{p}_j &= \frac{\partial L}{\partial q_j} \end{aligned} \right\}, \quad \dots(3)$$

and

Now from (1), we have

$$dH = \sum_{j=1}^n \frac{\partial H}{\partial q_j} dq_j + \sum_{j=1}^n \frac{\partial H}{\partial p_j} dp_j + \frac{\partial H}{\partial t} dt. \quad \dots(4)$$

Further, Hamiltonian H is given by

$$H = \sum_{j=1}^n p_j \dot{q}_j - L$$

$$\therefore dH = \sum_{j=1}^n \dot{q}_j dp_j + \sum_{j=1}^n p_j d\dot{q}_j - dL. \quad \dots(5)$$

From (2),

$$dL = \sum_{j=1}^n \frac{\partial L}{\partial q_j} dq_j + \sum_{j=1}^n \frac{\partial L}{\partial \dot{q}_j} d\dot{q}_j + \frac{\partial L}{\partial t} dt$$

Put in (5), we get

$$dH = \sum_{j=1}^n \dot{q}_j dp_j + \sum_{j=1}^n p_j d\dot{q}_j - \sum_{j=1}^n \frac{\partial L}{\partial q_j} dq_j - \sum_{j=1}^n \frac{\partial L}{\partial \dot{q}_j} d\dot{q}_j - \frac{\partial L}{\partial t} dt$$

Using (3), we get

$$dH = \sum_{j=1}^n \dot{q}_j dp_j + \sum_{j=1}^n p_j d\dot{q}_j - \sum_{j=1}^n \dot{p}_j dq_j - \sum_{j=1}^n p_j d\dot{q}_j - \frac{\partial L}{\partial t} dt$$

$$\text{or} \quad dH = \sum_{j=1}^n \dot{q}_j dp_j - \sum_{j=1}^n \dot{p}_j dq_j - \frac{\partial L}{\partial t} dt \quad \dots(6)$$

Comparing (4) and (6), we get

$$\dot{q}_j = \frac{\partial H}{\partial p_j}$$

$$\dot{p}_j = -\frac{\partial H}{\partial q_j}$$

$\dots(7)$

and
$$-\frac{\partial L}{\partial t} = \frac{\partial H}{\partial t}$$

Equations (7) is known as Hamilton's canonical equation of motion.

3.11 Principle of Least Action

If T is kinetic energy at time t of a conservative Holonomic system, then

$$\delta \int_{t_1}^{t_2} 2T \, dt = 0$$

i.e., $\int_{t_1}^{t_2} 2T \, dt$ has an extremum.

Proof : Since, we know that Lagrangian

$$L = T - V \quad \dots(1)$$

and $T + V = \text{Constant} = E$ (say)

where T is kinetic energy and V is potential energy of the system.

Now, by Hamilton's principle

$$\int_{t_1}^{t_2} \delta L \, dt = 0$$

$$\Rightarrow \int_{t_1}^{t_2} \delta (T - V) \, dt = 0 \quad [\text{By (1)}]$$

$$\text{or} \quad \int_{t_1}^{t_2} \delta [T - (E - T)] \, dt = 0 \quad [\text{By (2)}]$$

$$\text{or} \quad \int_{t_1}^{t_2} \delta (2T - E) \, dt = 0$$

$$\text{or} \quad \int_{t_1}^{t_2} \delta (2T) \, dt - \int_{t_1}^{t_2} \delta E \, dt = 0$$

or
$$\int_{t_1}^{t_2} \delta(2T) dt - 0 = 0 \quad [\because E \text{ is constant} \Rightarrow \delta E = 0]$$

or
$$\int_{t_1}^{t_2} \delta(2T) dt = 0$$

or

$$\delta \int_{t_1}^{t_2} 2T dt = 0$$

Problem Set

Problem 1 : Using Hamilton's principle, find the equation of one dimensional harmonic oscillator.

Solution : A system executing harmonic motion may be referred as harmonic oscillator.

e.g., A simple pendulum when the displacement of the motion is small, is an example of harmonic oscillator.

Now, the kinetic energy of harmonic oscillator $= \frac{1}{2} m\dot{x}^2$

Potential energy of harmonic oscillator

$$V = - \int F dx = \int kx dx = \frac{1}{2} kx^2$$

$\therefore$ The Lagrangian

$$\begin{aligned} L &= T - V \\ &= \frac{1}{2} m\dot{x}^2 - \frac{1}{2} kx^2 \end{aligned}$$

By Hamilton's principle

$$\delta \int_{t_1}^{t_2} L dt = 0$$

$$\Rightarrow \delta \int_{t_1}^{t_2} \left(\frac{1}{2} m \dot{x}^2 - \frac{1}{2} k x^2 \right) dt = 0$$

$$\Rightarrow \int_{t_1}^{t_2} \delta \left(\frac{1}{2} m \dot{x}^2 - \frac{1}{2} k x^2 \right) dt = 0$$

$$\Rightarrow \int_{t_1}^{t_2} (m \dot{x} \delta \dot{x} - k x \delta x) dt = 0$$

$$\text{or} \quad \int_{t_1}^{t_2} m \dot{x} \frac{d}{dt} (\delta x) dt - \int_{t_1}^{t_2} k x \delta x dt = 0$$

$$\text{or} \quad [m \dot{x} \delta x]_{t_1}^{t_2} - \int_{t_1}^{t_2} m \frac{d}{dt} (\dot{x}) \delta x dt - \int_{t_1}^{t_2} k x \delta x dt = 0$$

$$\Rightarrow 0 - \int_{t_1}^{t_2} m \frac{d}{dt} (\dot{x}) \delta x dt - \int_{t_1}^{t_2} k x \delta x dt = 0$$

[since $\delta x = 0$ at t_1 and t_2]

$$\text{or} \quad \int_{t_1}^{t_2} (m \ddot{x} + k x) \delta x dt = 0$$

Since δx is an arbitrary.

$$\therefore m \ddot{x} + k x = 0.$$

This is required equation of motion of one dimensional harmonic oscillator.

HIGHLIGHTS

- * Any family of curves $y = y(x, c)$ in a given region D of xy -plane is said to be form **proper field** if one and only one curve of family posses through every point of the region D .
- * A family of curves $y = y(x, c)$ is said to be form central field over domain D if :
 - (i) Curves cover D without self intersection
 - (ii) All curve passes through single point (x_0, y_0) .
- * A family of curves $y = y(x, c)$ passes through a single point (x_0, y_0) which is not in domain D . Then point (x_0, y_0) is called **centre** of pencil of curves.

- * A central field is called a field of extremals, if it is formed by a family of extremals.
- * To embed an arc AB of the extremal in a central field of extremals, it is sufficient that the conjugate point of A does not lie on arc AB . This is known as **Jacobi Condition**.
- * Legendre condition is sufficient condition to find an extremal of the functional.
- * The function

$$E(x, y, p, y') = F(x, y, y') - F(x, y, p) - (y' - p) F_p(x, y, p)$$

is called Weierstrass function.

- * Extremal is maximum if $E \leq 0$ and extremal is minimum if

$$E \geq 0.$$

This is **Legendre condition**.

○○○

Chapter **4**



Variational Method for Boundary Value Problems (Ordinary and Partial Differential Equation)

4.1 Introduction

The solution of Euler's equation with boundary conditions gives extremal of functional.

This approach gives a method of solving a boundary value problem approximately by assuming a trivial solution satisfying the given boundary conditions and extremizing the integral whose integrand is found from the given differential equation.

4.2 Rayleigh Ritz Method (For Ordinary Differential Equation)

Consider an ordinary differential equation

$$a_0(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_2(x) y = f(x) \quad \dots(1)$$

with boundary condition

$$y(x_1) = y_1, \quad y(x_2) = y_2.$$

In order to solve differential equation (1) by variational method, first, we construct $F(x, y, y')$ in such a way that Euler's equation of functional

$$I[y(x)] = \int_{x_1}^{x_2} F(x, y, y') dx \quad \dots(2)$$

becomes given differential equation (1).

Next, the proper choice of $y_0(x), y_1(x), y_2(x), \dots, y_n(x)$

we get

$$y(x) = y_0(x) + c_1 y_1(x) + \dots + c_n y_n(x). \quad \dots(3)$$

Out of these functions $y_0(x)$ is the simplest function which satisfy given boundary condition

$$i.e., \quad y_0(x_1) = y(x_1) = y_1$$

$$y_0(x_2) = y(x_2) = y_2$$

and other functions $y_1(x), y_2(x), \dots, y_n(x)$ are linearly independent function satisfy homogeneous boundary condition

$$y_k(x_1) = y_k(x_2) = 0; \quad k = 1, 2, \dots$$

$$i.e., \quad y_1(x_1) = y_1(x_2) = 0$$

$$y_2(x_1) = y_2(x_2) = 0$$

$$\dots \quad \dots \quad \dots$$

$$\dots \quad \dots \quad \dots$$

Now putting the value of $y(x)$ from (3) in (2), we get

$$I = I(c_1, c_2, \dots, c_n).$$

The constants $c_1, c_2, \dots, c_n$ are choosen in such a way that I is extremum.

Therefore by necessary condition for existence of extremal is

$$\frac{\partial I}{\partial c_1} = \frac{\partial I}{\partial c_2} = \frac{\partial I}{\partial c_3} = \dots = \frac{\partial I}{\partial c_n} = 0.$$



Solving these simultaneous equations, we will get $c_1, c_2, \dots, c_n$ and putting these value in (3) we get required solution of differential equation (1).

Remark : The selection of approximate solution may be done from solution

$$y(x) = (c_1 + c_2 x + c_3 x^2 + \dots + c_n x^n) (x - x^2)$$

one term approximation

$$y(x) = c_1 (x - x^2)$$

two term approximation

$$y(x) = (c_1 + c_2 x) (x - x^2)$$

which gives better approximation.

Obviously, approximate solution of ordinary differential equation are linearly independent.

Problem Set

Problem 1 : Solve the boundary value problem

$$y'' = 1$$

subject to boundary conditions

$$y(0) = 0, y(1) = 0$$

by Rayleigh-Ritz method.

Solution : The given differential equation is

$$y'' = 1 \quad \dots(1)$$

$$y(0) = 0, y(1) = 0.$$

First, we construct $F(x, y, y')$ in such a way that the Euler's equation of functional

$$I = \int_0^1 F(x, y, y') dx$$

becomes given differential equation $y'' = 1$, we choose

$$F = y'^2 + 2y$$

for which $y'' = 1$ is Euler's equation

because by Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 2 - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow 1 - y'' = 0$$

$$\Rightarrow y'' = 1.$$

Hence, the required variational problem is

$$I = \int_0^1 (y'^2 + 2y) dx. \quad \dots(2)$$

Next we assume the trivial solution

$$y(x) = c_0 + c_1 x + c_2 x^2.$$

$$\therefore y(0) = 0 \Rightarrow c_0 = 0 \quad \dots(3)$$

$$y(1) = 0 \Rightarrow c_1 + c_2 = 0 \Rightarrow c_2 = -c_1.$$

Therefore trivial solution becomes

$$y(x) = c_1 (x - x^2). \quad \dots(4)$$

$$\therefore y'(x) = c_1 (1 - 2x).$$

Putting value of y and y' in (2), we get

$$I = \int_0^1 [c_1^2 (1 - 2x)^2 + 2c_1 (x - x^2)] dx.$$

$$\begin{aligned} \therefore \frac{dI}{dc_1} &= \frac{d}{dc_1} \int_0^1 [c_1^2 (1 - 2x)^2 + 2c_1 (x - x^2)] dx \\ &= \int_0^1 \frac{\partial}{\partial c_1} \{c_1^2 (1 - 2x)^2 + 2c_1 (x - x^2)\} dx \\ &= \int_0^1 [2c_1 (1 - 2x^2) + 2(x - x^2)] dx \\ &= 2c_1 \left[-\frac{1}{2} \frac{(1 - 2x)^3}{3} \right]_0^1 + 2 \left[\left(\frac{x^2}{2} - \frac{x^3}{3} \right) \right]_0^1 \end{aligned}$$

$$\begin{aligned}
 &= 2c_1 \left(\frac{1}{6} + \frac{1}{6} \right) + 2 \left(\frac{1}{2} - \frac{1}{3} \right) \\
 &= \frac{2c_1}{3} + \frac{1}{3} = \frac{(2c_1 + 1)}{3}.
 \end{aligned}$$

$$\therefore \frac{dI}{dc_1} = \frac{(2c_1 + 1)}{3}.$$

The necessary condition for existence of extremal is

$$\frac{dI}{dc_1} = 0$$

$$\Rightarrow \frac{2c_1 + 1}{3} = 0$$

$$\Rightarrow 2c_1 + 1 = 0$$

$$\Rightarrow c_1 = -1/2.$$

Therefore (4) becomes

$$y(x) = -\frac{1}{2}(x - x^2)$$

$$\text{or } y(x) = \frac{1}{2}(x^2 - x).$$

This is exact solution of given boundary value problem.

Note : If we solve

$$\frac{d^2 y}{dx^2} = 1, y(0) = 0, y(1) = 0.$$

Integrate, we get

$$y(x) = \frac{x^2}{2} + c_1 x + c_2$$

$$y(0) = 0 \Rightarrow c_2 = 0$$

$$y(1) = 0 \Rightarrow \frac{1}{2} + c_1 + c_2 = 0$$

$$\Rightarrow c_1 = -1/2.$$

$$\therefore y(x) = \frac{x^2}{2} - \frac{1}{2}x$$

$$y(x) = \frac{1}{2}(x^2 - x).$$

Hence $y(x) = \frac{1}{2}(x^2 - x)$ is exact solution.

Problem 2 : Solve the boundary value problem

$$y'' - y + x = 0, \quad (0 \leq x \leq 1)$$

$$y(0) = y(1) = 0$$

by Rayleigh-Ritz method.

Solution : The given differential equation is

$$y'' - y + x = 0 \quad \dots(1)$$

with $y(0) = y(1) = 0.$

Its solution is equivalent to extremizing the functional

$$I = \int_0^1 F(x, y, y') dx$$

where $F(x, y, y') = 2xy - y^2 - y'^2.$

Hence, required variational problem is

$$\therefore I[y(x)] = \int_0^1 (2xy - y^2 - y'^2) dx. \quad \dots(2)$$

Next, assume the trivial solution is

$$y(x) = c_0 + c_1x + c_2x^2. \quad \dots(3)$$

Now, $y(0) = 0 \Rightarrow c_0 = 0$

$$y(1) = 0 \Rightarrow c_1 + c_2 = 0 \Rightarrow c_2 = -1.$$

$\therefore$ (3) becomes

$$y(x) = c_1(x - x^2). \quad \dots(4)$$

$$\therefore y'(x) = c_1(1 - 2x).$$

Putting value of y and y' in (2), we get

$$\begin{aligned}
 I &= \int_0^1 [c_1 \cdot 2x^2 (1-x) - c_1^2 x^2 (1-x)^2 - c_1^2 (1-2x)^2] dx \\
 &= \int_0^1 [2c_1 (x^2 - x^3) - c_1^2 (x^2 + x^4 - 2x^3) - c_1^2 (1 + 4x^2 - 4x)] dx \\
 &= \left[2c_1 \left(\frac{x^3}{3} - \frac{x^4}{4} \right) - c_1^2 \left(\frac{x^3}{3} + \frac{x^5}{5} - \frac{x^4}{2} + x + \frac{4}{3} x^3 - 2x^2 \right) \right]_0^1 \\
 I &= \frac{1}{6} c_1 - \frac{11}{30} c_1^2.
 \end{aligned}$$

$\therefore$ The necessary condition for existence of extremal is

$$\frac{dI}{dc_1} = 0$$

$$\Rightarrow \frac{1}{6} - \frac{11}{30} \cdot 2c_1 = 0$$

$$\Rightarrow c_1 = \frac{5}{22}.$$

Put in (4), we get

$$y(x) = \frac{5}{22} (x - x^2).$$

This is required approximate solution.

Problem 3 : Solve the boundary value problem

$$y'' + xy = -x$$

subject to boundary condition $y(0) = y(1) = 0$.

Solution : The given differential equation

$$y'' + xy = -x \quad \dots(1)$$

with $y(0) = y(1) = 0$.

Its solution is equivalent to extremizing the functional

$$I = \int_0^1 F(x, y, y') dx$$

where

$$F = y'^2 - xy^2 - 2xy.$$

$\therefore$ The required variational problem is

$$I = \int_0^1 (y'^2 - xy^2 - 2xy) dx. \quad \dots(2)$$

Let

$$y(x) = c_1 (x - x^2) \quad \dots(3)$$

be approximate solution satisfying given boundary condition.

$$\therefore y'(x) = c_1 (1 - 2x).$$

Putting value of y and y' in (2), we get

$$\begin{aligned} I &= \int_0^1 [c_1^2 (1 - 2x)^2 - xc_1^2 (x - x^2)^2 - 2xc_1 (x - x^2)] dx \\ &= \int_0^1 [c_1^2 (1 + 4x^2 - 4x - x^3 - x^5 + 2x^4) - 2c_1 (x^2 - x^3)] dx \\ &= c_1^2 \left[x + \frac{4}{3} x^3 - 2x^2 - \frac{x^4}{4} - \frac{x^6}{6} + \frac{2x^5}{5} \right]_0^1 - 2c_1 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 \\ &= c_1^2 \left(1 + \frac{4}{3} - 2 - \frac{1}{4} - \frac{1}{6} + \frac{2}{5} \right) - 2c_1 \left(\frac{1}{3} - \frac{1}{4} \right) \\ I &= c_1^2 \frac{19}{60} - c_1 \frac{1}{6}. \end{aligned}$$

$$\begin{aligned} \therefore \frac{dI}{dc_1} &= c_1 \frac{2 \times 19}{60} - \frac{1}{6} \\ &= \frac{38c_1}{60} - \frac{1}{6}. \end{aligned}$$

For existence of extremum

$$\frac{dI}{dc_1} = 0$$



$$\Rightarrow \frac{38c_1}{60} - \frac{1}{6} = 0$$

$$\begin{aligned} \Rightarrow c_1 &= \frac{60}{38 \times 6} = \frac{10}{38} \\ &= \frac{5}{19}. \end{aligned}$$

Putting value of c_1 in (3), we get

$$y(x) = \frac{5}{19} (x - x^2).$$

This is required approximate solution.

Problem 4 : Solve

$$xy'' + y' + y = x$$

subject to boundary condition $y(0) = 0$ and $y(1) = 1$.

Solution : The given differential equation is

$$xy'' + y' + y = x \quad \dots(1)$$

with $y(0) = 0, \quad y(1) = 1$

Its solution is equivalent to extremizing the function

$$I = \int_0^1 F(x, y, y') dx$$

where $F(x, y, y') = xy'^2 - y^2 + 2xy$.

$\therefore$ The required variational problem is

$$I = \int_0^1 (xy'^2 - y^2 + 2xy) dx. \quad \dots(2)$$

Let $y(x) = x + c (x - x^2) \quad \dots(3)$

be approximate solution which satisfying given boundary condition.

$\therefore y'(x) = 1 + c(1 - 2x)$

$$= 1 + c - 2cx.$$

Putting value of y and y' in (2), we get

$$\begin{aligned}
 I &= \int_0^1 [x(1+c-2cx)^2 - (x+cx-cx^2)^2 + 2x(x+cx-cx^2)] dx \\
 &= \int_0^1 [(1+c)^2 x + 4c^2 x^3 - 4c(1+c)x^2 - (1+c)^2 x^2 - c^2 x^4 \\
 &\quad + 2c(1+c)x^3 + 2c(1+c)x^2 - 2cx^3] dx \\
 &= \left[(1+c)^2 \frac{x^2}{2} + c^2 x^4 - 4c(1+c) \frac{x^3}{3} \right. \\
 &\quad \left. - (1+c)^2 \frac{x^3}{3} - c^2 \frac{x^5}{5} + 2c(1+c) \frac{x^3}{3} + 2c(1+c) \frac{x^3}{3} - \frac{cx^4}{4} \right]_0^1 \\
 I &= \frac{(1+c)^2}{2} \frac{1}{2} + c^2 - \frac{4}{3} c(1+c) - (1+c)^2 \frac{1}{3} - \frac{c^2}{5} + 2c(1+c) \frac{1}{3} \\
 &\quad + 2c(1+c) \frac{1}{3} - \frac{c}{2}
 \end{aligned}$$

Therefore for existence of extremal

$$\frac{dI}{dc} = 0$$

$$\Rightarrow c = \frac{5}{4}.$$

Putting value of c in (3), we get

$$y(x) = x + \frac{5}{4}(x - x^2)$$

$$\text{or } y(x) = \frac{1}{4}x(9 - 5x).$$

This is required approximate solution.

Note : For better approximation, we may choose

$$y(x) = x + (x - x^2)(c_1 + c_2 x)$$

and proceeding in the same way, we find

$$c_1 = \frac{85}{26}, c_2 = -\frac{35}{13}.$$

$$\begin{aligned} \therefore y(x) &= x + (x - x^2) \left(\frac{85}{26} - \frac{35}{13} x \right) \\ &= \frac{1}{26} (111 - 155x + 70x^2). \end{aligned}$$

Problem 5 : Solve $(1 - x^2) y'' - 2xy' + 2y = 0$

subject to boundary condition

$$y(0) = 0, y(1) = 1.$$

Solution : The given differential equation is

$$(1 - x^2) y'' - 2xy' + 2y = 0 \quad \dots(1)$$

with $y(0) = 0, y(1) = 0.$

Its solution is equivalent to extremizing the functional

$$I = \int_0^1 F(x, y, y') dx$$

where $F = (1 - x^2) y'^2 - 2y^2.$

Therefore, the required variational problem is

$$I = \int_0^1 [(1 - x^2) y'^2 - 2y^2] dx. \quad \dots(2)$$

Now let

$$y(x) = x + c(x - x^2) \quad \dots(3)$$

be approximate solution which satisfying the given boundary conditions.

$\therefore y'(x) = 1 + c(1 - 2x).$

Putting value of y and y' in (2), we get

$$I = \int_0^1 [(1 - x^2) \{1 + c - 2cx\}^2 - 2 \{x + cx - cx^2\}^2] dx.$$

Therefore, for existence of extremal $\frac{dI}{dc} = 0$ gives

$$c = 0.$$

Putting value of c in (3), we get

$$y(x) = x$$

This is required solution of given differential equation.

Note : $y(x) = x$ is exact solution of given differential equation.

Problem 6 : Find the approximate the smallest eigen value λ of

$$y'' + \lambda y = 0;$$

$$y(0) = 0, y(1) = 0$$

by Rayleigh-Ritz method.

Solution : The given differential equation is

$$y'' + \lambda y = 0 \quad \dots(1)$$

$$y(0) = 0, y(1) = 0.$$

Its solution is equivalent to extremizing the functional

$$I = \int_0^1 F(x, y, y') dx$$

where

$$F(x, y, y') = y'^2 - \lambda y^2.$$

$\therefore$ The required variational problem is

$$I = \int_0^1 (y'^2 - \lambda y^2). \quad \dots(2)$$

Let

$$y(x) = c(x - x^2) \quad \dots(3)$$

be approximate solution which satisfying given boundary condition

$$\therefore y'(x) = c(1 - 2x).$$

Putting value of y and y' in (2), we get

$$\begin{aligned}
 I &= \int_0^1 [c^2 (1-2x)^2 - \lambda c^2 (x-x^2)^2] dx \\
 &= \int_0^1 [c^2 (1+4x^2-4x) - \lambda c^2 (x^2+x^4-2x^3)] dx \\
 &= \int_0^1 c^2 [1+4x^2-4x - \lambda x^2 - \lambda x^4 + 2\lambda x^3] dx \\
 &= c^2 \left[x + \frac{4}{3} x^3 - 2x^2 - \frac{\lambda}{3} x^3 - \frac{\lambda}{5} x^5 + \lambda \frac{x^4}{2} \right]_0^1 \\
 &= c^2 \left[1 + \frac{4}{3} - 2 - \frac{\lambda}{3} - \frac{\lambda}{5} + \frac{\lambda}{2} \right] \\
 &= c^2 \left[\frac{1}{3} - \frac{1}{30} \lambda \right].
 \end{aligned}$$

$\therefore$ For existence of extremal

$$\frac{dI}{d\lambda} = 0$$

$$\Rightarrow 2c \left(\frac{1}{3} - \frac{1}{30} \lambda \right) = 0$$

$$\Rightarrow \frac{1}{3} - \frac{1}{30} \lambda = 0 \text{ for } c \neq 0$$

$$\Rightarrow \frac{\lambda}{30} = \frac{1}{3}$$

$$\Rightarrow \lambda = \frac{30}{3} = 10.$$

$\therefore \lambda = 10$ is least approximate eigen value of the given differential equation.

Problem 7 : Find the approximate solution of the problem of the minimum of the functional

$$I[y(x)] = \int_0^1 (y'^2 - y^2 - 2xy) dx$$

$y(0) = y(1) = 0$ and compare it with the exact solution.

Solution : Given that

$$I[y(x)] = \int_0^1 (y'^2 - y^2 - 2xy) dx \quad \dots(1)$$

$$y(0) = y(1) = 0.$$

Let $y(x) = c_1 (x - x^2)$... (2)

be approximate solution of (1) which satisfying given boundary conditions.

$$\therefore y'(x) = c(1 - 2x).$$

Putting value of y and y' in (1), we get

$$\begin{aligned} I &= \int_0^1 [c^2 (1 - 2x)^2 - c^2 (x - x^2)^2 - 2c (x^2 - x^3)] dx \\ &= \left[c^2 \left(x - 2x^2 + \frac{4}{3} x^3 - \frac{x^3}{3} - \frac{x^5}{5} + \frac{x^4}{2} \right) - 2c \left(\frac{x^3}{3} - \frac{x^4}{4} \right) \right]_0^1 \\ I &= \frac{3}{10} c^2 - \frac{1}{6} c. \end{aligned}$$

$$\begin{aligned} \therefore \frac{dI}{dc} &= \frac{3 \times 2}{10} c - \frac{1}{6} \\ &= \frac{3}{5} c - \frac{1}{6}. \end{aligned}$$

For existence of extremal

$$\begin{aligned} \frac{dI}{dc} &= 0 \\ \Rightarrow \frac{3}{5} c - \frac{1}{6} &= 0 \\ \Rightarrow c &= \frac{5}{18}. \end{aligned}$$

Putting value of c in (2), we get

$$y(x) = \frac{5}{18} (x - x^2).$$

This is required approximate solution.

Exact Solution

Here $F(x, y, y') = y'^2 - y^2 - 2xy$.

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow -2y - 2x - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow y + x + \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

$$\text{or } \frac{d^2 y}{dx^2} + y = -x$$

$$\text{or } (D^2 + 1) y = -x. \quad \dots(3)$$

Auxiliary equation is

$$m^2 + 1 = 0$$

$$\Rightarrow m = \pm i.$$

$$\therefore \text{C. F.} = c_1 \cos x + c_2 \sin x.$$

$$\text{Now, P. I.} = \frac{1}{D^2 + 1} (-x)$$

$$= -(1 + D^2)^{-1} x$$

$$= -[1 - D^2 + \dots] x$$

$$= -x.$$

Hence required solution is

$$y(x) = \text{C. F.} + \text{P.I.}$$

$$\text{i.e., } y(x) = c_1 \cos x + c_2 \sin x - x \quad \dots(4)$$

Now

$$y(0) = 0 \Rightarrow c_1 = 0$$

$$y(1) = 0 \Rightarrow c_1 \cos 1 + c_2 \sin 1 - 1 = 0$$

$$\Rightarrow 0 + c_2 \sin 1 - 1 = 0$$

$$\Rightarrow c_2 = \frac{1}{\sin 1}.$$

Putting value of c_1 and c_2 in (4), we get

$$y(x) = \frac{\sin x}{\sin 1} - x.$$

This is exact solution.

Now, compare the approximate solution with exact solution for some value of x :

x	0	0.2	0.4	0.6	0.8	1
Approximate solution	0	0.044	0.067	0.067	0.044	0
Exact solution	0	0.036	0.063	0.071	0.052	0

4.3 Galerkin's Method

Galerkin's method is another method to solve any type of boundary value problems either linear or non-linear.

In this method, the boundary conditions are taken homogeneous *i.e.*, of the form

$$a(x) \frac{d^2 y}{dx^2} + b(x) \frac{dy}{dx} + c(x) y = f(x)$$

$$y(x_1) = 0 \text{ and } y(x_2) = 0. \quad \dots(1)$$

If the boundary condition are not homogeneous, it can be made homogeneous by choosing an approximate transformation

$$u = y - y_0 - \frac{y_1 - y_0}{x_1 - x_0} (x - x_0)$$

with conditions that

$$u_r(x_1) = 0 \text{ and } u_r(x_2) = 0; \quad r = 1, 2, \dots, n.$$

Further let

$$y(x) = c_1 u_1(x) + c_2 u_2(x) + \dots + c_n u_n(x) \quad \dots(2)$$

be an approximate solution.

To find the constants c_i 's; $i = 1, 2, \dots, n$.

We use the residue function defined as

$$R(x, c_r) = \left[a(x) \frac{d^2}{dx^2} + b(x) \frac{d}{dx} + c(x) \right] \sum_{r=1}^n c_r u_r(x) - f(x) \quad \dots(3)$$

$$\text{when} \quad \int_{x_1}^{x_2} R(x, c_r) u_r(x) dx = 0; \quad r = 1, 2, \dots, n \quad \dots(4)$$

We can find the value of all constants c_i 's using (3) and then putting value of these constants in (1), we get required solution.

Problem Set

Problem 1 : Solve the boundary value problem

$$\frac{d^2 y}{dx^2} + y = e^x$$

$$y(0) = y(\pi / 2) = 0$$

by Galerkin's method.

Solution : The given differential equation is

$$\frac{d^2 y}{dx^2} + y = e^x$$

$$\text{with} \quad y(0) = y(\pi / 2) = 0. \quad \dots(1)$$

Here given boundary conditions are homogeneous.

Let approximate solution is

$$y(x) = c_1 u_1(x) = c_1 \sin 2x \quad \dots(2)$$

which satisfy given boundary condition.

Now we find constant c_1 .

For consider, corresponding residue function by

$$\begin{aligned} R(x, c_1) &= \left(\frac{d^2}{dx^2} + 1 \right) c_1 \sin 2x - e^x \\ &= -4c_1 \sin 2x + c_1 \sin 2x - e^x \end{aligned}$$

where
$$\int_0^{\pi/2} (-4c_1 \sin 2x + c_1 \sin 2x - e^x) \sin 2x \, dx = 0.$$

On simplification, we get

$$c_1 = -\frac{8}{15\pi} (e^{\pi/2} + 1).$$

Put in (2), we get

$$y(x) = -\frac{8}{15\pi} (e^{\pi/2} + 1) \sin 2x.$$

This is required approximation solution.

Problem 2 : Solve the boundary value problem

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = 2x^2$$

$$y(0) = 0, \quad y(1) = 1$$

by Galerkin's method.

Solution : The given differential equation is

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = 2x^2$$

$$y(0) = 0, \quad y(1) = 1. \quad \dots(1)$$

Here, the boundary conditions are not homogeneous.

Therefore, we choose transformation

$$u = y - y_0 - \frac{y_1 - y_0}{x_1 - x_0} (x - x_0)$$

$$\Rightarrow u = y - 0 - \frac{1 - 0}{1 - 0} (x - 0) \quad [\because x_0 = 0, y_0 = 0, x_1 = 1, y_1 = 1]$$

$$\Rightarrow u = y - x$$

$$\therefore u' = y' - 1 \text{ and } u'' = y''.$$

Putting all these value in (1), we get

$$x^2 \frac{d^2 u}{dx^2} + x \frac{du}{dx} - u = 2x$$

$$\text{with } u(0) = 0, u(1) = 0. \quad \dots(2)$$

Here the boundary conditions are homogeneous.

$$\text{Let } u(x) = c_1 u_1(x) = c_1 x(x - 1) \quad \dots(3)$$

be approximate solution.

Now, to find constant c_1 consider the residue function is

$$\begin{aligned} R(x, c_1) &= \left(x^2 \frac{d^2}{dx^2} + x \frac{d}{dx} - 1 \right) c_1 u_1(x) - 2x^2 \\ &= \left(x^2 \frac{d^2}{dx^2} + x \frac{d}{dx} - 1 \right) c_1 (x^2 - x) - 2x^2 \\ &= x^2 \cdot 2c_1 + xc_1 (2x - 1) - c_1 (x^2 - x) - 2x^2 \end{aligned}$$

$$\text{where } \int_0^1 [x^2 \cdot 2c_1 + xc_1 (2x - 1) - c_1 (x^2 - x) - 2x^2] x(x - 1) dx = 0$$

$$\Rightarrow \int_0^1 (3c_1 x - 2x) x^2 (x - 1) dx = 0$$

$$\Rightarrow (3c_1 - 2) \int_0^1 x^3 (x - 1) dx = 0$$

$$\Rightarrow (3c_1 - 2) \left[\frac{x^5}{5} - \frac{x^4}{4} \right]_0^1 = 0$$

$$\Rightarrow (3c_1 - 2) \left[\frac{1}{5} - \frac{1}{4} \right] = 0$$

$$\Rightarrow (3c_1 - 2) \frac{-1}{20} = 0$$

$$\Rightarrow 3c_1 - 2 = 0$$

$$\Rightarrow c_1 = \frac{2}{3}.$$

Put in (3), we get

$$u(x) = \frac{2}{3} x (x - 1)$$

since $u = y - x$

$$\Rightarrow y = u + x$$

$$\therefore y = \frac{2}{3} x (x - 1) + x.$$

This is required approximate solution.

4.4 Partial Differential Equation (By Rayleigh-Ritz Method)

Consider boundary value problem

$$u_{xx} + u_{yy} + qu = r$$

in the region R bounded by closed curve C .

Let $u(x)$ be prescribed on C .

Then, the corresponding variation problem is to extremize the functional is

$$\iint_R (u_x^2 + u_y^2 - qu^2 + 2ru) dx dy$$

for which we take the trivial function such that it satisfy the given boundary conditions.

Problem Set

Problem 1 : Solve the Poisson's equation

$$u_{xx} + u_{yy} = -1$$

in a square defined by $|x| \leq 1, |y| \leq 1$ and $u = 0$ when $x = \pm 1, y = \pm 1$.

Solution : Given that

$$u_{xx} + u_{yy} = -1. \quad \dots(1)$$

Compare with

$$u_{xx} + u_{yy} + qu = r,$$

we get

$$q = 0, \quad r = -1.$$

Hence, the corresponding variational problem is to extremize the functional is

$$I = \iint_R (u_x^2 + u_y^2 - qu^2 + 2ru) \, dx \, dy$$

i.e.,

$$I = \iint_R (u_x^2 + u_y^2 - 2u) \, dx \, dy \quad \dots(2)$$

where R is square defined by $|x| \leq 1, |y| \leq 1$.

Let

$$u(x, y) = c(1 - x^2)(1 - y^2) \quad \dots(3)$$

be trivial function which satisfies the given boundary conditions.

$\therefore$

$$u_x = -2cx(1 - y^2); \quad u_y = -2cy(1 - x^2).$$

Putting value of u, u_x and u_y in (2), we get

$$\begin{aligned} I &= \int_{-1}^1 \int_{-1}^1 [4c^2 x^2 (1 - y^2) + 4c^2 y^2 (1 - x^2)^2 \\ &\quad - 2c(1 - x^2)(1 - y^2)] \, dx \, dy \\ &= \frac{32}{45} (8c^2 - 5c) \quad \text{[On simplification]} \end{aligned}$$

Now, for existence of extremal

$$\frac{dI}{dc} = 0$$

$$\Rightarrow \frac{32}{45} (16c - 5) = 0$$

$$\Rightarrow 16c - 5 = 0$$

$$\Rightarrow 16c = 5$$

$$\Rightarrow c = \frac{5}{16}.$$

Putting value of c in (3), we get

$$u(x, y) = \frac{5}{16} (1 - x^2) (1 - y^2).$$

This is required approximate solution.

Problem 2 : Solve the Poisson's equation in a circle

$$u_{xx} + u_{yy} = -1, \quad x^2 + y^2 \leq 1$$

when $u = 0$ on $x^2 + y^2 = 1$.

Solution : The corresponding variational problem is to extremize the functional is

$$I = \iint_R (u_x^2 + u_y^2 - 2u) \, dx \, dy \quad \dots(1)$$

where R is the region $x^2 + y^2 = 1$ i.e., interior and boundary of circle $x^2 + y^2 = 1$.

$$\text{Let } u(x, y) = c(1 - x^2 - y^2) \quad \dots(2)$$

be approximate solution which satisfy the given boundary condition ($u = 0$ on circle $x^2 + y^2 = 1$).

$$\therefore u_x = -2cx, \quad u_y = -2cy.$$



Putting value of u, u_x, u_y in (1), we get

$$\begin{aligned} I &= \iint_R [4c^2 x^2 + 4c^2 y^2 - 2c(1 - x^2 - y^2)] dx dy \\ &= \iint_R [4c^2 (x^2 + y^2) - 2c(1 - x^2 - y^2)] dx dy. \end{aligned}$$

Put $x = r \cos \theta, y = r \sin \theta$

$$\Rightarrow r^2 = x^2 + y^2 \text{ and } dx dy = r d\theta dr$$

$$\begin{aligned} \therefore I &= \int_{r=0}^{r=1} \int_{\theta=0}^{\theta=2\pi} [4c^2 r^2 - 2c(1 - r^2)] r d\theta dr \\ &= \int_0^1 [4c^2 r^2 - 2c(1 - r^2)] r dr \cdot 2\pi \\ &= 2\pi \int_0^1 [4c^2 r^3 - 2c(r - r^3)] dr \\ &= 2\pi \left[4c^2 \frac{r^4}{4} - 2c \left(\frac{r^2}{2} - \frac{r^4}{4} \right) \right]_0^1 \\ &= 2\pi \left[c^2 - 2c \left(\frac{1}{2} - \frac{1}{4} \right) \right] \\ I &= 2\pi \left[c^2 - \frac{1}{2} c \right]. \end{aligned}$$

$$\therefore \frac{dI}{dc} = 2\pi \left(2c - \frac{1}{2} \right).$$

Therefore for existence of extremal

$$\frac{dI}{dc} = 0$$

$$\Rightarrow 2\pi \left(2c - \frac{1}{2} \right) = 0$$

$$\Rightarrow 2c - \frac{1}{2} = 0$$

$$\Rightarrow c = \frac{1}{4}.$$

Putting value of c in (2), we get

$$u(x, y) = \frac{1}{4} (1 - x^2 - y^2).$$

This is required approximate solution.

Problem 3 : Solve the Poisson's equation

$$u_{xx} + u_{yy} = -1$$

in an equilateral triangle bounded by $x = \pm a$ and $y = \pm \frac{x}{\sqrt{3}}$ given that $u = 0$ on the boundary of the triangle.

Solution : The corresponding variational problem is to extremize the functional is

$$I = \iint_R (u_x^2 + u_y^2 - 2u) dx dy \quad \dots(1)$$

where R is the region bounded by $x = a$ and $y = \pm \frac{x}{\sqrt{3}}$.

$$\text{Let } u(x, y) = c (x^2 - 3y^2) (a - x) \quad \dots(2)$$

be approximate solution which satisfies given boundary conditions.

$$\therefore u_x = c (2ax - 3x^2 + 3y^2)$$

$$\text{and } u_y = -6cy (a - x).$$

Putting value of u, u_x, u_y in (1), we get

$$I = \iint_R [c^2 (2ax - 3x^2 + 3y^2)^2 + 36c^2 y^2 (a - x)^2 - 2c (x^2 - 3y^2) (a - x)] dx dy$$

$$= 2c^2 \int_{x=0}^{x=a} \int_{y=0}^{y=\frac{x}{\sqrt{3}}} [(2ax - 3x^2 + 3y^2)^2 + 36c^2 y^2 (a - x)^2] dx dy$$

$$- 4c \int_{x=0}^{x=a} \int_{y=0}^{y=\frac{x}{\sqrt{3}}} (x^2 - 3y^2) (a - x) dx dy$$

$$I = \frac{2a^5}{15\sqrt{3}} (2c^2 a - c) \quad \text{(On simplification)}$$

Therefore for existence of extremal

$$\frac{dI}{dc} = 0$$

$$\Rightarrow \frac{2a^5}{15\sqrt{3}} (4ca - 1) = 0$$

$$\Rightarrow 4ca - 1 = 0$$

$$\Rightarrow c = \frac{1}{4a}.$$

Putting value of c in (2), we get

$$u(x, y) = \frac{1}{4a} (x^2 - 3y^2) (a - x).$$

This is required approximate solution.

Problem 4 : Solve the Laplace equation

$$u_{xx} + u_{yy} = 0$$

in a square defined by $|x| \leq 1, |y| \leq 1$, when $u = 0$ on $|x| = 1$ and $u = 1 - x^2$ on $|y| = 1$.

Solution : The given partial differential equation is

$$u_{xx} + u_{yy} = 0$$

Compare with

$$u_{xx} + u_{yy} + qu = r$$

we get

$$q = 0, r = 0.$$

Hence, the corresponding variational problem is to extremize the functional is

$$I = \iint_R (u_x^2 + u_y^2 - qu^2 + 2ru) dx dy$$

$$i.e., \quad I = \iint_R (u_x^2 + u_y^2) dx dy \quad \dots(1)$$

$$[\because q = 0, r = 0]$$

Let $u(x, y) = (1 - x^2) [1 + c (1 - y^2)]$... (2)

be trivial function, which satisfy the given boundary condition.

$$\therefore u_x = -2x [1 + c (1 - y^2)]$$

$$u_y = -2yc (1 - x^2).$$

Therefore (1) becomes

$$I = \int_{-1}^1 \int_{-1}^1 [4x^2 \{1 + 2c (1 - y^2) + c^2 (1 - y^2)^2\} + 4y^2 c^2 (1 - 2x^2 + x^4)] dx dy.$$

Now $\frac{dI}{dc} = 0$

$$\Rightarrow \int_{-1}^1 \int_{-1}^1 [8x^2 (1 - y^2) + 8x^2 c (1 - y^2)^2 + 8y^2 c (1 - 2x^2 + x^4)] dx dy = 0$$

Evaluating the integral, we get

$$c = -\frac{5}{8}.$$

Putting value of c in (1), we get

$$u(x, y) = (1 - x^2) \left[1 - \frac{5}{8} (1 - y^2) \right].$$

This is required approximate solution.

Problem 5 : Find the estimate of the least eigen value of

$$u_{xx} + u_{yy} + \lambda u = 0$$

in the region R bounded by the circle $x^2 + y^2 = 1$.

Given that $u = 0$ on the boundary.

Solution : Here, the corresponding variational problem is to extremize the functional is

$$I = \iint_R (u_x^2 + u_y^2 - \lambda u^2) dx dy \quad \dots (1)$$

where R is region bounded by $x^2 + y^2 \leq 1$.

Let $u(x, y) = c (1 - x^2 - y^2)$

be trivial solution which satisfies the boundary condition *i.e.*, $u = 0$ at $x^2 + y^2 = 1$.

$$\therefore u_x = -2cx, \quad u_y = -2cy.$$

Putting value of u, u_x, u_y in (1), we get

$$I = \iint_R [4c^2 (x^2 + y^2) - \lambda c^2 (1 - x^2 - y^2)^2] dx dy$$

$$\therefore \frac{dI}{dc} = 0$$

$$\Rightarrow \iint_R [8c (x^2 + y^2) - 2\lambda c (1 - x^2 - y^2)^2] dx dy = 0$$

$$\Rightarrow \lambda = \frac{4 \iint_R (x^2 + y^2) dx dy}{\iint_R (1 - x^2 - y^2)^2 dx dy}$$

Put $x = r \cos \theta$, $y = r \sin \theta$.

$$\therefore dx dy = r d\theta dr.$$

$$\therefore \lambda = \frac{4 \int_{r=0}^1 \int_{\theta=0}^{2\pi} r^2 r d\theta dr}{\int_{r=0}^1 \int_{\theta=0}^{2\pi} (1 - r^2)^2 d\theta dr}$$

$$\therefore \lambda = \frac{2\pi}{2\pi \left(\frac{1}{6}\right)}$$

$$= 6.$$

Hence $\lambda = 6$ is required least eigen value.

Problem 6 : Find the estimate of the least eigen value of

$$u_{xx} + u_{yy} + \lambda u = 0$$

in the region R bounded by $|x| \leq 1$, $|y| \leq 1$ and $u = 0$ at $x = \pm 1$ and at $y = \pm 1$.

Solution : Here, the variational problem is to extremize the functional is

$$I = \iint_R (u_x^2 + u_y^2 - \lambda u^2) dx dy \quad \dots(1)$$

where R is region bounded by $|x| \leq 1, |y| \leq 1$.

Now let

$$u(x, y) = c(1 - x^2)(1 - y^2)$$

be trivial solution which satisfies the given boundary condition

$$i.e., \quad u = 0 \text{ at } x = \pm 1, y = \pm 1.$$

$$\therefore \quad u_x = -2cx(1 - y^2)$$

$$u_y = -2cy(1 - x^2).$$

Putting value of u, u_x, u_y in (1), we get

$$\begin{aligned} \therefore \quad I &= \iint_R [4c^2 x^2 (1 - y^2)^2 + 4c^2 y^2 (1 - x^2)^2 \\ &\quad - \lambda c^2 (1 - x^2)^2 (1 - y^2)^2] dx dy \\ I &= \int_{-1}^1 \int_{-1}^1 [4c^2 x^2 (1 - y^2)^2 + 4c^2 y^2 (1 - x^2)^2 \\ &\quad - \lambda c^2 (1 - x^2)^2 (1 - y^2)^2] dx dy \end{aligned}$$

$$\therefore \quad \frac{dI}{dc} = 0$$

$$\Rightarrow \int_{-1}^1 \int_{-1}^1 [8cx^2(1 - y^2)^2 + 8cy^2(1 - x^2)^2 - 2\lambda c(1 - x^2)^2(1 - y^2)^2] dx dy = 0$$

On simplification,

$$\Rightarrow \left[\frac{256}{45} - \frac{256\lambda}{225} \right] = 0$$

$$\Rightarrow \quad \lambda = 5.$$

This is required least eigen value.

4.5 Kantorovich Method

In order to solve partial differential equation, Kantorovich method is another more efficient method in comparison of Rayleigh-Ritz method.

In this method, a trivial solution

$$\begin{aligned} u(x, y) &= f_1(x) u_1(x, y) + f_2(x) u_2(x, y) + \dots + f_n u_n(x, y) \\ &= \sum_{i=1}^n f_i(x) u_i(x, y) \end{aligned}$$

is taken.

Here, $u_i(x, y)$, $\forall_i$ satisfying the given boundary condition and $f_i(x)$ are unknown function of x .

Problem Set

Problem 1 : By Kantorovich Method : Solve the Poisson equation

$$u_{xx} + u_{yy} = -1$$

in a square defined by $|x| \leq 1$, $|y| \leq 1$, where $u = 0$ at $x = \pm 1$ and $y = \pm 1$.

Solution : Here, the corresponding variation problem is to extremize the functional is

$$I = \iint_R (u_x^2 + u_y^2 - 2u) dx dy \quad \dots(1)$$

where R is square defined by $|x| \leq 1$, $|y| \leq 1$ i.e., $-1 \leq x \leq 1$, $-1 \leq y \leq 1$.

$$\text{Let } u(x, y) = (1 - y^2) f(x) \quad \dots(2)$$

be trivial solution

where $f(x)$ is unknown function to be determined such that $f(\pm 1) = 0$.

$$\therefore u_x = (1 - y^2) f'(x), \quad u_y = -2y f(x).$$

Putting value of u, u_x, u_y in (1), we get

$$I = \int_{-1}^1 \int_{-1}^1 [(1-y^2)^2 \{f'(x)\}^2 + 4y^2 \{f(x)\}^2 - 2(1-y^2)f(x)] dx dy.$$

On simplification, we get

$$I = \frac{8}{3} \int_{-1}^1 \left(\frac{2}{5} f'^2 + f^2 - f \right) dx.$$

Consider
$$F = \frac{2}{5} f'^2 + f^2 - f.$$

By Euler's equation

$$\frac{\partial F}{\partial f} - \frac{d}{dx} \left(\frac{\partial F}{\partial f'} \right) = 0$$

$$\Rightarrow 2f - 1 - \frac{d}{dx} \left(\frac{2}{5} \cdot 2f' \right) = 0$$

$$\Rightarrow 2f - 1 - \frac{4}{5} f'' = 0$$

or
$$f'' - \frac{5}{2} f + \frac{5}{4} = 0$$

i.e.,
$$\frac{d^2 f}{dx^2} - \frac{5}{2} f + \frac{5}{4} = 0$$

$$\Rightarrow \frac{d^2 f}{dx^2} - \frac{5}{2} f = -\frac{5}{4}. \quad \dots(3)$$

Auxiliary equation is

$$m^2 - \frac{5}{2} = 0$$

$$\Rightarrow m = \pm \sqrt{\frac{5}{2}}.$$

$$\therefore \text{C. F.} = A \cosh \sqrt{\frac{5}{2}} x + B \sinh \sqrt{\frac{5}{2}} x$$

$$\begin{aligned} \text{P. I.} &= \frac{1}{D^2 - \frac{5}{2}} \left(-\frac{5}{4} \right) = -\frac{5}{4} \frac{1}{D^2 - \frac{5}{2}} e^{0x} \\ &= -\frac{5}{4} \times \frac{1}{-5/2} \end{aligned}$$

$$\text{P. I.} = \frac{1}{2}.$$

Hence solution of (3) is

$$f(x) = \frac{1}{2} + A \cosh \left(\sqrt{\frac{5}{2}} x \right) + B \sinh \left(\sqrt{\frac{5}{2}} x \right). \quad \dots(4)$$

Now using boundary condition

$$f(\pm 1) = 0$$

$$\Rightarrow f(x) = \frac{1}{2} \left[\frac{1 - \cosh h \left(\sqrt{\frac{5}{2}} x \right)}{\cosh h \sqrt{\frac{5}{2}}} \right].$$

Therefore, from (2), required solution is

$$u(x, y) = \frac{1}{2} (1 - y^2) \left[1 - \frac{\cosh h \left(\sqrt{\frac{5}{2}} x \right)}{\cosh h \sqrt{\frac{5}{2}}} \right].$$

From this we have

$$u(0, 0) = 0.303.$$

Using Rayleigh-Ritz method

$$u(0, 0) = 0.313.$$

But the exact value is

$$u(0, 0) = 0.295.$$

Hence, these value shows that Kantorovich method is superior than Rayleigh-Ritz method.

Problem 2 : Solve $u_{xx} + u_{yy} = 0$

in a square defined by

$$|x| \leq 1, |y| \leq 1$$

where $u(\pm 1, y) = 1 - y^2$ **and** $u(x, \pm 1) = 0$.

Solution : Here, the corresponding variational problem is to extremize the functional

$$I = \iint_R (u_x^2 + u_y^2) dx dy \quad \dots(1)$$

where R is square defined by $|x| \leq 1, |y| \leq 1$.

$$\text{Let} \quad u(x, y) = (1 - y^2) f(x) \quad \dots(2)$$

where $f(x)$ is unknown function to be determined such that $f(\pm 1) = 0$.

$$\therefore \quad u_x = (1 - y^2) f'(x); \quad u_y = -2y f(x).$$

Putting value of u, u_x, u_y in (2), we get

$$I = \int_{-1}^1 \int_{-1}^1 [(1 - y^2) \{f'(x)\}^2 + 4y^2 \{f(x)\}^2] dx dy.$$

On simplification, we get

$$I = \frac{8}{3} \int_{-1}^1 \left(\frac{2}{5} f'^2 + f^2 \right) dx$$

and the corresponding Euler's equation is

$$-\frac{4}{5} f'' + 2f = 0$$

or
$$f'' - \frac{5}{2} f = 0. \quad \dots(3)$$

Auxiliary equation is

$$m^2 - \frac{5}{2} = 0$$

$$\Rightarrow m^2 = \frac{5}{2}$$

$$\Rightarrow m = \pm \sqrt{5/2}$$

Hence solution of (2) is

$$f(x) = A \cosh \left(\sqrt{\frac{5}{2}} \right) x + B \sinh \left(\sqrt{\frac{5}{2}} \right) x.$$

Using boundary condition

$$f(\pm 1) = 1,$$

we get
$$f(x) = \frac{\cosh \left(\sqrt{\frac{5}{2}} \right) x}{\cosh \left(\sqrt{\frac{5}{2}} \right)}.$$

Therefore (2) becomes

$$u(x, y) = (1 - y^2) \frac{\cosh (\sqrt{5/2}) x}{\cosh (\sqrt{5/2})}.$$

This is required solution.

Exercise 4.1

1. Solve $y'' + y = -x$

subject to boundary conditions $y(0) = y(1) = 0$, by Rayleigh-Ritz method.

2. Solve $y'' + (1 + x^2) y + 1 = 0$

with boundary condition $y(\pm 1) = 0$ by Rayleigh-Ritz method.

3. Find the approximate solution by Rayleigh-Ritz method of differential equation $y'' + x^2 y = x$ with $y(0) = y(1) = 0$.

4. Find approximately the smallest eigen value λ of

$$y'' + \lambda y = 0; y(1) = y(-1) = 0.$$

5. Find approximately the smallest eigen value λ of

$$(1 + x) y'' + y' + \lambda y = 0$$

$$y(0) = y(1) = 0.$$

6. Solve

$$y'' + y = -\frac{x^3}{5}; y(0) = 0, y(1) = 1$$

by Rayleigh-Ritz method.

7. Solve the boundary value problem

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = 2x^2$$

$$y(0) = 0, y(1) = 1$$

by Galerkin's method.

8. Solve the Poisson equation in an ellipse

$$u_{xx} + u_{yy} = -1, \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$$

when $u = 0$ on $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by Rayleigh-Ritz method.

9. Solve the Poisson equation

$$u_{xx} + u_{yy} = -1$$

in an equilateral triangle bounded by

$$x = a \quad \text{and} \quad y = \pm \frac{x}{\sqrt{3}}$$

given that $u = 0$ on the boundary of triangle, by Kantorovich method.

Answers

- | | |
|----|--|
| 1. | $y(x) = \frac{5}{18} (x - x^2)$ |
| 2. | $y(x) = \frac{35}{38} (1 - x^2)$ |
| 3. | $y(x) = \frac{7}{136} x (1 - x)$ |
| 4. | $\lambda = 2 \cdot 5$ |
| 5. | $\lambda = 15$ |
| 6. | $y(x) = \frac{(13x - 3x^2)}{10}$ |
| 7. | $y(x) = x + \frac{2}{3} x (x - 1)$ |
| 8. | $u(x, y) = \frac{1}{2} \frac{a^2 b^2}{a^2 + b^2} \left(1 - \frac{x^2}{a^2} - \frac{y^2}{b^2} \right)$ |
| 9. | $u(x, y) = \frac{(a - x)}{4a} (x^2 - 3y^2).$ |

HIGHLIGHTS

- * The German mathematician **W. Ritz** gave variational approach to solve boundary value problem for ordinary and partial differential equation in 1908.

This is also known as **Rayleigh-Ritz** method.

- * The variational problem corresponding to given ordinary differential equation is

$$I = \int_{x_0}^{x_1} F(x, y, y') dx$$

where, we construct $F(x, y, y')$ in say that the corresponding Euler's equation becomes original differential equation.

- * Rayleigh-Ritz method gives problem approximate solution of boundary value problems.
- * The variational problem of partial differential equation

$$u_{xx} + u_{yy} + qu = r$$

is
$$\iint_R (u_x^2 + u_y^2 - qu^2 + 2ru) dx dy$$

where R is region bounded by curve.

- * The Galerikin's method can be used to solve any type of boundary value problem of ordinary and partial differential equation, either linear or non-linear.
- * Kantorovich method is more efficient method to solve boundary value problem of partial differential equation.

Objective Type Questions

(For NET, GATE and other Competitive Examinations)

Multiple Choice Questions

1. The variational problem of extremizing the functional

$$I[y(x)] = \int_0^{2\pi} \left[\left(\frac{dy}{dx} \right)^2 - y^2 \right] dx, \quad y(0) = 1$$

$y(2\pi) = 1$ has :

[C.S.I.R.-Net 2011 (June)]

- (a) A unique solution
 - (b) Exactly two solution
 - (c) An infinite number of solutions
 - (d) No solution.
2. In the Ritz method, seeking an extremum of the functional

$$I[y] = \int_{x_0}^{x_1} F\left(x, y, \frac{dy}{dx}\right) dx$$

$$y(x_0) = a, \quad y(x_1) = b.$$

[C.S.I.R.-Net 2011 (June)]

The co-ordinate function/or the admissible function $\phi_i(x)$; $i = 1, 2, \dots$ defined on $[x_0, x_1]$ must be :

- (a) Linearly independent
- (b) Continuous
- (c) Smooth
- (d) Linearly independent, smooth and the functional be considered not along admissible curves $y = y(x)$ but only along possible linear combinations of admissible functions.

3. Consider the functional

$$I[y] = \int_{1/2}^{x_1} \sqrt{1 + y'^2} dx$$

subject to the condition that the left end of the extremal is fixed at the point $\left(\frac{1}{2}, \frac{1}{4}\right)$ and the right end (x_1, y_1) be movable along the straight line $y = x - 5$. Let c_1 and c_2 be arbitrary constants, then :

[C.S.I.R. NET-2011 (Dec.)]

- (a) The general solution of the Euler's equation satisfies $2c_1 + 4c_2 = 1$
 (b) The transversality condition that holds at R the right end yields

$$\sqrt{1 + c_1^2} + (1 - c_1) \frac{c_1}{\sqrt{1 + c_1^2}} = 0$$

- (c) The relation $c_1 x_1 + c_2 = x_1 - 5$ is true.
 (d) The extremal is $y = -x + \frac{3}{4}$.

4. The variational problem of extremizing the functional

$$I[y(x)] = \int_1^3 y(3x - y) dx$$

$$y(3) = 4\frac{1}{2}, \quad y(1) = 1$$

has :

[C.S.I.R.-NET 2012 (June)]

- (a) A unique solution (b) Exactly two solutions
 (c) An infinite number of solution (d) No solution.
5. An approximate solution $z = z_0(x, y)$ to the problem of extremizing the functional

$$I\{z(x, y)\} = \iint_D \left[\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 - 2z \right] dx dy$$

where D is the square $-1 \leq x \leq 1, -1 \leq y \leq 1$ and $z = 0$ on the boundary of the square, is of the form :

[C.S.I.R. NET-2012 (June)]

- (a) $z_0 = \sum_{i=1}^n \alpha_i \phi_i(x, y)$ where α_i is constant and functions ϕ_i is linearly independent in D .
- (b) $z_0 = \alpha_1 \phi_1(x, y) + \alpha_2 \phi_2(x, y)$ where α_1 and α_2 are constants.
- (c) $z_0 = \alpha \phi(x, y)$ where α is constant and ϕ is continuous in D .
- (d) $z_0 = \frac{(x^2 - 1)(y^2 - 1)}{16}$.

6. Consider the functional

$$J(y) = \int_a^b F(x, y, y') dx$$

where $F(x, y, y') = y + y'$ for admissible function y . Then J has :

[C.S.I.R.-NET 2013 (Dec.)]

- (a) No extremals
- (b) Several extremal
- (c) $y(x) = e^{-x}$ as an extremal
- (d) $y(x) = \text{constat}$ as an extremal.

7. An extremal of the functional

$$J(y) = \int_a^b \sqrt{1 + |y'(x)|^2} dx. \quad [\text{C.S.I.R. NET-2013 (Dec.)}]$$

- (a) Is the straight line connecting $(a, y(a))$ and $(b, y(b))$
- (b) Is a solution to the differential equation $y' = c\sqrt{1 + y'^2}$ for some constant
- (c) Is a solution to $y'' = 0$
- (d) Does not exist.

8. The extremum of the functional

$$I = \int_0^1 \left[\left(\frac{dy}{dx} \right)^2 + 12xy \right] dx$$

satisfying the conditions $y(0) = 0$ and $y(1) = 1$ is attained on the curve :

[GATE-2005]

- (a) $y = \sin^2 \frac{\pi x}{2}$
- (b) $y = \sin \frac{\pi x}{2}$
- (c) $y = x^3$
- (d) $y = \frac{1}{2} \left[x^3 + \sin \frac{\pi x}{2} \right]$.

9. The extremal for the functional

$$I = \int_0^{\pi/2} (xy + y^2 - 2y^2 y') dx$$

with $y(0) = 0$, $y(\pi/2) = 1$ is :

- (a) $y = x$ (b) $y = \frac{x}{2}$
 (c) $y = -\frac{x}{2}$ (d) Problem is meaningless.

10. The functional

$$V[y(x)] = \int_0^2 [y'^2 + 6xy + x^3] dx$$

$y(0) = 0$, $y(2) = 2$ can be extremized on the curve :

[GATE-2003]

- (a) $y = x$ (b) $2y = x^3$
 (c) $y = x^3 - 6x$ (d) $2y = x^3 - 2x$.

11. An extremal of the functional

$$I[y(x)] = \int_a^b F\left(x, y, \frac{dy}{dx}\right) dx$$

$$y(a) = y_1, \quad y(b) = y_2$$

satisfies Euler's equation which in general :

[GATE-2002]

- (a) Is a second order linear differential equation.
 (b) Is a non-linear ordinary differential equation of order greater than two.
 (c) Admits a unique solution satisfying the conditions

$$y(a) = y_1, \quad y(b) = y_2.$$

- (d) May not admit a solution satisfy the conditions

$$y(a) = y_1, \quad y(b) = y_2.$$

12. Extremal $y = y(x)$ for the functional problem

$$I[y(x)] = \int_0^1 (y + y')^2 dx$$

satisfies the differential equation :

[GATE-2003]

- (a) $y'' + y = 0$ (b) $y'' - y = 0$
 (c) $y'' + y' = 0$ (d) $y' + y = 0$.

13. Extremal for the variational problem

$$V[y(x)] = \int_1^2 (y^2 + x^2 y'^2) dx$$

satisfies the differential equation :

[GATE-2004]

- (a) $xy'' - 2xy' + y = 0$ (b) $2xy' - y = 0$
 (c) $x^2 y'' + 2xy' - y = 0$ (d) $x^2 y'' - y = 0$.

14. Which of the following is linear functional ?

- (a) $I[y(x)] = \int_a^b (y' + 2y^2) dx$
 (b) $I[y(x)] = \int_a^b (y'^2 + 2y^2) dx$
 (c) $I[y(x)] = \int_a^b (y'^2 + 2y) dx$
 (d) $I[y(x)] = \int_a^b (y' + 2y) dx$.

15. A necessary conditions for functional

$$I[y(x)] = \int_a^b f(x, y, y') dx$$

to be an extremum is :

- (a) $\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$ (b) $\frac{\partial f}{\partial y} + \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$
 (c) $\frac{\partial f}{\partial y'} - \frac{d}{dx} \left(\frac{\partial f}{\partial y} \right) = 0$ (d) $\frac{\partial f}{\partial y'} + \frac{d}{dx} \left(\frac{\partial f}{\partial y} \right) = 0$.

16. The distance between the curves $y_1(x) = x$ and $y_2(x) = x^2$ on interval $[0, 1]$ is :

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{3}{4}$ (d) 1.

17. Extremal for the variational problem

$$I[y(x)] = \int_0^{\log^2} (e^{-x} y'^2 - e^x y^2) dx$$

satisfies the differential equation

- (a) $y'' + y' + e^{2x} y = 0$ (b) $y'' - y' + e^{2x} y = 0$
(c) $y'' + y' + e^x y = 0$ (d) $y'' + y' + e^{-x} y = 0$.

18. Extremal for the variational problem

$$I[y(x)] = \int_{x_0}^{x_1} y'(1 + x^2 y') dx$$

is solution of differential equation :

- (a) $y'' + 2y = 0$ (b) $xy'' + y' = 0$
(c) $xy'' + 2y' = 0$ (d) $y'' + 2xy' = 0$.

19. The functional

$$I[y(x)] = \int_a^b (y'^2 + 12xy) dx$$

can be extremized on the curve :

- (a) $y(x) = x$ (b) $y(x) = x^3 + c_1 x + c_2$
(c) $y(x) = x^2 + x$ (d) $y(x) = x^3 - 1$.

20. Extremal of the functional

$$I[y(x)] = \int_1^2 \frac{x^3}{y'^2} dx$$

$y(1) = 0$ and $y(2) = 3$ is :

- (a) $y = x^2 - 1$ (b) $y = x^3 - 1$
(c) $y = x - 1$ (d) $y = x^3 - x$.

21. Extremal of the functional

$$I[y(x)] = \int_0^1 (y'^2 + 4y^2) dx$$

$$y(0) = e^2, y(1) = 1$$

lies on the curve :

(a) $y = e^{2+2x}$

(b) $y = e^{2-2x}$

(c) $y = e^{2x}$

(d) None of these.

22. The functional

$$J[y(x)] = \int_0^2 [y'^2 + 6xy + x^3] dx$$

$$y(0) = 0, y(2) = 2$$

can be extremized on the curve :

(a) $y = x$

(b) $2y = x^3$

(c) $y = x^3 - 6x$

(d) $2y = x^3 - 2x$.

23. Extremals of the functional

$$\int_0^{\pi/2} (y''^2 - y^2 + x^2) dx :$$

(a) One parameter family of curves

(b) Two parameter family of curves

(c) Three parameter family of curves

(d) Four parameter family of curves.

24. Extremal of the functional

$$I[y(x)] = \int_{x_0}^{x_1} (2xy + y''^2) dx$$

lies on the :



- (a) Five parameter family of curves
- (b) Six parameter family of curves
- (c) Seven parameter family of curves
- (d) None of these.

25. Geodics on a plane are :

- (a) Parabola
- (b) Straight lines
- (c) Ellipse
- (d) Cycloid.

26. The possible value of α for which the variational problem

$$I[y(x)] = \int_0^1 (3y^2 + 2x^3 y') dx$$

$y(\alpha) = 1$ has extremal are :

- (a) $-1, 0$
- (b) $0, 1$
- (c) $-1, 1$
- (d) $-1, 0, 1$

27. The number of extremals of the functional

$$\int_0^\pi (2yz - 2y^2 + y'^2 - z'^2) dx$$

satisfying the conditions $y(0) = 0, y(\pi) = 1, z(0) = 0, z(\pi) = -1$ is :

- (a) 1
- (b) 2
- (c) 0
- (d) Infinite.

28. The extremal for the functional

$$I[y(x)] = \int_{x_0}^{x_1} (xy' + y'^2) dx$$

are given by the following family of curves :

[GATE-2004]

- (a) $y = c_1 + c_2 x + \frac{x^2}{4}$
- (b) $y = 1 + c_1 x + c_2 \frac{x^2}{4}$
- (c) $y = c_1 + x + c_2 \frac{x^2}{4}$
- (d) $y = c_1 + c_2 x - \frac{x^2}{4}$

29. The extremum for the variational problem

$$I[y(x)] = \int_0^{\pi/8} (y'^2 + 2yy' - 16y^2) dx$$

$$y(0) = 0, \quad y(\pi/8) = 1$$

occurs for the curve :

- (a) $y = \sin 4x$ (b) $y = \sqrt{2} \sin 2x$
 (c) $y = 1 - \cos 4x$ (d) $y = \frac{1 - \cos 8x}{2}$.

30. The extremum value of the variational problem

$$\int_0^{\pi/2} (y'^2 - y^2) dx;$$

$$y(0) = 0, \quad y(\pi/2) = 1 \text{ is :}$$

- (a) $\frac{1}{5}$ (b) 2
 (c) 1 (d) 0.

31. Let $J[y(x)]$ be functional defined by

$$J[y(x)] = \int_0^{\pi/2} \left[\left(\frac{dy}{dx} \right)^2 - y^2 \right] dx$$

$$y(0) = 0, \quad y(\pi/2) = 1$$

where the unknown function $y(x)$ possesses two derivatives everywhere in $\left(0, \frac{\pi}{2}\right)$, then : [GATE-2006]

- (a) The functional has an extremum which can not be achieved in the class of continuous function.
 (b) The corresponding Euler's equation does not have a unique solution satisfying the given boundary condition.
 (c) J is non-linear.
 (d) J is linear.

32. The extremal of the functional

$$I = \int_a^b \frac{\sqrt{1 + y'^2}}{x} dx$$

is :

- (a) Family of catenary (b) Family of parabolas
(c) Family of circles (d) None of these.

33. The functional

$$I[y(x)] = \int_0^1 (xy + y^2 - 2y^2 y') dx$$

$y(0) = 1, y(1) = 2$ has :

- (a) One extremal (b) No extremal
(c) Infinite extremal (d) More than one but finite extremal.

34. The functional

$$\int_0^1 (y'^2 + x^2) dx$$

with $y(1) = 1$ achieves its :

- (a) Weak maximum on all its extremals.
(b) Weak minimum on all its extremals.
(c) Weak maximum on some, but not on all of its extremals.
(d) Weak minimum on some but not all of its extremals.

35. The functional

$$I[y(x)] = \int_0^1 \left(x + 2y + \frac{1}{2} y'^2 \right) dx$$

$y(0) = 0, y(1) = 0$ has :

- (a) Weak minimum on the function $y = x^2 - x$
(b) Strong minimum on the function $y = x^2 - x$
(c) Weak maximum on the function $y = x^2 - x$
(d) Strong maximum on the function $y = x^2 - x$.

36. The weak minimum of the functional

$$I[y(x)] = \int_1^2 \frac{x^3}{y'^2} dx$$

$$y(1) = 1, \quad y(2) = 4$$

attained on the curve :

- (a) $y = x^3$ (b) $y = x^2$
 (c) $y = x^3 - x$ (d) None of these.

37. In which situation the variational problem ?

$$\int_{x_1}^{x_2} f(x, y, y') dx$$

$$y(x_1) = y_1, \quad y(x_2) = y_2$$

becomes meaningless :

- (a) When Euler's equation reduces into identity.
 (b) When $y(x)$ exists but it is not satisfy the given boundary condition.
 (c) When $f(x, y, y') = M(x, y) + N(x, y)$.
 (d) All of the above.

38. Euler's equation for the functional

$$I[z(x, y)] = \iint_D \left[\left(\frac{\partial z}{\partial x} \right)^2 - \left(\frac{\partial z}{\partial y} \right)^2 \right] dx dy$$

is :

- (a) $\frac{\partial^2 z}{\partial y^2} - \frac{\partial^2 z}{\partial x^2} = 0$ (b) $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$
 (c) $\frac{\partial^2 z}{\partial y^2} = 0$ (d) $\frac{\partial^2 z}{\partial x^3} + \frac{\partial^2 z}{\partial y^3} = 0$.

39. Euler's equation for the functional

$$\int_{x_1}^{x_2} [a(x) y'^2 + 2b(x) y + y' + c(x) y^2] dx :$$

- (a) First order linear differential equation.
- (b) Second order linear differential equation.
- (c) Second order non-linear differential equation.
- (d) A linear differential equation of order more than two.

40. Extremal $y = y(x)$ for the variational problem

$$I = \int_0^1 (1 + y''^2) dx$$

satisfies the ordinary differential equation is :

- (a) Homogeneous linear differential equation of fourth order.
- (b) Non-homogeneous linear differential equation of fourth order.
- (c) Homogeneous non-linear differential equation of fourth order.
- (d) Homogeneous linear differential equation of order more than fourth order.

41. The Euler's equation for a functional $\int_a^b F(x, y) dx$ is :

- (a) $F_{y'} = c_1$
- (b) $F_y - y' F_{y'} = c_1$
- (c) $F_y = 0$
- (d) None of these.

42. The extremals of the functional

$$I [y(x), z(x)] = \int_0^{\pi/2} (y'^2 + z'^2 + 2yz) dx$$

are solution of simultaneous differential equation :

- (a) $y'' - z = 0, z'' - y = 0$
- (b) $y'' + z = 0, z'' + y = 0$
- (c) $y'' - z = 0, z'' + y = 0$
- (d) $y'' + z = 0, z'' - y = 0$.

43. Euler's equations for the functional $\int_1^2 (y'^2 + z^2 + z'^2) dx$ are given by :

(a) $y'' = 0, z'' - z = 0$ (b) $y'' - y = 0, z'' = 0$

(c) $y'' + y = 0, z'' + z = 0$ (d) None of these.

44. The extremals of the functional

$$I[y(x), z(x)] = \int_0^{\pi/2} (y'^2 + z'^2 + 2yz) dx$$

$$y(0) = 0, y(\pi/2) = 1; z(0) = 0, z(\pi/2) = -1$$

occurs on the curves :

(a) $y = \cos x, z = \sin x$ (b) $y = \sin x, z = -\sin x$

(c) $y = \cos x, z = \cos x$ (d) $y = \cos x, z = -\sin x$.

45. Extremal for the functional

$$I = \int_1^2 (y'^2 + z^2 + z'^2) dx$$

satisfying the conditions

$$y(1) = 1, y(2) = 2$$

$$z(1) = 0, z(2) = 1 \text{ is :}$$

(a) $y(x) = x, z(x) = \frac{\sinh(x-1)}{\sinh(1)}$

(b) $y(x) = x, z(x) = \frac{\sin(x-1)}{\sin 1}$

(c) $y(x) = x, z(x) = \frac{\cosh(x-1)}{\cosh(1)}$

(d) None of these.

46. Extremal of the isometric problem $\int_{x_1}^{x_2} y'^2 dx$ subject to $\int_{x_1}^{x_2} y dx = c$ is :

(a) $y = \lambda x^2 + ax + b$ (b) $y = \lambda x^3 + ax + b$

(c) $ax + b$ (d) None of these.

47. The shortest distance between two points in a plane is :

- (a) Circle (b) Parabola
(c) Ellipse (d) Straight line.

48. Extremals of the functional $\int_{x_1}^{x_2} y \sqrt{1 + y'^2} dx$ is attained on the :

- (a) Catenary (b) Parabola
(c) Circle (d) Ellipse.

49. The corresponding variational problem of boundary value problem

$$\frac{d^2 y}{dx^2} + \lambda^2 y = 0$$

$$y(0) = 0 = y(l) \text{ is :}$$

- (a) $I[y(x)] = \int_0^1 (y'^2 - \lambda^2 y^2) dx$
(b) $I[y(x)] = \int_0^1 (y'' - \lambda^2 y^2) dx$
(c) $I[y(x)] = \int_0^1 (y'^2 + \lambda^2 y^2) dx$
(d) None of these.

50. Solution of the Poisson equation in the square $u_{xx} + u_{yy} = -1$; $|x| \leq 1$, $|y| \leq 1$, $u = 0$ at $x = \pm 1$ and $y = \pm 1$ is :

- (a) $u(x, y) = \frac{1}{16} (1 - x^2) (1 - y^2)$
(b) $u(x, y) = \frac{5}{16} (1 - x^2) (1 - y^2)$
(c) $u(x, y) = \frac{5}{16} (1 + x^2) (1 + y^2)$
(d) None of these.

Answers

Objective Type Questions

1.	(c)	2.	(a), (b), (c), (d)			3.	(a), (b), (c)			4.	(d)
5.	(a), (d)		6.	(d)	7.	(a), (b)		8.	(c)	9.	(d)
10.	(d)	11.	(d)	12.	(b)	13.	(c)	14.	(d)	15.	(a)
16.	(a)	17.	(b)	18.	(c)	19.	(b)	20.	(a)	21.	(b)
22.	(d)	23.	(d)	24.	(b)	25.	(b)	26.	(c)	27.	(d)
28.	(d)	29.	(a)	30.	(d)	31.	(c)	32.	(c)	33.	(b)
34.	(b)	35.	(b)	36.	(b)	37.	(d)	38.	(a)	39.	(b)
40.	(a)	41.	(c)	42.	(a)	43.	(a)	44.	(b)	44.	(b)
45.	(a)	46.	(d)	47.	(d)	48.	(a)	49.	(a)	50.	(b)

Hints and Solutions

1. (c) See problem 8 on page 30.

3. (a), (b), (c)

$$I[y] = \int_{x_1}^{x_2} \sqrt{1 + y'^2} \, dx$$

$$\therefore F(x, y, y') = \sqrt{1 + y'^2}$$

$$\Rightarrow \frac{\partial F}{\partial y'} = \frac{y'}{\sqrt{1 + y'^2}}$$

$\therefore$ By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 0 - \frac{d}{dx} \left(\frac{y'}{\sqrt{1+y'^2}} \right) = 0$$

$$\Rightarrow \frac{d}{dx} \left(\frac{y'}{\sqrt{1+y'^2}} \right) = 0.$$

Integrating, we get

$$\frac{y'}{\sqrt{1+y'^2}} = c$$

$$\Rightarrow y'^2 = c^2 (1 + y'^2)$$

$$\Rightarrow y'^2 (1 - c^2) = c^2$$

$$\Rightarrow y' = \frac{c}{\sqrt{1-c^2}} = c_1 \text{ (say)}$$

$$\therefore y' = c_1.$$

Integrating

$$y = c_1 x + c_2 \quad \dots(1)$$

Since extremal is fixed at $\left(\frac{1}{2}, \frac{1}{4}\right)$, therefore

$$y\left(\frac{1}{2}\right) = \frac{1}{4}$$

$$\Rightarrow \frac{1}{4} = c_1 \cdot \frac{1}{2} + c_2$$

$$\text{or} \quad 2c_1 + 4c_2 = 1. \quad \dots(2)$$

$\therefore$ Option (a) is true.

Since, the right end of extremal moves along the line

$$y = x - 5$$

$$\Rightarrow y(5) = 0$$

$$\Rightarrow 0 = c_1 \cdot 5 + c_2. \quad \dots(3)$$



Solving (1) and (2), we get

$$c_1 = -\frac{1}{18} \text{ and } c_2 = \frac{5}{18}.$$

Putting these value in (1), we get

$$y = -\frac{x}{18} + \frac{5}{18}$$

$$\Rightarrow y = -\frac{x+5}{18}$$

which is required extremal.

Since the extremal $y = c_1x + c_2$ movable along the straight line $y = x - 5$.

$$\therefore y_1 = c_1x_1 + c_2 \text{ and } y_1 = x_1 - 5$$

$$\Rightarrow c_1x_1 + c_2 = x_1 - 5.$$

$\therefore$ Option (c) is true.

Now here $\psi(x) = x - 5$, $y = c_1x + c_2$ and $y' = c_1$ and $F = \sqrt{1 + y'^2}$.

Therefore by transversality condition

$$F - (y' - \psi') \frac{\partial F}{\partial y'} = 0$$

$$\Rightarrow \sqrt{1 + y'^2} - (y' - 1) \cdot \frac{y'}{\sqrt{1 + y'^2}} = 0$$

$$\Rightarrow \sqrt{1 + y'^2} - \frac{y'(y' - 1)}{\sqrt{1 + y'^2}} = 0.$$

Put $y' = c_1$, we get

$$\sqrt{1 + c_1^2} + (1 - c_1) \frac{c_1}{\sqrt{1 + c_1^2}} = 0.$$

$\therefore$ Option (b) is true.

4. (d) We have

$$F = y(3x - y) = 3xy - y^2.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow (3x - 2y) - \frac{d}{dx} (0)$$

$$\Rightarrow 3x - 2y = 0$$

$$\Rightarrow y = \frac{3}{2} x.$$

This is not extremal, because it does not satisfy boundary conditions.

6. (d) $F(y, y') = y + y'.$

By necessary condition for extremal

$$F - y' F_{y'} = \text{constant}$$

$$\Rightarrow y + y' - y' \times 1 = \text{constant}$$

$$\Rightarrow y = \text{constant}.$$

$$\therefore y(x) = \text{constant as an extremal}$$

$\therefore$ Option (d) is true.

7. (a), (b)

$$F = \sqrt{1 + (y'(x))^2}$$

$$= \sqrt{1 + y'^2(x)}.$$

$$\therefore \frac{\partial F}{\partial y} = 0 \text{ and } \frac{\partial F}{\partial y'} = \frac{y'}{\sqrt{1 + y'^2}}.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$



$$\Rightarrow 0 - \frac{d}{dx} \left(\frac{y'}{\sqrt{1+y'^2}} \right) = 0$$

$$\Rightarrow \frac{d}{dx} \left(\frac{y'}{\sqrt{1+y'^2}} \right) = 0.$$

Integrating, we get

$$\frac{y'}{\sqrt{1+y'^2}} = c = \text{constant}$$

$$y' = c\sqrt{1+y'^2}. \quad \dots(1)$$

$\therefore$ Option (b) is true.

Now from (1)

$$y'^2 = c^2 (1 + y'^2)$$

$$\Rightarrow (1 - c^2) y'^2 = c^2$$

$$\Rightarrow y'^2 = \frac{c^2}{1 - c^2}$$

$$\Rightarrow y' = \frac{c}{\sqrt{1 - c^2}} = c_1 \text{ (say).}$$

Integrating, we get

$$y = c_1 x + c_2.$$

This is straight line joining $(a, y(a))$ and $(b, y(b))$.

$\therefore$ Option (a) is true.

8. (c) $F = y'^2 + 12xy.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 12x - \frac{d}{dx}(2y') = 0$$

$$\Rightarrow 6x - \frac{d^2 y}{dx^2} = 0$$

$$\Rightarrow \frac{d^2 y}{dx^2} = 6x.$$

Integrating, we get

$$\frac{dy}{dx} = 3x^2 + c_1.$$

Again integrating, we get

$$y = x^3 + c_1 x + c_2 \quad \dots(1)$$

$$y(0) = 0 \Rightarrow c_2 = 0$$

$$y(1) = 1 \Rightarrow c_1 + c_2 + 1 = 1$$

$$\Rightarrow c_1 = 0.$$

$\therefore$ (1) becomes

$$y = x^3.$$

$\therefore$ Option (c) is true.

9. (d) $F = xy + y^2 - 2y^2 y'.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow x + 2y - 4yy' - \frac{d}{dx}(-2y)^2 = 0$$

$$\Rightarrow x + 2y - 4yy' + 4yy' = 0$$

$$\Rightarrow x + 2y = 0$$



$$\Rightarrow y = -\frac{x}{2}.$$

$$\therefore y(0) = 0, y(\pi/2) = -1/2.$$

$\therefore y = -x/2$ does not satisfy given boundary condition. Hence the problem is meaningless.

$\therefore$ Option (d) is true.

$$\begin{aligned} 12. \quad (b) \quad F &= (y + y')^2 \\ &= y^2 + y'^2 + 2yy'. \end{aligned}$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 2y + 2y' - \frac{d}{dx} (2yy') = 0$$

$$\Rightarrow y + y' - y'' - y' = 0$$

$$\Rightarrow y'' - y = 0.$$

$\therefore$ Option (b) is true.

$$13. \quad (c) \quad F = y^2 + x^2 y'^2.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 2y - \frac{d}{dx} (x^2 2y') = 0$$

$$\Rightarrow y - \frac{d}{dx} (x^2 y') = 0$$

$$\text{or} \quad y - x^2 y'' - 2xy' = 0$$

$$\Rightarrow x^2 y'' + 2xy' - y = 0.$$

$\therefore$ Option (c) is true.

14. (d) $I[y(x)] = \int_a^b (y' + 2y) dx.$

$$\begin{aligned} \therefore I[c_1 y_1(x) + c_2 y_2(x)] &= \int_a^b [(c_1 y_1(x) + c_2 y_2(x))' + 2(c_1 y_1(x) + c_2 y_2(x))] dx \\ &= \int_a^b [c_1 y_1'(x) + c_2 y_2'(x) + 2c_1 y_1(x) + 2c_2 y_2(x)] dx \\ &= c_1 \int_a^b [y_1'(x) + 2y_1(x)] dx + c_2 \int_a^b [y_2'(x) + 2y_2(x)] dx \\ &= c_1 I[y_1(x)] + c_2 I[y_2(x)]. \end{aligned}$$

$$\therefore I[c_1 y_1(x) + c_2 y_2(x)] = c_1 I[y_1(x)] + c_2 I[y_2(x)]$$

$\Rightarrow I$ is linear.

$\therefore$ Option (d) is true.

17. (b) $F = e^{-x} y'^2 - e^x y^2.$

By Euler's equation

$$\begin{aligned} \frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) &= 0 \\ \Rightarrow -e^x 2y - \frac{d}{dx} (e^{-x} 2y') &= 0 \\ \Rightarrow e^x y + e^{-x} y'' - e^{-x} y' &= 0 \\ \Rightarrow y'' - y' + e^{2x} y &= 0. \end{aligned}$$

$\therefore$ Option (b) is true.

18. (c) $F = y'(1 + x^2 y')$

$$= y' + x^2 y'^2.$$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 0 - \frac{d}{dx} (1 + 2x^2 y') = 0$$

$$\Rightarrow \frac{d}{dx} (1 + 2x^2 y') = 0$$

$$\Rightarrow 0 + 2x^2 y'' + 4xy' = 0$$

$$\Rightarrow xy'' + 2y' = 0.$$

$\therefore$ Option (c) is true.

19. (b) $F = y'^2 + 12xy.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 12x - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow \frac{d^2 y}{dx^2} = 6x.$$

Integrating, we get

$$y = x^3 + c_1 x + c_2.$$

$\therefore$ Option (b) is true.

20. (a) See problem (4) on page 24.

21. (b) $F = y'^2 + 4y^2.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 8y - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow 4y - \frac{d^2 y}{dx^2} = 0$$

or $\frac{d^2 y}{dx^2} - 4y = 0.$

Auxiliary equation is

$$m^2 - 4 = 0$$

$$\Rightarrow m = \pm 2.$$

$\therefore$ Solution is

$$y = c_1 e^{2x} + c_2 e^{-2x}. \quad \dots(1)$$

Now, $y(0) = e^2 \Rightarrow c_1 + c_2 = e^2. \quad \dots(2)$

$$y(1) = 1 \Rightarrow c_1 e^2 + c_2 e^{-2} = 1. \quad \dots(3)$$

Solving (2) and (3), we get

$$c_1 = 0, \quad c_2 = e^2.$$

Putting value of c_1 and c_2 in (1), we get

$$y = e^2 e^{-2x} = e^{2-2x}.$$

$\therefore$ Option (b) is true.

22. (d) $F = y'^2 + 6xy + x^3.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 6x - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow 3x - \frac{d}{dx} \left(\frac{dy}{dx} \right) = 0$$

$$\Rightarrow \frac{d^2 y}{dx^2} - 3x = 0$$

$$\Rightarrow \frac{d^2 y}{dx^2} = 3x.$$

Integrating, we get

$$y = \frac{x^3}{2} + c_1 x + c_2 \quad \dots(1)$$

$$\left. \begin{array}{l} y(0) = 0 \Rightarrow c_2 = 0 \\ y(2) = 2 \Rightarrow 2c_1 + c_2 = -2 \end{array} \right\} \Rightarrow \begin{array}{l} c_1 = -1 \\ c_2 = 0 \end{array}$$

Therefore, (1) becomes

$$y = \frac{x^3}{2} - x$$

$$\Rightarrow 2y = x^3 - 2x.$$

$\therefore$ Option (d) is true.

23. (d) $F = y''^2 - y^2 + x^2.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) = 0$$

$$\Rightarrow -2y - \frac{d}{dx} (0) + \frac{d^2}{dx^2} (2y'') = 0$$

$$\Rightarrow \frac{d^4 y}{dx^4} - y = 0.$$

Auxiliary equation is

$$m^4 - 1 = 0$$

$$\Rightarrow (m^2 - 1)(m^2 + 1) = 0$$

$$\Rightarrow m = \pm 1, \pm i.$$

$\therefore$ Solution is

$$y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x.$$

This is required extremal which contain four constants.

Hence extremal is four parameter family of curves.

$\therefore$ Option (d) is true.

24. (b) $F = 2xy + y'''^2.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) - \frac{d^3}{dx^3} \left(\frac{\partial F}{\partial y'''} \right) = 0$$

$$\Rightarrow 2x - \frac{d}{dx} (0) + \frac{d^2}{dx^2} (0) - \frac{d^3}{dx^3} (2y''') = 0$$

$$\Rightarrow \frac{d^6 y}{dx^6} = x.$$

Since this is six order differential equation therefore its solution contain six arbitrary constants.

Hence extremal is six parameter family of curves.

$\therefore$ Option (b) is true.

26. (c) $F = 3y^2 + 2x^3 y'.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 6y - \frac{d}{dx} (2x^3) = 0$$

$$\Rightarrow 6y - 2 \times 3x^2 = 0$$

$$\Rightarrow y = x^2$$

$$\begin{aligned} \therefore y(\alpha) = 1 &\Rightarrow \alpha^2 = 1 \\ &\Rightarrow \alpha = \pm 1. \end{aligned}$$

$\therefore$ Option (c) is true.

27. (d) See problem 1 on page 69, we get

$$y = (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x \quad \dots(1)$$

and $z = (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x - 2c_2 \sin x + 2c_4 \cos x \quad \dots(2)$

$$\therefore y(0) = 0 \Rightarrow c_1 = 0$$

$$y(\pi) = 1 \Rightarrow -c_2 \cdot \pi = 1 \Rightarrow c_2 = -\frac{1}{\pi}.$$



Therefore (1) becomes

$$y = -\frac{1}{\pi} x \cos x + (c_3 + c_4 x) \sin x.$$

This gives infinite extremal.

Similarly, (2) gives infinite extremal.

$\therefore$ Option (d) is true.

28. (d) $F = xy' + y'^2.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 0 - \frac{d}{dx} (x + 2y') = 0$$

$$\Rightarrow 1 + 2 \frac{d^2 y}{dx^2} = 0$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -\frac{1}{2}.$$

Integrating, we get

$$y = c_1 + c_2 x - \frac{x^2}{4}.$$

$\therefore$ Option (d) is true.

29. (a) $F = y'^2 + 2yy' - 16y^2.$

By Euler's equation equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 2y' - 32y - \frac{d}{dx} (2y' + 2y) = 0$$

$$\Rightarrow y' - 16y - y'' - y' = 0$$

$$\Rightarrow y'' + 16y = 0$$

i.e., $\frac{d^2 y}{dx^2} + 16y = 0.$

Auxiliary equation is

$$m^2 + 16 = 0$$

$$\Rightarrow m = \pm 4i.$$

$\therefore$ Solution is

$$y = c_1 \cos 4x + c_2 \sin 4x. \quad \dots(1)$$

Now,

$$y(0) = 0 \Rightarrow c_1 = 0$$

$$y(\pi/8) = 1 \Rightarrow c_2 \sin \pi/2 = 1$$

$$\Rightarrow c_2 = 1.$$

Therefore (1) becomes

$$y = \sin 4x.$$

$\therefore$ Option (a) is true.

$$30. \quad (d) \quad I = \int_0^{\pi/2} (y'^2 - y^2) dx. \quad \dots(1)$$

Here $F = y'^2 - y^2.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow -2y - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow \frac{d^2 y}{dx^2} + y = 0.$$

$\therefore$ The solution is

$$y = c_1 \cos x + c_2 \sin x$$

$$y(0) = 0 \Rightarrow c_1 = 0$$

$$y(\pi/2) = 1 \Rightarrow c_2 = 1.$$

$\therefore$ Required extremal is

$$y = \sin x.$$

$$\therefore y' = \cos x.$$

Therefore from (1) extremum value is

$$\begin{aligned}
 &= \int_0^{\pi/2} (\cos^2 x - \sin^2 x) dx \\
 &= \int_0^{\pi/2} \cos 2x dx \\
 &= \left[\frac{\sin 2x}{2} \right]_0^{\pi/2} \\
 &= \frac{1}{2} [0 - 0] \\
 &= 0.
 \end{aligned}$$

$\therefore$ Option (d) is true.

32. (c) See Example (5) on page 13.

33. (b) $F = xy + y^2 - 2y^2 y'.$

By Euler's equation

$$\begin{aligned}
 &\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0 \\
 \Rightarrow &x + 2y - 4yy' - \frac{d}{dx} (-2y^2) = 0 \\
 \Rightarrow &x + 2y - 4yy' + 4yy' = 0 \\
 \Rightarrow &x + 2y = 0 \\
 \Rightarrow &y = -\frac{x}{2}.
 \end{aligned}$$

This extremal does not satisfy given boundary condition.

$$i.e., \quad y(1) = -\frac{1}{2} \neq 2$$

$\Rightarrow$ This gives no extremal.

$\therefore$ Option (b) is true.

34. (b) $F = y'^2 + x^2.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$$\Rightarrow 0 - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow \frac{d^2 y}{dx^2} = 0.$$

Integrating, we get

$$y = c_1 x + c_2$$

$$y(1) = 1 \Rightarrow c_1 + c_2 = 1$$

$$\Rightarrow c_2 = -c_1.$$

$$\therefore y = c_1 x - c_1$$

$$\Rightarrow y' = c_1.$$

Now $F = y'^2 + x^2$

$$\begin{aligned} F_{y'} &= 2y' \\ &= 2c_1 > 0. \end{aligned}$$

$\therefore$ Weak minimum on all its extremals.

$\therefore$ Option (b) is true.

35. (b) See the problem (3) on page 158.

38. (a) $F = z_x^2 - z_y^2.$

By Euler's equation

$$\frac{\partial F}{\partial z} - \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial z_x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial z_y} \right) = 0$$

$$0 - \frac{\partial}{\partial x} (2z_x) - \frac{\partial}{\partial y} (-2z_y) = 0$$

$$\Rightarrow \frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = 0$$

$\therefore$ Option (a) is true.

40. (a) $F = 1 + y''^2.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) = 0$$

$$\Rightarrow 0 - \frac{d}{dx} (0) + \frac{d^2}{dx^2} (2y'') = 0$$

$$\Rightarrow \frac{d^4 y}{dx^4} = 0.$$

This is homogeneous linear differential equation of fourth order.

$\therefore$ Option (a) is true.

42. (a) $F = y'^2 + z'^2 + 2yz = 0.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

and $\frac{\partial F}{\partial z} - \frac{d}{dx} \left(\frac{\partial F}{\partial z'} \right) = 0$

$$\Rightarrow 2z - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow y'' - z = 0$$

and $2y - \frac{d}{dx} (2z') = 0$

$$\Rightarrow z'' - y = 0.$$

$\therefore$ Option (a) is true.

43. (a) $F = y'^2 + z^2 + z'^2.$

By Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

and
$$\frac{\partial F}{\partial z} - \frac{d}{dx} \left(\frac{\partial F}{\partial z'} \right) = 0$$

$\Rightarrow 0 - \frac{d}{dx} (2y') = 0$

$\Rightarrow y'' = 0$

and
$$2z - \frac{d}{dx} (2z') = 0$$

$\Rightarrow z'' - z = 0.$

$\therefore$ Option (a) is true.

44. (b) See problem ((2) on page 71.

49. (a) Consider
$$I[y(x)] = \int_0^1 (y'^2 - \lambda^2 y^2) dx. \quad \dots(1)$$

Here
$$F = y'^2 - \lambda^2 y^2.$$

Then by Euler's equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0$$

$\Rightarrow -2\lambda^2 y - \frac{d}{dx} (2y') = 0$

$\Rightarrow \lambda^2 y + \frac{d^2 y}{dx^2} = 0$

i.e.,
$$\frac{d^2 y}{dx^2} + \lambda^2 y = 0$$

Hence Euler's equation of functional (1) gives given differential equation.

$\therefore$ Option (a) is true.

50. (b) See the problem (1) on page 197.